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TOPIC: PROPERTIES OF MATTER

KINETIC THEORY OF MATTER

THE THREE STATES OF MATTER

What is matter?

This word is used to cover all the substances and materials from which the physical universe is composed.

Matter is defined as anything that occupies space and has mass. There are many millions of different substances known, and all of them can be categorised as solids, liquids or gases. These are what we call the three states of matter.

SOLIDS

A solid at a given temperature, has a definite volume and shape which may be affected by changes in temperature. Solids usually increase slightly in size when heated (expansion) and usually decrease in size if cooled (contraction).

LIQUIDS

A liquid, at a given temperature, has a fixed volume and will take up the shape of any container into which it is poured. Like a solid, a liquid's volume is slightly affected by the changes in temperature.

GASES

A gas, at a given temperature, has neither a definite (fixed) shape nor a definite volume. It will take up the shape of any container into which it is placed and will spread out evenly within it. The volume of a gas is affected quite markedly by changes in temperature.

Liquids and gases, unlike solids, are relatively compressible. This means that their volume can be reduced by the application of pressure. Gases are much more compressible than liquids.

THE KINETIC THEORY MATTER

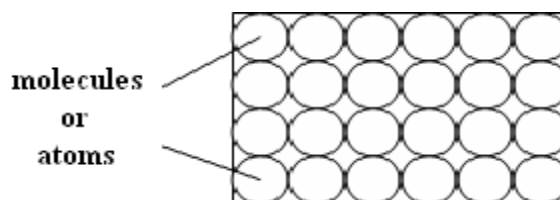
The theory helps to explain the way in which matter behaves. The evidence is consistent with the idea that all matter is made up of

tiny particles. This theory explains the physical properties of matter in terms of the movement of its constituent particles. The main points of the theory are:

- All matter is made up of tiny, moving particles invisible to the naked eye. Different substances have different types of particles (atoms, molecules or ions) which have different sizes.
- The particles move all the time. The higher the temperature, the faster they move on average.
- Heavier particles move more slowly than lighter ones at a given temperature.

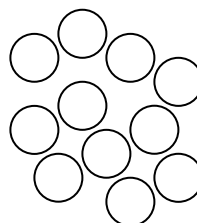
EXPLAINING THE STATES OF MATTER

SOLIDS



- There are strong attractive forces between the particles which hold them close together.
- The particles have little freedom of movement and can only vibrate about a fixed position.
- The particles are arranged in a regular manner, which explains why many solids form crystals.

LIQUIDS

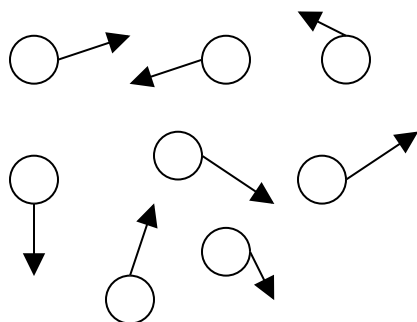


- The particles are still close together but

they move around in a random way and often collide with one another.

- The forces of attraction between the particles in a liquid are weaker than those in a solid.
- The particles are irregularly packed (arranged).
- The particles in a liquid have more energy on average than the particles in the solid substance.

GASES



- The particles are relatively far apart. They are free to move anywhere within the container in which they are held.
- Particles move randomly at very high velocities (speeds), much more rapidly than those in a liquid.
- Particles exert virtually no forces of attraction on each other because they are relatively far apart.

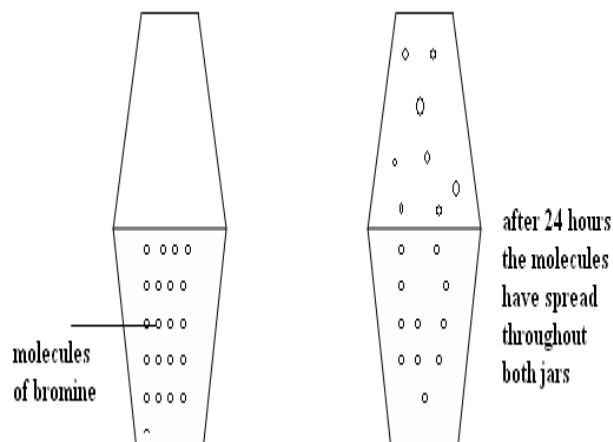
DIFFUSION

If one passes by a cosmetic counter they usually smell the perfumes. For this to happen gas particles must be leaving open perfume bottles and be spreading out through the air in the store. This spreading out of a gas is called **diffusion**. Diffusion takes place in a haphazard and random way.

Definitions:

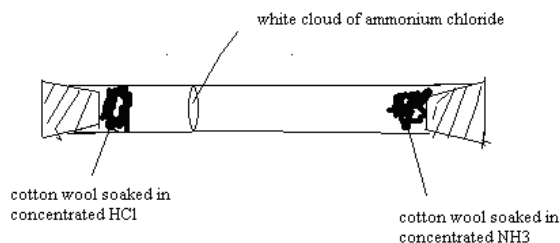
- Diffusion is the process by which different substances mix as a result of the random motions of their particles.
- Diffusion is the spreading of a substance of its own accord due to molecular motion. All gases diffuse to fill the space available to them. For example, the red-brown fumes of gaseous bromine spread evenly

throughout gas jars from the liquid present in the lower gas jar.



After 24 hours the bromine fumes have diffused throughout both gas jars.

Gases diffuse at different rates. The speed of diffusion depends on the speed of its molecules and is greater for lighter molecules. For instances, if one piece of cotton wool is soaked in concentrated ammonia solution and another is soaked in concentrated hydrochloric acid and these are put at opposite ends of a dry glass tube, then after a few minutes a white cloud ammonium chloride appears.



This shows the position at which the two gases meet and react. The white cloud forms in the position shown because the ammonia particles are lighter than the hydrogen chloride particles (released from the hydrochloric acid) and so move faster. Generally, lighter particles move faster than heavier ones at a given temperature.

Diffusion also takes place in liquids but it is much slower process than in gases. This is because the particles of a liquid move much more slowly.

When diffusion takes between a liquid and a

gas it is known as **intimate mixing**.

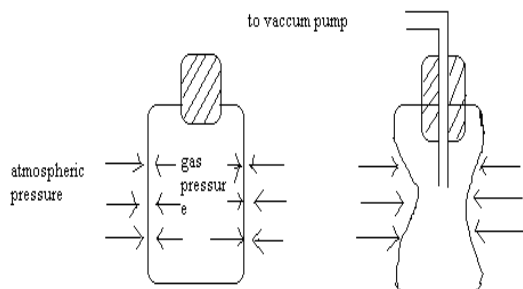
BROWNIAN MOTION

When fine pollen grains are on the surface of water, they move about in a random way. It was discovered that the pollen grains move because the much smaller and faster-moving water particles are constantly colliding with them. This random motion of visible particles (i.e. pollen grains) caused by much smaller, invisible ones (water molecules) is called **Brownian motion**.

THE GAS LAWS

PRESSURE OF THE GAS

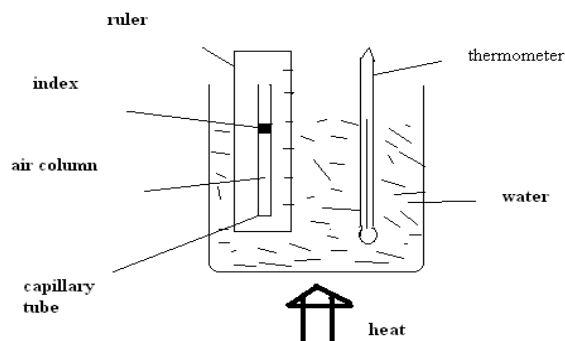
Air has weight and because of this weight the atmosphere exerts a large pressure at sea level, about $100000\text{N/m}^2 = 10^5 = 100\text{kPa}$. This pressure acts equally in all directions. A gas in a container exerts a pressure on the walls of the container. If air is removed from the can by a vacuum pump, the can collapses because the air pressure outside is greater than inside.



The can collapses because there is unbalanced pressure acting on the walls of the can i.e. no gas pressure acting on the wall of the container.

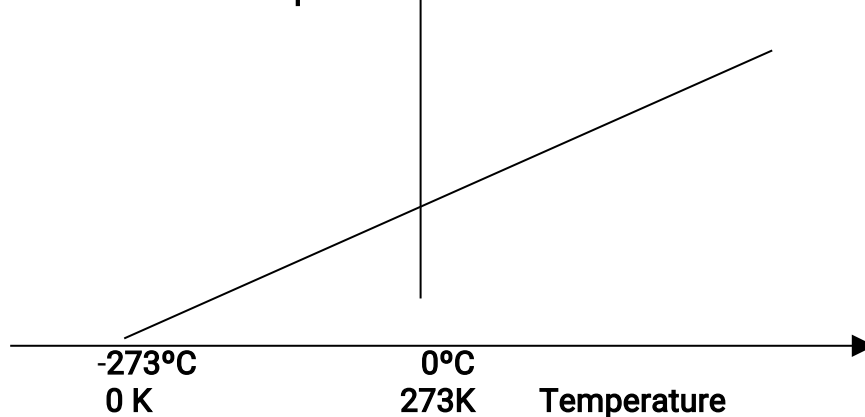
EFFECT ON VOLUME OF A GAS (FIXED MASS) IF TEMPERATURE IS CHANGED (AT CONSTANT PRESSURE)

Arrange the apparatus as shown in the figure below.



The initial readings of the index (lower end) and thermometer are noted. At different temperatures, the readings of the index are recorded. Put the results in a table. Plot a graph of volume on the y-axis and temperature (in $^{\circ}\text{C}$) on the x-axis.

Volume or pressure



The volume-temperature graph for a gas is a straight line. It shows that gases expand uniformly with temperature i.e. equal temperature rises (increases) cause equal volume increases.

The graphs do not pass through the Celsius

temperature origin ($^{\circ}\text{C}$). If they are produced backwards they cut the temperature axis at about -273°C . This temperature is called **absolute zero**, because it is believed it is the lowest temperature possible. At absolute zero molecular motion ceases (stops) and a substance has no internal energy.

Definitions:

- **Absolute zero** is the lowest possible temperature for all substances.
- **Absolute temperature** is a temperature measured with respect to absolute zero on the Kelvin scale.

The conversions from $^{\circ}\text{C}$ to K are made by adding 273. For example

$$0^{\circ}\text{C} = 273 \text{ K}$$

$$15^{\circ}\text{C} = 273 + 15 = 288\text{K}$$

$$100^{\circ}\text{C} = 273 + 100 = 373\text{K}$$

Kelvin or absolute temperatures are represented by the letter T and if θ stands for a Celsius scale temperature then, in general:

$$T = 273 + \theta$$

The conversions from K to $^{\circ}\text{C}$ are made by subtracting 273. For example,
 $373\text{K} = 373 - 273 = 100^{\circ}\text{C}$

CHARLES' LAW:

The volume of a fixed mass of gas is directly proportional to its absolute temperature if the pressure is kept constant.

$$V \propto T \text{ or } V = \text{constant} \times T$$

$$V/T = \text{constant}$$

$$\text{Therefore, } V_1/T_1 = V_2/T_2$$

Worked example:

A gas of volume 2m^3 at 27°C is cooled to -123°C at constant pressure. What is the new volume?

Solution:

$$V_1 = 2\text{m}^3, \quad T_1 = (27 + 273) = 300\text{K}, \quad T_2 = -123 + 273 = 150\text{K}, \quad V_2 = ?$$

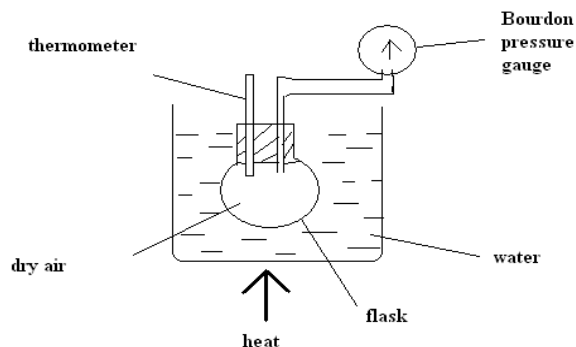
$$\text{Substituting in } V_1/T_1 = V_2/T_2$$

$$V_2 = \frac{2 \times 150}{300} = 1\text{m}^3$$

RELATIONSHIP BETWEEN PRESSURE AND

TEMPERATURE AT CONSTANT VOLUME

In an experiment (shown in the figure below), the air is enclosed in a sealed flask in order to keep its volume constant. The temperature of the air is increased in stages by heating the water around the flask. At each stage, the pressure of the air is measured on the Bourdon gauge which is attached to the flask by a short length of tubing.



The readings are used to plot a graph showing how the pressure of the air varies with temperature. A graph plotted shows a straight line passing through the origin (0 K). The graph shows all usual signs of a simple proportion i.e.

- Doubling the absolute temperature doubles its pressure
- Dividing the pressure by absolute temperature always produces the same value

These findings can be summed up in Pressure law.

PRESSURE LAW:

The pressure of a fixed mass of gas is directly proportional to its absolute temperature, provided the volume of the gas is kept constant.

$$P \propto T \text{ or } P = \text{constant} \times T$$

$$P/T = \text{constant}$$

$$\text{Therefore, } P_1/T_1 = P_2/T_2$$

Worked example:

A gas in a fixed container is at a pressure of 1200mmHg and a temperature of 27°C . What will be its pressure if it is heated to a

temperature of 177°C?

$$P_1 = 1200\text{mmHg}, \quad T_1 = (27 + 273) = 300\text{K},$$

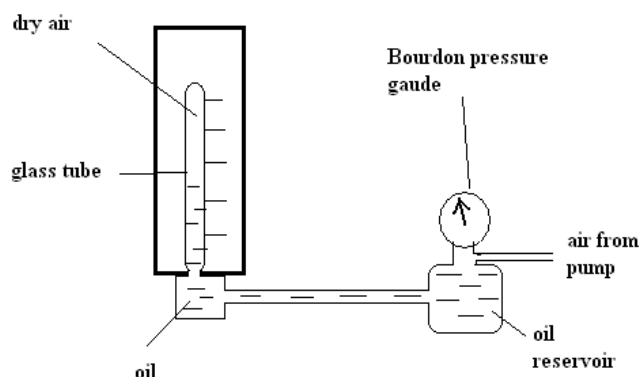
$$T_2 = 177 + 273 = 450\text{K}, \quad P_2 = ?$$

Substituting in $P_1/T_1 = P_2/T_2$

$$P_2 = \frac{1200 \times 450}{300} = 1800\text{mmHg}$$

RELATIONSHIP BETWEEN PRESSURE AND VOLUME AT CONSTANT TEMPERATURE

An experiment is arranged as shown in the diagram below.



The volume of the air is reduced in stages by pumping outside air into the oil reservoir. This forces more oil up into the glass tube. The pressure of the air is measured for several different volumes. When the readings are plotted on the graph it is discovered that pressure and volume have an inverse proportion. This inverse proportion can be recognised in two ways

- (i) when the volume is reduced, the pressure increases
- (ii) when the volume is halved, the pressure is doubled, and so on.

These findings can be expressed in the form of Boyle's law.

BOYLE'S LAW:

The pressure of a fixed mass of gas is inversely proportional to its volume provided the temperature of the gas is kept constant.

$$P \propto \frac{1}{V} \quad \text{or} \quad PV = \text{constant}$$

$$P_1 \times V_1 = P_2 \times V_2$$

Worked example:

If a certain quantity of gas has volume 30cm^3 at a pressure of $1 \times 10^5 \text{ Pa}$, what is its volume when the pressure is $2 \times 10^5 \text{ Pa}$?

$$V_1 = 30\text{cm}^3, \quad P_1 = 1 \times 10^5 \text{ Pa}, \quad P_2 = 2 \times 10^5 \text{ Pa}, \quad V_2 = ?$$

$$\text{Substituting in} \quad P_1 \times V_1 = P_2 \times V_2$$

$$V_2 = \frac{30 \times 1 \times 10^5}{2 \times 10^5} = 15\text{cm}^3$$

THE COMBINED GAS EQUATION

By combining the gas laws it is possible to show the relationship between the pressure, volume and temperature, of a fixed mass of gas as;

$$\frac{PV}{T} = \text{constant}$$

$$\text{Or} \quad \frac{P_1 \times V_1}{T_1} = \frac{P_2 \times V_2}{T_2}$$

Worked example:

A quantity of gas occupies a volume of 4m^3 . The pressure of the gas is 3 atmospheres when its temperature is 27°C . What will be its pressure if it is compressed into half the volume and heated to a temperature of 127°C ?

$$P_1 = 3\text{atm}, \quad V_1 = 4\text{m}^3, \quad T_1 = (273 + 27) = 300\text{K},$$

$$V_2 = 2\text{m}^3, \quad T_2 = (273 + 127) = 400\text{K}$$

$$\text{Substituting in the formula; } \frac{P_1 \times V_1}{T_1} = \frac{P_2 \times V_2}{T_2}$$

$$\frac{3 \times 4}{300} = \frac{P_2 \times 2}{400}$$

$$P_2 = 8\text{atm}$$

Work:

1. A container holds gas at 27°C. To what temperature must it be heated for its pressure to double? Assume the volume of the gas remains unchanged and the container does not leak.
2. A gas of volume 2m³ at 27°C is heated to 327°C. What is its new volume?
3. If a certain quantity of gas has volume 30cm³ at a pressure of 1 x 10⁵ Pa, what is its volume when pressure is 5 x 10⁵ Pa?
4. The volume of the cylinder is 0.10m³. It contains helium gas at a pressure of 1.0 x 10⁷ Pa. the balloon seller fills each balloon to a volume of 1.0 x 10⁻²m³ and a pressure of 1.2 x 10⁵ Pa.
 - a. Explain in terms of particles, how the helium in the cylinder produces pressure.
 - b. Calculate the total volume that the helium gas will occupy at a pressure of 1.2 x 10⁵ Pa. (Assuming no change in the temperature of helium).
 - c. Calculate the number of balloons of volume 1.0 x 10⁻²m³ that the balloon seller can fill using the gas.
5. A gas syringe contains 50 cm³ of oxygen gas at 20°C. If the temperature was increased to 45°C, what would be the volume occupied by this gas, assuming constant pressure throughout?
6. A bicycle pump contains 50cm³ of air at a pressure of 1 x 10⁵ Pa. What would be the volume of the air if the pressure was increased to 2.1 x 10⁵ Pa at constant temperature?
7. If the volume of a gas collected at 60°C and 1 x 10⁵ Pa pressure was 70cm³, what would be the volume at a temperature of

0°C

and pressure of 4 x 10⁵ Pa?

8. A gas occupies a volume of 2m³ when its pressure is 1140mmHg and its temperature is 27°C. What volume would it occupy at standard temperature and pressure (0°C and 760mmHg)?

PRESSURE IN LIQUIDS

Pressure is the force (or thrust) acting on a unit area.

Pressure is calculated by dividing the force or thrust acting at right angles to a surface by the area over which it acts:

$$\text{Pressure} = \frac{\text{force}}{\text{Area}}$$

If force is measured in Newton (N) and area in square metres (m²), pressure is measured in N/m² or Pascal (Pa).

$$1 \text{ N/m}^2 \equiv 1 \text{ Pa}$$

For example,

If a 100N force acts on an area of 1m², the pressure = 100N/1m² = 100N/m² or 100Pa.

If a 100N force acts on an area of 2m², the pressure = 100N/2m² = 50N/m² or 50 Pa.

The larger the force and the smaller the area, the greater is the pressure.

Worked examples:

1. The average wind pressure on the wall is 100Pa. if the area of the wall is 6m², what is the total force acting?

$$P = \frac{F}{A}$$

$$F = P \times A = 100\text{Pa} \times 6\text{m}^2 = 600\text{N}$$

2. A concrete block measuring 0.5m x 1m x 2m

has a mass of 2600 Kg. What is the maximum pressure it can exert on the ground?

$$\begin{aligned} \text{Downward force (weight)} &= 2600 \times 10 \\ &= 2600 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Area over which this force acts} &= 0.5\text{m} \times 1\text{m} \\ &= 0.5\text{m}^2 \end{aligned}$$

$$\text{Pressure} = \frac{26000\text{N}}{0.5\text{m}^2} = 52000 \text{ Pa}$$

LIQUID PRESSURE

- Pressure in a liquid increases with depth. This is the case because the farther down you go the greater the weight of the liquid above. In the **figure (i)** below, water spurts out fastest and furthest from the lowest hole.

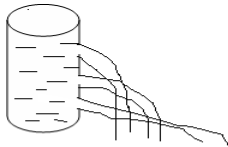


Figure (i)

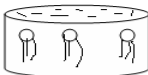


Figure (ii)



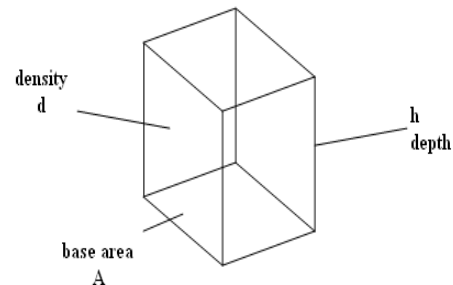
Figure (iii)

- Pressure at one level (depth) acts equally in all directions. The can of water in the figure below has similar holes all round it at the same level. Water comes out equally fast and equally far from each hole as shown in the **figure (ii)** above. Hence the pressure exerted by the water at this depth is the same in all directions.
- A liquid finds its own level. The liquid is at the same level in each tube in the figure below and this confirms that the pressure at the foot of a liquid column depends only on the vertical depth of the liquid and not on the tube width or shape. This illustrated in **figure (iii)** above.
- Pressure depends on the density of the liquid. The denser the liquid, the greater the pressure at any given depth.

CALCULATING PRESSURE IN A LIQUID

The pressure at any point in a liquid can be

calculated provided you know the depth beneath the surface and the density of the liquid. The container in the figure below has a base area A . It is filled to a depth h with a liquid density d . To calculate the pressure acting on the base, you need to know the weight of liquid pressing down the base.



$$\begin{aligned} \text{Volume of liquid} &= \text{base area} \times \text{depth} = Ah \\ \text{Mass of liquid} &= \text{density} \times \text{volume} = dAh \\ \text{Weight of liquid} &= \text{mass} \times g \text{ (10)} = 10dAh \\ \text{Therefore, force on base} &= 10dAh \\ \text{This force is acting on an area } A: \\ \text{Therefore, pressure} &= \frac{\text{force}}{\text{Area}} = \frac{10dAh}{A} \end{aligned}$$

$$\text{Pressure} = 10hd$$

This pressure acts equally in all directions at depth h and depends only on h and density d . its value will be in Pascal (Pa) if h is in metres m and density d in Kg/m^3 .

Work:

- A girl is wearing a pair of shoes. The area of contact for a shoe is 160cm^2 and she weighs 550N . Calculate the pressure exerted on the ground by the girl.
- A pressure of 10Pa acts on an area of 3.0m^2 . What is the force acting on the area?
- In a hydraulic press a force of 20N is applied to a piston of area 0.20m^2 . What is
 - the pressure transmitted through the liquid?
 - the force on the other piston whose

area is 1.0m^2 ?

4. What is the pressure 100m below the surface of sea water of density 1150Kg/m^3 ?

MEASURING PRESSURE DIFFERENCES

MANOMETER

A manometer is a useful instrument for measuring pressure differences. It consists of a U-tube containing a suitable liquid; i.e. a low-density liquid (such as oil) when measuring small pressure differences and a high – density liquid (such as mercury) for large pressure differences.

The pressure p_1 of the gas supply is greater than atmospheric pressure p_0 and this pressure is transmitted to the liquid in the left-hand limb of the tube. This forces the liquid round the tube into the right-hand limb until there is a 'head' of liquid h above the lowest surface level.

Equating pressures at this level gives

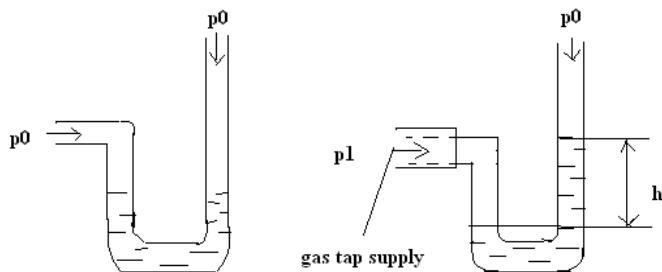
$p_1 = h\rho g + p_0$ and so excess pressure of the gas above the atmospheric pressure is given by

$$\Delta p = p_1 - p_0 = h\rho g$$

$$\Delta p = h\rho g$$

Note: h is the 'head' of the liquid, ρ is the density of the liquid and g is acceleration due

to gravity (10ms^{-2}).



Manometer to measure the excess pressure of the gas supply

Worked example:

Calculate atmospheric pressure in Pa when a mercury barometer supports a column of mercury 76cm high. (Density of mercury = 13600kgm^{-3})

$$\Delta p = ? \quad h = 0.76\text{m} \quad \rho = 13600\text{kgm}^{-3}$$

$$g = 10\text{ms}^{-2}$$

$$\Delta p = h\rho g$$

$$= 0.76 \times 13600 \times 10 \text{ Pa}$$

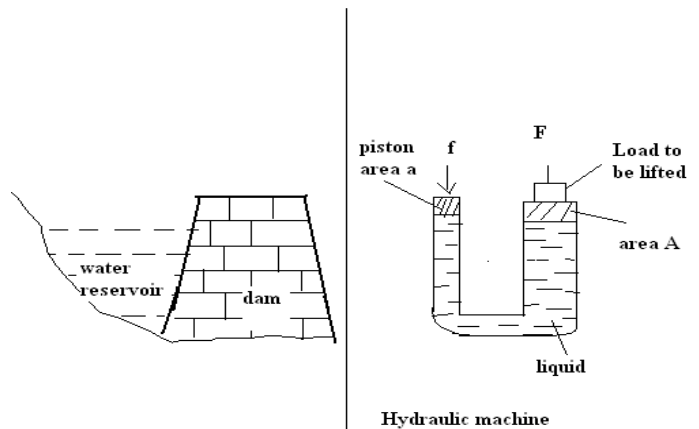
$$= 103360 \text{ Pa}$$

APPLICATIONS OF LIQUID PRESSURE

1. WATER SUPPLY SYSTEM

A town's water supply often comes from a reservoir on high ground. Water flows from it through pipes to any tap or storage tank that is below the level of water in the reservoir. The lower the place supplied, the greater the water pressure. In a very tall building it may be necessary first to pump the water to a large tank in the roof.

Reservoirs for water supply or for hydroelectric power stations are often made in mountainous regions by building a dam at one end of a valley. The dam must be thicker at the bottom than at the top due to large water pressure at the bottom.



1. HYDRAULIC MACHINES

Liquids are almost incompressible (i.e. their volume cannot be reduced by squeezing) and they 'pass on' any pressure applied to them. Use is made of these facts in hydraulic machines.

Referring to the hydraulic machine in the figure above:

Suppose a downward force f acts on a piston of area a . The pressure transmitted through the liquid is

$$\text{Pressure} = \frac{f}{a}$$

This pressure acts on a second piston of larger area A producing an upward force F

area

$$F = \text{pressure} \times A$$

$$= \frac{f}{a} \times A$$

$$= f \times \frac{A}{a}$$

Since A is larger than a , F must be larger than f and the hydraulic machine (system) is a force multiplier; the multiplying factor is $\frac{A}{a}$

Hydraulic machines are used in garages to lift cars, hydraulic fork-lift trucks and hydraulic car brakes use the same principle.

EXPANSION AND CONTRACTION OF SOLIDS

In general, when matter is heated it expands and when cooled it contracts. According to the kinetic theory, the molecules of solids are in constant vibrations. When heated they vibrate faster and force each other a little farther apart. The linear (length) expansion of solids is small and for the effect to be noticed the solid must be long and/or the temperature change large.

USES OF EXPANSION

1. 'Shrink – fitting' of axles into gear wheels. The axles are cooled in liquid nitrogen at very low temperature i.e. -196°C and they contract (shrink) until the gear wheels can be slipped on to them. On

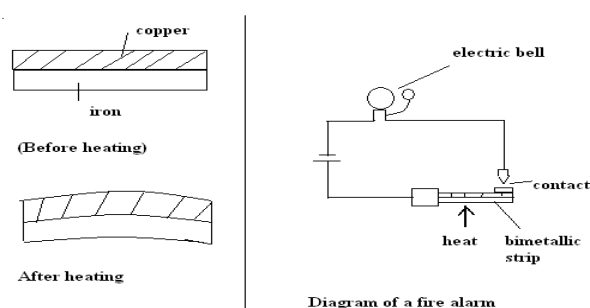
regaining normal temperature the axles expand to give a very tight fit.

2. A tight metal lid can be removed from a glass jar by immersing the lid in hot water so that it expands.

3. Bimetallic strip – if two different metals e.g. copper and iron are riveted together so that they can not move separately, they form a bimetallic strip. When heated, copper expands more than iron and to allow this, the strip bends towards the metal that expands less.

Bimetallic strips have many uses:

(a) fire alarm



Heat from the fire makes the bimetallic strip bend and complete the electrical circuit, so ringing the alarm bell.

(b) a thermostat in an iron

A thermostat keeps the temperature of an appliance constant. When, for example, the iron reaches the required temperature the strip bends and breaks the circuit at the contacts and switches off the heater. After cooling a little the strip remakes contact and turns the heater.

PRECAUTIONS AGAINST EXPANSION

- Gaps are left between lengths of railway lines to allow expansion in summer.
- Small gaps are left between brick walls to allow expansion.

TOPIC: ELEMENTS AND CHEMICAL BONDING

STRUCTURE OF AN ATOM

Everything you see around you is made out of tiny particles called atoms. These atoms are made up of even smaller 'subatomic' particles. The most important of these are electrons, protons and neutrons. The protons and neutrons are found in the centre of the atom, which is called the nucleus. The neutrons have no charge and protons are positively charged. The nucleus occupies only a very small volume of the atom but is very dense.

The rest of the atom surrounding the nucleus is where electrons are mostly likely to be found. The electrons are negatively charged and move around very quickly in electron shells or energy levels. The electrons are held within the atom by an electrostatic force of attraction between themselves and the positive charge of the protons in the nucleus.

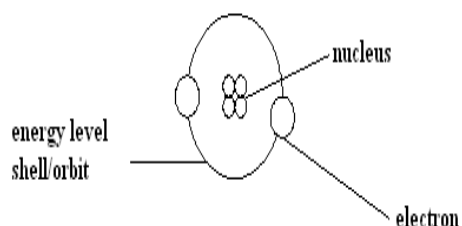


Diagram of the structure of an atom and its parts.

Particle	Symbol	Relative mass (amu)	Relative charge
Proton	p	1	+1
Neutron	n	1	0
Electron	e	1/1837	-1

ATOMIC NUMBER AND MASS NUMBER

The number of protons in the nucleus of an atom is called the atomic number. Each element has its own atomic number. For example, the helium atom has an atomic number of 2, since it has two protons in its nucleus.

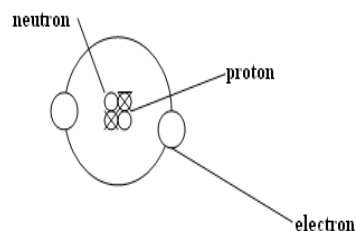


Diagram of the structure of helium atom.

The total number of protons and neutrons found in the nucleus of an atom is called the mass number (or nucleon number).

Mass number = atomic number (No. of protons) + No. of neutrons

The atomic number and mass number of an element are usually written in the following shorthand way:



(4 is the mass number, 2 is the atomic number and He is the symbol of the element i.e. helium).

Number of neutrons = mass number – atomic number.

For example, the number of neutrons in one atom of magnesium ${}^{24}_{12}\text{Mg}$ is $(24 - 12) = 12$.

ELECTRON CONFIGURATION AND THE PERIODIC TABLE.

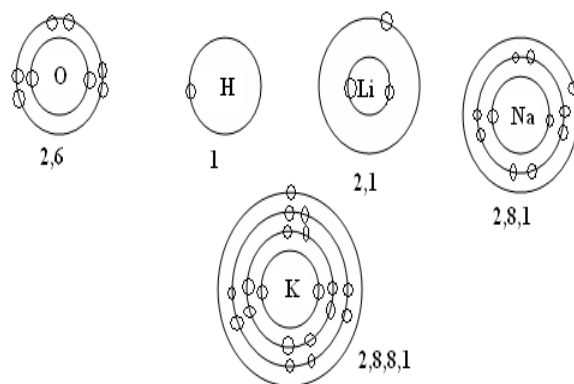
The way in which the electrons are distributed is called the electron configuration or electron structure.

Each of the electron energy levels can hold only a certain number of electrons as follows:

- First energy level holds up to 2 electrons
- Second energy level holds up to 8 electrons
- Third energy level holds up to 18 electrons. The third energy level can be occupied by a maximum of 18 electrons. However, when eight electrons have occupied this level a certain stability is given to the atom and the next two electrons go into the fourth energy level, and then the remaining ten electrons complete the third energy level.

The electrons fill the energy levels starting from the energy level nearest to the nucleus, which has the lowest energy. When this is full (with two electrons) the next electron goes into second energy level. When this energy level is full with eight electrons, then the electrons begin to fill the third and fourth energy levels as stated above.

For example, an oxygen atom (${}^{16}_8\text{O}$) has an atomic number of 8 and therefore has eight electrons. The electron configuration for oxygen can be written in a short hand way as 2,6



The electron configurations and structures for oxygen, hydrogen, lithium, sodium and potassium.

ELECTRON ARRANGEMENT IN THE FIRST 20 ELEMENTS

Element	Symbol	Atomic number	Number of electrons	Electron structure
Hydrogen	H	1	1	1
Helium	He	2	2	2
Lithium	Li	3	3	2,1
Beryllium	Be	4	4	2,2
Boron	B	5	5	2,3
Carbon	C	6	6	2,4
Nitrogen	N	7	7	2,5
Oxygen	O	8	8	2,6
Fluorine	F	9	9	2,7
Neon	Ne	10	10	2,8
Sodium	Na	11	11	2,8,1
Magnesium	Mg	12	12	2,8,2
Aluminium	Al	13	13	2,8,3
Silicon	Si	14	14	2,8,4
Phosphorus	P	15	15	2,8,5
Sulphur	S	16	16	2,8,6
Chlorine	Cl	17	17	2,8,7
Argon	Ar	18	18	2,8,8
Potassium	K	19	19	2,8,8,1
Calcium	Ca	20	20	2,8,8,2

THE PERIODIC TABLE

The periodic table was devised in 1869 by

Russian Dmitri Mendeleev, who was a professor of chemistry at St. Petersburg University. His periodic table was based on the chemical and physical properties of the elements that had been discovered at that time.

The vertical columns are called **groups** and the horizontal rows are called **periods**.

Those elements with similar chemical properties are found in the same columns or groups. There are eight groups of elements. Some of the groups have been given special names.

- Group 1: The alkali metals **except H**
- Group 2: The alkaline earth metals
- Group 4: The semiconductors or metalloids
- Group 7: The halogens
- Group 8(0): The gases or noble gases

I	II	III	IV	V	VI	VII	VIII
H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca					Br	

A group is a vertical column of the periodic table containing elements with similar properties with the same number of electrons in their outer energy levels. They have an increasing number of inner energy levels as you descend the group.

Periods are horizontal rows of the periodic table. Within a period the atoms of all the elements have the same number of occupied energy levels but have an increasing number of electrons in the outer energy level. For example, calcium is an element found in group 2 and period 4. What it means is that calcium has four energy levels (shells) and the outermost (fourth) shell occupies 2 electrons.

GROUP 1: ALKALI METALS PROPERTIES

- They are good conductors of electricity and heat
- They are soft metals

- They are metals with low densities
- They have shiny surfaces when freshly cut with a knife
- They burn in oxygen or air to form solid oxides
- They react vigorously with water to give an alkaline solution of the metal hydroxide as well as producing hydrogen gas.
- They react vigorously with halogens to form metal halides.

GROUP 2: ALKALINE EARTH METALS PROPERTIES

- They are harder than those in group 1
- They are silvery-grey in colour when pure and clean
- They are good conductors of heat and electricity
- They burn in oxygen or air to form solid oxides
- They react with water, but they do so much less vigorously than the elements in group 1.

GROUP 7: THE HALOGENS PROPERTIES

- These elements are coloured and darker going down the group
- They exist as diatomic molecules, for example Cl₂, Br₂ and I₂
- They show a gradual change from a gas (Cl₂), through a liquid (Br₂), to a solid (I₂).
- They form molecular compounds with other non metallic elements, for example HCl.
- They react with hydrogen to produce hydrogen halides.
- They react with metals to produce ionic metal halides.

GROUP 0: THE NOBLE GASES PROPERTIES

- They are colourless gases
- They exist as individual atoms, for example He, Ne and Ar
- They are very unreactive

STABLE ELECTRON CONFIGURATION (NOBLE GASES)

Element	Symbol	Electron configuration
Helium	He	2
Neon	Ne	2,8
Argon	Ar	2,8,8

Helium, neon and argon make up a most unusual group of non metals called the noble gases.

These gases are chemically unreactive because they have electron configurations which are stable and very difficult to change. They are so stable that other elements attempt to attain these electron configurations during chemical reactions.

PERIODICITY

Going across the periodic Table i.e. from Li to Ne, the number of shells remains the same but the number of protons increases. This means the effective nuclear charge (the charge from the protons) increases as we go across the periodic table and this in turn makes the atomic size to decrease across the periodic table.

Going down the periodic table i.e. from F to Cl the number of shells increases but the number of electrons in the outer shells remains the same. This means the effective nuclear charge decreases as we go down the group because of the shielding effect of the increasing number of shells makes the atomic size of the elements to increase as we go down the periodic table.

BOND FORMATION

Atoms are joined together by bonds and the process of forming these bonds is called bonding. Elements react to form compounds so as to achieve full electron

energy levels. This means an atom has to lose or gain or share electrons to gain stability.

There are three main types of bonding namely; ionic (electrovalent), covalent and metallic.

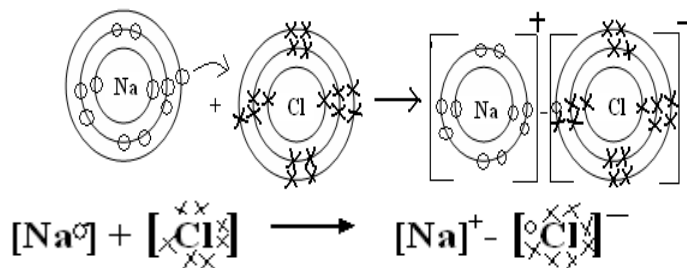
(a) IONIC (ELECTROVALENT) BONDING

Ionic bonds are usually found in compounds that contain metals combined with non-metals. When this type of bond is formed, electrons are transferred from the metal atoms to the non-metal atoms during the chemical reaction. For example, sodium and chlorine combine to make sodium chloride.

Sodium + chlorine —→ sodium chloride

(Na) (Cl) (NaCl)

Sodium has just one electron in its outer energy level (Na 2,8,1), chlorine has seven electrons in its outer energy level (Cl 2,8,7). When these two elements react, the outer electron of each sodium atom is transferred to the outer energy level of a chlorine atom. In this way both the atoms obtain full outer energy levels and become 'like' the nearest noble gas. The sodium atom becomes a sodium ion (positive) while chlorine atom becomes a chloride ion (negative). Only the outer electrons are important in bonding.

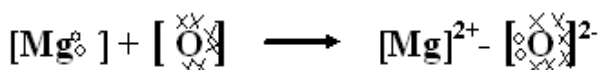


Dot and cross diagram for the reaction between sodium and chlorine

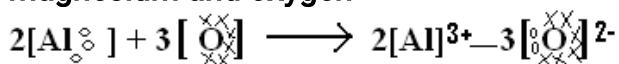
The charges on the sodium and chloride ions are equal but opposite. They balance each other and the resulting formula for

sodium chloride is **NaCl**. These oppositely charged ions attract each other and are bonded to one another by strong electrostatic forces. This type of bonding is called ionic or electrovalent bonding.

Definition: Ionic bond is the electrical attraction between oppositely charged ions which are produced when electrons are transferred from one atom (metal) to another (non-metal) during a chemical reaction. Other examples include:



Dot and cross diagram of reaction between magnesium and oxygen



Dot and cross diagram of reaction between aluminium and oxygen.

PROPERTIES OF IONIC COMPOUNDS

Ionic compounds have the following properties.

- They are usually solids at room temperature, with high melting points. This is due to the strong electrostatic forces holding the crystal lattice together. A lot of energy is therefore needed to separate the ions and melt the substance.
- They are usually hard substances.
- They usually cannot conduct electricity when solid, because the ions are not free to move. They conduct only in molten and aqueous forms.
- They mainly dissolve in water.

Work:

Draw the dot and cross diagrams for the following reactions and suggest the molecular formulae of the compounds formed;

(a) lithium reacting with fluorine

(b) magnesium reacting with chlorine

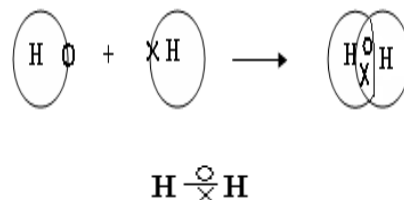
(c) sodium reacting with oxygen

(d) aluminium reacting with chlorine

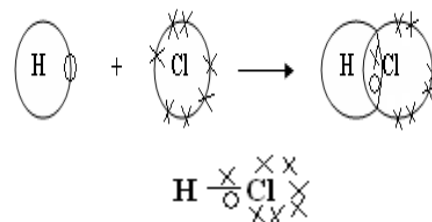
(b) COVALENT BONDING

Another way in which atoms can gain the stability of the noble gas electron configuration is by sharing the electrons in their outer energy levels. This occurs between non-metal atoms, and the bond formed is called a covalent bond. The simplest example of this type of bonding is hydrogen molecule, H_2 .

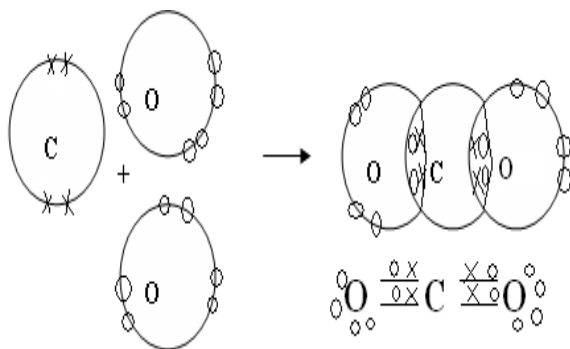
Each hydrogen atom in the molecule has one electron. In order to obtain a full outer energy level and gain the electron configuration of the noble gas each of the hydrogen atoms must have two electrons. To do this, the two hydrogen atoms allow their outer energy levels to overlap. A molecule of hydrogen is formed, with two hydrogen atoms sharing a pair of electrons. This shared pair of electrons is known as a **single covalent bond**.



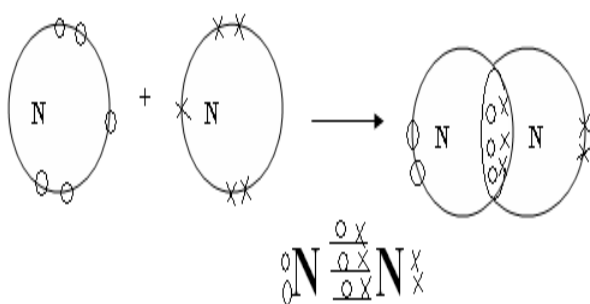
Dot and cross diagram of a molecule of hydrogen (H_2)



This is an example of a **single covalent bond**.
Dot and cross diagram of a molecule of hydrogen chloride (HCl)



This is an example of **double covalent bond**.
Dot and cross diagram of a molecule of carbon dioxide (CO_2)



This is an example of a **triple covalent bond**.
Dot and cross diagram of a molecule of nitrogen (N_2)

PROPERTIES OF COVALENT COMPOUNDS

Covalent compounds have the following properties.

- As simple molecular substances, they are usually gases, liquids or solids with low melting and boiling points. The melting points are low because of the weak intermolecular forces of attraction which exist between simple molecules. Giant molecular substances have higher melting points, because the whole structure is held together by strong covalent bonds within the giant molecule.
- Generally, they do not conduct electricity when molten or dissolved in water. This is because they do not contain ions. However, some molecules actually react with water to form ions.

- Generally, they do not dissolve in water. However, water is an excellent solvent and can interact with and dissolve some covalent molecules better than others.

Work:

Draw the dot and cross diagrams of the following molecules and state whether the type of bonding is single or double or triple covalent;

- (i) Cl_2 (ii) NH_3 (iii) H_2O (iv) CH_4 (v) O_2 (vi) C_2H_4 (vii) CH_3OH

POLAR AND NON POLAR COVALENT BONDS

In a diatomic molecule of hydrogen or chlorine the two identical nuclei must attract the pair of electrons equally strongly. In a molecule of HCl , the chlorine nucleus attracts the electrons more strongly than does the hydrogen nucleus.

Electronegativity is the measure of how strongly an atom in a compound attracts electrons in a bond. Fluorine is the most electronegative of all the elements. So in H-F molecule the fluorine takes more than its fair share of the bonded electron-pair. Consequently both atoms become slightly charged and this makes the hydrogen fluoride molecule **polar**.

$\delta+$ $\delta-$

-

$\text{H} - \text{F}$

H is partially positively charged and F is partially negatively charged.

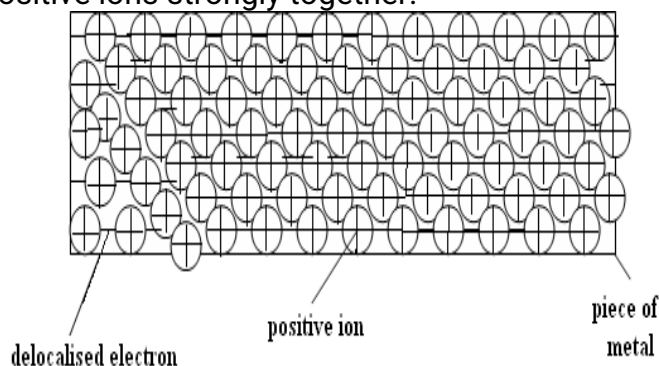
A hydrogen molecule is said to be **non-polar** because the nuclei of the two hydrogen atoms attract the pair of electrons equally strongly.

(c) METALLIC BOND (BONDING IN METALS)

This is the bond that holds the atoms together in a metal. When atoms are in gaseous state, their electrons are localised in the shells (energy levels). This means electrons are inside the electron shells.

In a solid state, metal atoms take up fixed positions as a result their electrons are

delocalised. This means the valence electrons are set free from the electron shells of each metal atom. Electrons are now able to move freely and become delocalised. The metal atoms having lost their electrons have a positive charge. These charged atoms are called ions. So in the solid metal, delocalised electrons are free to move freely between all the ions of the metal. The ions are in fixed positions and act a bit like stationary 'island' in a sea of 'electrons'. This 'sea of electrons' (negative) attracts the positive ions strongly together.



A metallic bond is the attraction between fixed metal ions in a piece of metal and the free delocalised electrons that move and surround them.

PROPERTIES OF METALS

- Since the atoms are held together very strongly, they are close together and a lot of energy is needed to pull them apart. This explains why the melting points and densities of most metals are high.
- Electric current in a metal is caused by electrons moving through the metal. This happens because the outer electrons in the metal atoms easily become delocalised (detached) from the atoms to form the 'sea' of electrons. These electrons will then move away from the negative terminal in a circuit and towards the positive terminal. The moving electrons can also take heat energy through the metal and so metals are good conductors of heat.
- It is the 'sea' of electrons that makes

the metallic bond strong but not rigid and so it will deform, not crush when it is hit.

The table below gives some physical properties of metals.

Property (physical)	Explanation
• They are strong and hard • They have high melting and boiling points • They are insoluble in water	Strong metallic bonds hold atoms together in the solid. Large energy is needed to separate metal atoms.
• They are malleable – can be hammered into different shapes • They are ductile – can be drawn out into wires	Metallic bonds are 'flexible' and not fixed in one direction only (atoms can slide across each other).
• They conduct electricity and heat	Free 'delocalised' electrons carry electrical current and heat energy through the metal lattice.
• They have high densities	Metal atoms are pulled together into a small space by the strong metallic bond.

CHEMICAL PROPERTIES

1. Metals react with oxygen to form oxides. For example, zinc reacts with oxygen to form zinc oxide.
2. Metals react with acids to form salts and hydrogen gas.

USES OF METALS

1. Steel – engine blocks, drain covers
2. Carbon steel – car bodies, railway lines, knives, razor blades
3. Transmission of electric current i.e. Cu

4. Pots (kitchen utensils), furniture

HALOGENS

PROPERTIES OF HALOGENS

Fluorine, chlorine, bromine and iodine are all poisonous, often corrosive, non-metallic elements. The properties include;

1. They are coloured and the colours darken going down the group i.e. Cl_2 – pale green, Br_2 – red brown, I_2 – purple black etc.
2. They exist as diatomic molecules, for example Cl_2 , Br_2 and I_2 .
3. They show a gradual change from a gas (Cl_2) through a liquid (Br_2), to a solid (I_2).
4. They form molecular compounds with other non-metallic elements, for example HCl .
5. They react with hydrogen to produce the hydrogen halides i.e.
 $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$.
6. They react with metals to produce ionic metal halides i.e.
 $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$.

TRENDS AS WE GO DOWN GROUP VII

- Halogens react by attracting electrons. As the atoms increase in size the attraction decreases and so reactivity decreases down the group.
- Intermolecular forces increase as the mass of the molecule increases and so intermolecular attraction and therefore melting/boiling points increase down the group.
- The atomic radius increases as we go down group VII (halogens) because number of shells/energy levels increases down the group.
- Effective nuclear charge decreases down the group because the increasing numbers of shells shield the nuclear charge from the nucleus (protons).
- As you go down the group the outermost electrons are further from the nucleus. They will, therefore, be held less securely, and the resulting reactivity of group 7 will decrease

down the group.

DISPLACEMENT REACTIONS

If chlorine reacts with potassium bromide the less reactive halogen (bromine) is displaced by the more reactive halogen, chlorine as shown in the equation below;

Potassium bromide + chlorine $\longrightarrow$ potassium chloride + bromine



The reactivity of halogens decreases on going down the group;

Decreasing reactivity $\longrightarrow$



Chlorine is more reactive than bromine because chlorine is being more strongly attracted into the outer energy level of the smaller atom. The attractive force on it will be greater than in the case of bromine, since the outer energy level of chlorine is closer to the nucleus.

This simply tells us that when bromine (Br_2) is mixed with hydrogen chloride (HCl), there will be no chemical reaction taking place. This is the case because bromine (Br_2) can not displace chlorine since it is less reactive.

USES OF HALOGENS

The halogens and halogenic compounds are used in a multitude of different ways.

- Fluorine is used in the form of fluorides in drinking water and tooth paste because it reduces tooth decay by hardening the enamel on teeth.
- Chlorine is used to make PVC (polyvinyl chloride) plastic as well as household bleaches. It is also used to kill bacteria and viruses in drinking water.
- Bromine is used to make disinfectants, medicines and fire retardants.
- Iodine is used in medicines and disinfectants and also as a photographic chemical.

SULPHUR

Sulphur is a non-metallic element which has a very important role in the chemical industry.

It is a yellow solid which is found in large quantities but in various forms throughout the world.

SOURCES OF SULPHUR

- Sulphur is found in metal ores such as copper pyrites (CuFeS_2) and zinc blende (ZnS).
- Sulphur is also found in volcanic regions of the world.
- Natural gas and oil contain sulphur and its compounds.

Sulphur obtained from these sources is known as 'recovered sulphur' and it is an important source of the element.

ALLOTROPES OF SULPHUR

When an element can exist in more than one physical form in the same state it is said to exhibit allotropy. Each of the different physical forms is called an allotrope.

Definition: Allotropes are different structural forms of the same element having the same physical state.

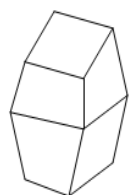
Examples of elements that exist as allotropes are carbon and sulphur. The allotropes of carbon are graphite, buckminsterfullerene and diamond. The main allotropes of sulphur are called rhombic sulphur and monoclinic sulphur. Both of these solid forms of sulphur are made up of S_8 molecules. The fact that there are two different allotropes of sulphur is due to the way in which these S_8 molecules pack together. In rhombic sulphur the molecules are packed more closely than in the monoclinic form.



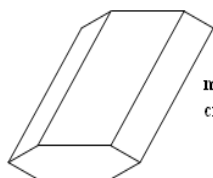
molecules closely packed in rhombic sulphur



molecules loosely packed in monoclinic sulphur



rhombic crystal



monoclinic crystal

Although sulphur is insoluble in water, it will

dissolve in an organic solvent such as methylbenzene. If a solution of sulphur in methylbenzene is heated and allowed to cool then crystals of monoclinic sulphur are produced. When the temperature of the solution falls below 96°C , rhombic sulphur crystals are produced. Rhombic sulphur is stable below 96°C and monoclinic sulphur is stable above 96°C . This temperature is called the **transition temperature**.

PROPERTIES OF SULPHUR

- It is a yellow, brittle solid at room temperature. Does not conduct electricity. It is insoluble in water.
- Sulphur will react with both metals and non-metals.
 - (i) it reacts with magnesium metal to form magnesium metal to form magnesium sulphide.

Magnesium + sulphur $\longrightarrow$ magnesium sulphide

(Mg) (S) (MgS)

- (ii) it reacts with oxygen to produce sulphur dioxide gas.

Sulphur + oxygen $\longrightarrow$ sulphur dioxide

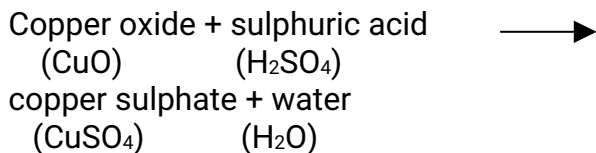
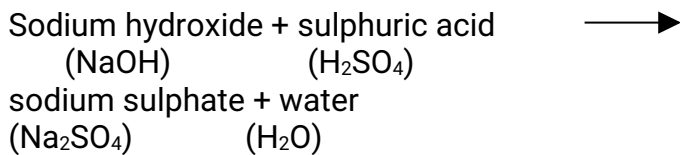
(S) (O_2) (SO_2)

USES OF SULPHUR

- The majority of sulphur is used to produce perhaps the most important industrial chemical known as sulphuric acid (H_2SO_4).
- Sulphur is also used to vulcanise rubber, a process which makes the rubber harder and increases its elasticity.
- Sulphur is also used in the manufacturing of matches, fireworks and fungicides.
- Sulphur is also used as sterilising agent and in medicines.

SULPHATES

Sulphuric acids (H_2SO_4) react with bases such as sodium hydroxide (NaOH) and copper (II) oxide (CuO) to produce normal salts called sulphates and water.



USES OF SULPHATES

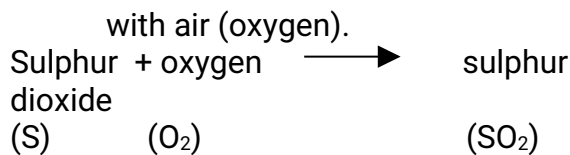
Below is a table that shows a few sulphates (salts) and their uses;

Sulphate (salt)	Formula	Use
Iron (II) sulphate	FeSO_4	In 'iron' tablets
Ammonium sulphate	$(\text{NH}_4)_2\text{SO}_4$	Fertiliser
Barium sulphate	BaSO_4	Used in diagnostic medical X-ray studies
Calcium sulphate	$\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$	'Plaster of Paris' used to set bones, popularly known as POP
Magnesium sulphate	MgSO_4	In medicine it is used as a laxative – substance that promotes emptying of the bowels

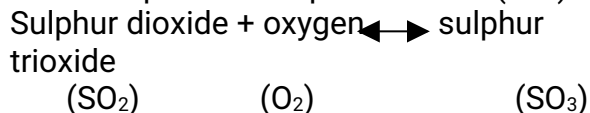
SULPHURIC ACID

The major use of sulphur is in the production of sulphuric acid. The process by which sulphuric acid is produced is known as the **Contact Process**. The process has the following stages:

- Sulphur dioxide is first produced, primarily by the reaction of sulphur

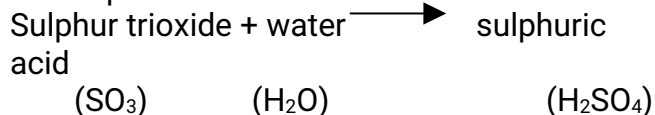


- Clean gases of sulphur dioxide (SO_2) are heated to a temperature of approximately 450°C and fed into a reaction vessel, where they are passed over a catalyst of vanadium (V) oxide (V_2O_5). This catalyses the reaction between sulphur dioxide and oxygen to produce sulphur trioxide (SO_3).



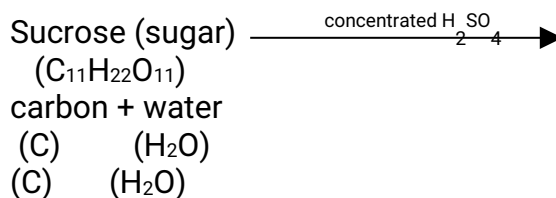
This reaction is reversible.

- If this sulphur trioxide is added directly to water, sulphuric acid is produced. This reaction, however, is very violent and a thick mist is produced.



CONCENTRATED SULPHURIC ACID

Concentrated sulphuric acid is a very powerful **dehydrating agent**. This means it will take water from a variety of substances. One such substance is cane sugar or sucrose.



Concentrated sulphuric acid will also remove water from hydrated copper (II) sulphate crystals leaving only anhydrous copper (II) sulphate. If a few drops of concentrated sulphuric acid are added to blue hydrated copper sulphate crystals, they slowly turn white as the water of crystallisation is removed by the acid. Eventually, only white powder – anhydrous copper sulphate remains.

Hydrated copper (II) sulphate
anhydrous + water
($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)

concentrated H_2SO_4
2 4

It is used mainly as the raw material for the production of many substances. These include;

- (a) metal salts (sulphates)
- (b) fertilisers
- (c) chemicals
- (d) paints and pigments
- (e) detergents and soaps
- (f) artificial fibres
- (g) dyes
- (h) steel manufacturing
- (i) plastics and many more.

copper (II) sulphate + water
(CuSO_4) (H_2O)

CAUTION: Concentrated acid should be treated very carefully, because it will also remove water from flesh! It is a very corrosive substance and should always be handled with care.

USES OF SULPHURIC ACID

TOPIC: CHEMICAL REACTIONS I

STOICHIOMETRY

Chemical reaction is the rearrangement of different atoms to form new substances. For example, carbon reacts with oxygen to form carbon dioxide.

What happens in a chemical reaction can be put and expressed in writing as a chemical equation. A chemical equation is an equation that shows the reactants and products of a chemical reaction.

A chemical reaction can only take place if there are collisions between the reactant molecules. These collisions must have enough speed (kinetic energy) to break the old bonds and so cause the formation of new bonds.

When writing chemical equations the following rules must be observed;

- (i) Make sure the correct formulae for reactants and products are written.
- (ii) Balance the number of atoms on both sides of the equation by multiplying molecules, not by adding new atoms.
- (iii) In the brackets after each reactant and product write down its physical state solid (s), gas (g), liquid (l) and an aqueous [water] solution (aq).

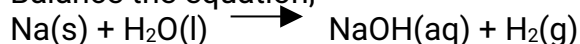
BALANCING CHEMICAL EQUATIONS

When balancing chemical equations it must

be observed that there must be equal number of different atoms on either side of the equation. This is done because in a chemical reaction no atom is lost nor gained nor created. When balancing a chemical equation do not change the formula of a compound or element.

Example 1:

Balance the equation;

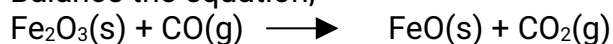


It is observed from the equation there is a total of 2 hydrogen atoms in reactant's side and 3 hydrogen atoms in the product's side. If a coefficient 2 is put in front of NaOH the total number of hydrogen atoms becomes 4 in product's side. To balance off, a coefficient 2 is put on both Na and H_2O . The balanced equation becomes;



Example 2:

Balance the equation;

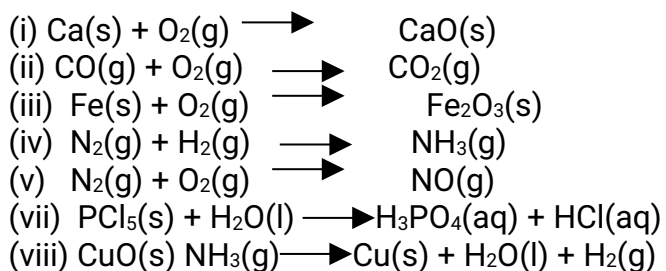


To balance the number of Fe atoms on both sides, there is need to put a coefficient 2 on FeO. This also means that the whole equation is balanced since the number of O atoms is 4 on both sides and there is 1 C atom on either side.



Work:

(a) Balance the following chemical equations;



(b) When dilute hydrochloric acid (HCl) is added to marble chips (calcium carbonate, CaCO_3), carbon dioxide gas (CO_2), is given off and a solution of calcium chloride (CaCl_2) is produced plus water.

(i) Write down the formulae of reactants

and products.

(ii) Balance the atoms on both sides of the equation.

UNITS OF MEASUREMENT

Avogadro's constant: 6.02×10^{23}

This is the same number of carbon atoms contained in exactly 12g of the carbon-12 isotope.

A mole is an amount of substance containing 6.02×10^{23} particles.

Mole Calculations:

Mass (in grams) = number of moles x mass of 1 mole of the element

Number of moles = $\frac{\text{mass of an element}}{\text{Mass of 1 mole of that element}}$

Work:

- Calculate the mass of (a) 2 moles (b) 0.25 mole of iron (Fe = 56)
- Calculate the number of moles of aluminium present in (a) 108g and (b) 13.5g of the element (Al = 27)

Relative Atomic Mass (RAM)

This is the mass one atom of an element i.e. the RAM of Al = 27.

Relative Molecular Mass (RMM)

This is the mass of one molecule calculated from the sum of RAMs of the atoms involved.

Relative Formula Mass (RFM)

This is the sum of the relative atomic masses (RAMs) of all those elements shown in the chemical formula of the substance.

Example 1:

Calculate the RFM of ethanol ($\text{C}_2\text{H}_5\text{OH}$).

RAMs: C = 12, H = 1 and O = 16

$$\text{C} = 2 \times 12 = 24$$

$$\text{H} = 6 \times 1 = 6$$

$$\text{O} = 1 \times 16 = 16$$

$$\text{RFM of } \text{C}_2\text{H}_5\text{OH} = 46$$

Example 2:

Calculate the mass of 3 moles of CO_2 .

$$\text{C} = 1 \times 12 = 12$$

$$\text{O} = 2 \times 16 = 32$$

$$\text{RFM of } \text{CO}_2 = 44$$

Mass of compound = No. of moles x RFM

$$= 3 \times 44$$

$$= 132\text{g}$$

Note:

No. of moles of an element = $\frac{\text{mass of element}}{\text{RAM}}$

RAM

No. of moles of compound = $\frac{\text{mass of a substance}}{\text{RFM}}$

RFM

MOLECULAR AND EMPIRICAL FORMULA

A molecular formula is a chemical formula of a molecule or compound that gives the total number of atoms present in the molecule or compound. For example, the molecular formula of butane is C_4H_{10} . This means there are 14 atoms in total in this compound i.e. 4 C atoms and 10 H atoms.

The empirical formula of a compound is the simplest formula that shows the lowest ratio of the atoms that make up the compound. For instance, glucose has a molecular formula $C_6H_{12}O_6$ and its empirical formula is CH_2O . It is found by dividing each of the subscripts by 6 to get the lowest ratio i.e. $C_{6/6}H_{12/6}O_{6/6} = CH_2O$.

Example 1:

Find the empirical formula of the compound that has the following percentage composition by mass of elements: C = 40%, H = 6.67%, O = 53.33%

Solution

Element	atomic prop.	ratio of atoms	simplest atom ratio
C	40/12	3.33	3.33/3.33 = 1
H	6.67/1	6.67	6.67/3.33 = 2
O	53.33/16	3.33	3.33/3.33 = 1

The empirical formula is $C_1H_2O_1 = CH_2O$.

Example 2:

In an experiment an unknown organic compound was found to contain 0.12g of carbon and 0.02g of hydrogen.

(a) Calculate the empirical formula of the compound.

(b) If the RFM of the compound is 56, calculate the molecular formula.

Solution

(a) Element	number of moles	simplest mole ratio
C	0.12/12 = 0.01	0.01/0.01 = 1
H	0.02/1 = 0.02	0.02/0.01 = 2

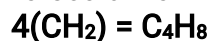
Empirical formula is $C_1H_2 = CH_2$

(b) RFM of compound = 56

RFM of CH_2 = 14

Ratio = $56/14 = 4$

Therefore, the molecular formula is



Work:

1. In an experiment 2.6g of chromium were heated in an excess of chlorine gas. It was found that 7.93g of chromium chloride was formed. If RFM of chromium chloride is 158, (a) calculate the empirical formula (b) molecular formula of chromium chloride.

2. Find the empirical formula of a substance that has the following percentage composition: C = 42.87%, H = 2.36%, N = 16.67%, O = 38.10% (RAMs C = 12, H = 1, N = 14, O = 16)

3. Work out the empirical formula of the compound with the following percentage composition: P = 20.19%, O = 10.42%, Cl = 69.39%.

4. When magnesium ribbon is heated strongly, it burns very brightly to form the white powder called magnesium oxide. The following data were collected from the experiment;

Mass of crucible	=
14.63g	
Mass of crucible and magnesium	=
14.87g	
Mass of crucible and magnesium oxide	=
15.03g	
Mass of magnesium used	=
Mass of oxygen which reacted with magnesium	=

(a) Calculate mass of magnesium and mass of oxygen used in the reaction.

(b) Determine the empirical formula of magnesium oxide.

5. Calculate the empirical formula of an organic compound containing 92.3% carbon and

7.7% hydrogen, by mass. The RFM of the organic compound is 78. What is its molecular formula?

CONCENTRATION/MOLARITY

Concentration of a solution can be expressed in g/l or g/dm^3 and moles/ dm^3 or moles/litre.

Molarity of a solution is the concentration expressed in moles/dm³ or moles/litre.

Molarity of a solution = number of moles of a solute

Volume of a solution in dm³

Formulae for molarity calculations

1. Concentration (in g/dm³) = molarity x RFM
2. Number of moles of a substance = volume x molarity
3. Mass of a substance (in grams) = number of moles x RFM

Example 1:

What is the concentration in g/dm³ of 0.4M HCl?

Solution:

RFM of HCl = 1 + 35.5 = 36.5g/mole

Molarity of HCl = 0.4M = 0.4 moles/dm³

Therefore, concentration = molarity x RFM
= 0.4 X 36.5g/dm³
= 14.6g/dm³

Example 2:

How many moles are contained in a 250cm³ solution of 0.1M HNO₃?

Solution:

Volume = 250/1000dm³ = 0.25dm³

Molarity = 0.1M = 0.1moles/dm³

Therefore, number of moles = volume x molarity
= 0.25 x 0.1
= 0.025

Example 3:

Calculate the molarity of a solution made by dissolving 5.85g of NaCl in 4dm³ of solution.

Solution:

RFM of NaCl = 23 + 35.5 = 58.5

Number of moles = mass/RFM = 5.85/58.5 = 0.1

Therefore, molarity = number of moles/volume
= 0.1/4 moles/dm³
= 0.025 moles/dm³

Work:

1. What is the concentration in g/dm³ of 0.08M Na₂CO₃?
2. How many moles are contained in the following solutions: (a) 4 dm³ of 0.5M HNO₃ (b) 250cm³ of 2M Na₂CO₃?
3. How many grams of Ca(OH)₂ are there in a solution of volume 2.5dm³ and a molarity of 2M?
4. Calculate the molarity of a solution made by dissolving 6.3g of HNO₃ in 250cm³ of solution.

STANDARD SOLUTION

This is a solution of known concentration or molarity.

A 0.5M HCl solution is said to be a standard solution because its concentration (molarity) is known i.e. 0.5 moles/dm³.

Preparation of standard solution:

In order to make up a standard solution of known molarity (concentration) it is necessary to be able to measure out:

- (a) exact masses of solutes and
- (b) exact volumes of solutions.

To do this we must use an accurate weighing balance, such as the triple beam balance or electronic balance, which can measure masses to an accuracy of ±0.01g. To measure volumes of liquids accurately we use volumetric apparatus such as pipette, measuring cylinder and burette.

How to prepare 0.1M Na₂CO₃ solution of volume 1dm³.

Number of moles required = volume x molarity

$$= 1\text{dm}^3 \times 0.1\text{mol/dm}^3 = 0.1$$

Mass of Na₂CO₃ required = number of moles x RFM

RFM of Na₂CO₃ = 106g/mol

Therefore, mass of Na₂CO₃ required = 0.1mole x 106g/mol

$$= 10.6\text{g}$$

Using a triple beam balance an exact mass of 10.6g Na_2CO_3 is measured. This mass is put in volumetric flask of volume 1dm^3 . Distilled water is slowly added to the solid Na_2CO_3 while stirring using a glass rod until the solid dissolves completely. The distilled water is added until the volume of 1dm^3 is reached. The solution formed is 0.1M Na_2CO_3 .

Work:

1. Explain how you can prepare a 500cm^3 of 0.4M $\text{Ca}(\text{OH})_2$ solution.
2. You are asked to prepare a solution of 250cm^3 of 1.5M NaOH . Explain the procedure.
3. Calculate mass of KOH that needs to be used to prepare 500cm^3 of 2M solution in water.

TITRATION

Titration is the gradual addition of one liquid to another. A titration usually involves the slow addition of reacting volumes of liquids taking part in a chemical reaction. The best known example of this is **neutralisation reaction**.

Neutralisation reaction is the reaction between an acid and a base to form salt and water. A simple acid-base titration involves the addition of an acid solution to an alkali (a base) solution until the reaction is complete. When enough acid has been added to the base to neutralise it we say that we had reached the **end point** or **neutral point**. To know exactly when the end point has been reached, an **indicator** is used.

An indicator is a substance used to show whether a substance is acidic or alkaline (basic).

The end point is shown by a change of colour from acid to base when enough of the acid solution has been added to the base to cause neutralisation. Below is a table showing indicators and their colours in acid and alkaline solution.

	acid solution	alkaline solution
Blue litmus	Red	Blue
Methyl orange	Pink	Yellow
Methyl red	Red	Yellow
Phenolphthalein	Colourless	Pink
Red litmus	Red	Blue

The following formula is used to calculate the molarity and volume of solutions:

$$\frac{M_1 \times V_1}{N_1} = \frac{M_2 \times V_2}{N_2}$$

(Base) (Acid)

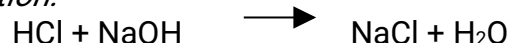
M_1 = molarity of a base, M_2 = molarity of an acid, V_1 = volume of a base, V_2 = volume of an acid, N_1 = number of moles of a base and N_2 = number of moles of an acid.

Note: Number of moles of a base and an acid are determined from a well balanced chemical equation.

Example 1:

25.0cm^3 of 0.1M NaOH were neutralised by 20.0cm^3 of HCl of molarity M_2 from the burette. Calculate the molarity of the acid (M_2).

Solution:



$M_1 = 0.1\text{M}$, $V_1 = 25.0\text{cm}^3$, $N_1 = 1$, $V_2 = 20.0\text{cm}^3$, $N_2 = 1$, $M_2 = ?$

Substituting in the formula:

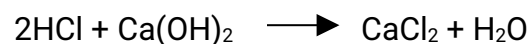
$$\frac{M_1 \times V_1}{N_1} = \frac{M_2 \times V_2}{N_2}$$

$$\frac{0.1 \times 25.0}{1} = \frac{M_2 \times 20.0}{1}$$

Therefore, $M_2 = 0.125\text{M}$.

Example 2:

What volume in cm^3 of 0.3M HCl will neutralise 50.5cm^3 of 0.1M $\text{Ca}(\text{OH})_2$?



Solution:

$M_1 = 0.3\text{M}$, $V_1 = 50.5\text{cm}^3$, $N_1 = 2$, $M_2 = 0.1\text{M}$, $N_2 = 1$ and $V_2 = ?$

Substituting in the formula:

Indicator	Colour in	Colour in
-----------	-----------	-----------

$$\frac{M_1 \times V_1}{N_1} = \frac{M_2 \times V_2}{N_2}$$

$$\frac{0.1 \times 50.5}{1} = \frac{0.3 \times V_2}{2}$$

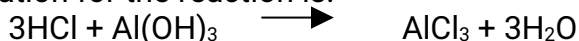
Therefore, $V_2 = 33.67 \text{ cm}^3$ (to 2 d.p.)

Work:

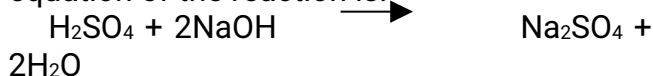
1. The molarity of a solution of HCl is found by titrating it against 0.25M Na_2CO_3 solution, using methyl orange as the indicator. The result was that 25cm^3 of 0.25M Na_2CO_3 solution required 20.8cm^3 of the acid of unknown molarity M_2 . Work the concentration of the acid in (a) moles/ dm^3 (b) grams/ dm^3 . The balanced equation of the reaction is:



2. What volume in cm^3 of 0.3M HCl will neutralise 10cm^3 of $\text{Al}(\text{OH})_3$? The balanced equation for the reaction is:



3. 22.4cm^3 of a solution containing 0.10mol dm^{-3} of sulphuric acid just neutralised 25.0cm^3 of a sodium hydroxide solution. What is the concentration of this sodium hydroxide solution? The balanced equation of the reaction is:



4. In a titration involving 24.0cm^3 potassium hydroxide solution against a solution containing 1 mol dm^{-3} of sulphuric acid, 28.0cm^3 of the acid was found to just neutralise the alkali (base) completely.

(a) Write a word and balanced chemical

equation for the reaction.

(b) Name the suitable indicator for the

titration and state the colour change

you would observe.

(c) Calculate the concentration of the

alkali (base) in mol dm^{-3} .

Many substances exist as gases. If we want to find the number of moles of a gas we can do this by measuring the volume rather than the mass. Chemists have shown by experiment that: ***One mole of any gas occupies a volume of approximately 24dm^3 (24 litres) at room temperature and pressure (rtp).***

At standard temperature and pressure (STP) the volume of one mole of any gas is approximately 22.4dm^3 . Therefore, it is relatively easy to convert volumes of gases into moles and moles of gases into volumes using the following relationships:

Number of moles of a gas
= $\frac{\text{volume of the gas (in } \text{dm}^3 \text{ at rtp)}}{24\text{dm}^3}$

Volume of a gas (in dm^3 at rtp)
= number of moles $\times 24\text{dm}^3$

Note: If it is at rtp we use 24dm^3 and if it is at stp we use 22.4dm^3 .

Example 1:

Calculate the number of moles of NH_3 gas in a volume of 72dm^3 of the gas measured at rtp.

Solution:

Volume of NH_3 gas = 72dm^3

Using the formula:

Number of moles of a gas
= $\frac{\text{volume of the gas (in } \text{dm}^3 \text{ at rtp)}}{24\text{dm}^3}$
= $\frac{72\text{dm}^3}{24\text{dm}^3}$
= 3

Example 2:

Calculate the volume of CO_2 gas occupied by 0.5 moles of the gas at STP.

Solution:

Number of moles of CO_2 gas = 0.5

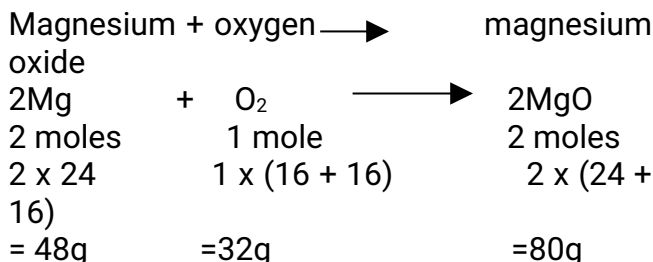
Using the formula:

MOLES AND GASES

Volume of a gas (in dm³ at rtp)
 = number of moles x 22.4dm³ (at STP)
 = 0.5 x 22.4 dm³
 = 11.2 dm³

REACTING MASSES

Consider the reaction below:

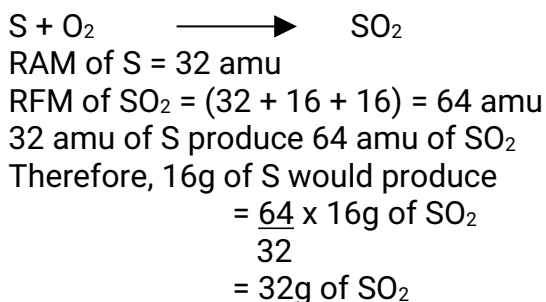


You will notice that the total mass of reactants is equal to the total mass of product. This is true for any chemical reaction and it is known as the **law of conservation of mass**.

Example:

Calculate the mass of sulphur dioxide produced by burning 16g of sulphur in excess of oxygen.

Solution:

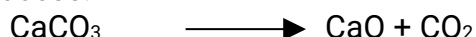


Work:

(a) Calculate the mass of copper required to produce 159g of copper (II) oxide when heated in excess oxygen. (RAMs Cu = 63.5, O = 16)

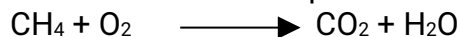


(b) Lime (CaO) is produced by heating limestone (CaCO₃). The equation for the process:



Calculate the amount of lime produced when 10 tonnes of limestone are heated. (RAMs Ca = 40, O = 16, C = 12)

(c) When methane burns in air, the following chemical reaction takes place:



(i) Balance the equation.

(ii) If 18g of water were produced, find the mass of oxygen used.

(d) 0.048g of magnesium was reacted with excess dilute hydrochloric acid at rtp. The hydrogen gas given off was collected.

(i) Write a well balanced equation for the reaction.

(ii) How many moles of Mg were used?

(iii) Calculate the number of moles of hydrogen and hence the volume of this gas produced. (RAMs Mg = 24, H = 1, Cl = 35.5)

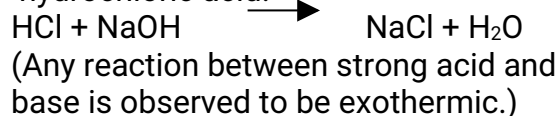
HEATS OF REACTION

When a chemical reaction takes place heat energy is given out, causing the container to get hot or heat energy can be taken in causing the container to get cold.

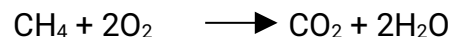
An **exothermic reaction** is a reaction that gives out heat energy to the surrounding.

Examples of exothermic reactions:

(a) The reaction between NaOH and hydrochloric acid.



(b) Combustion reaction i.e.



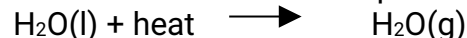
An **endothermic reaction** is a reaction that takes in (absorbs) heat energy from the surroundings.

Examples of endothermic reactions:

(a) Dissolving of ammonium nitrate in water;



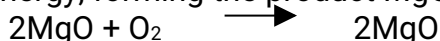
(b) Evaporation of water to water vapour.



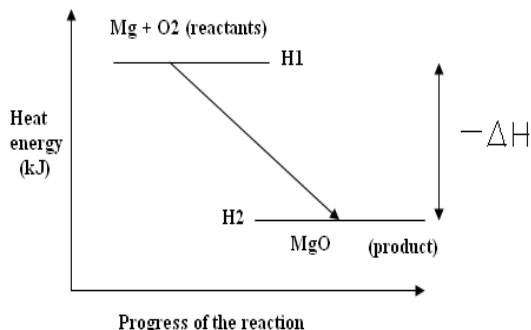
HEAT ENERGY LEVEL DIAGRAMS

Consider the reaction between magnesium and oxygen to form magnesium oxide. The reactants, Mg burned in oxygen, begin at a high position of energy. During burning, the

reaction moves naturally to a low position of heat energy, forming the product MgO.



The heat energy level diagram for this exothermic reaction looks like this:



Heat of reaction is the difference between the final heat energy and the initial heat energy. The symbol for heat of reaction is ΔH . In the case above,

Heat of reaction (ΔH)

= final heat energy – initial heat energy

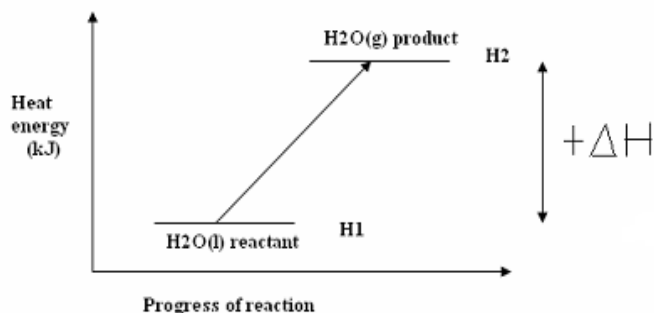
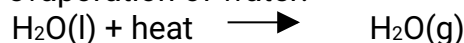
$$= H_1 - H_2$$

Heat of reaction is **negative** in exothermic reaction because final heat energy is less than the initial heat energy as such the difference is negative.

Explanation: When high energy reactants move to low energy products, the change in heat energy, ΔH , is called the **Heat of Reaction** and is negative for all exothermic reactions.

Heat energy level diagram for an endothermic reaction.

Consider the evaporation of water.



The heat of reaction (ΔH) = $H_2 - H_1$ and the result is positive since final heat energy (H_2) is greater than the initial heat energy (H_1).

The positive sign in the diagram means the reaction moves uphill and heat energy is taken in from the surroundings. Therefore, the surroundings lose heat and get cold. Heat is gained by the reaction and the heat of reaction is positive for an endothermic reaction.

Work:

Sulphur is burnt in a plentiful supply of air. The heat is given out from the reaction. Given that the initial heat energy is 28 kJ and the final heat energy is 10 kJ.

- Write a well balanced equation for the reaction.
- State whether the reaction is exothermic or endothermic. Give a reason for your choice.
- Prepare well-labelled energy level diagram for the reaction.
- Calculate the heat of reaction.

BOND BREAKING AND BOND FORMATION

The breaking of old bonds needs energy putting into the reaction. For instance, in the reaction between carbon and oxygen to form carbon dioxide, the kinetic energy of the oxygen molecules is used up as they get near to the carbon atoms because of the repulsions. Kinetic energy is used up in breaking the old bonds.

Formation of new bonds takes place naturally and no energy is needed.

A chemical reaction requires a collision to cause the breaking, followed by the making,

of the chemical bonds. The energy required or needed in breaking the old bonds is and uses up energy. The making up of new bonds in forming the products gives out energy (exothermic). This energy has a negative value

TOPIC: CHEMICAL REACTIONS II

ELECTRON TRANSFER REACTIONS REDUCTION AND OXIDATION (REDOX) REACTIONS

OXIDATION

The word 'oxidation' can be explained as

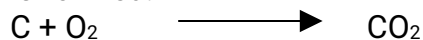
- (i) the increase in the oxygen content

positive (endothermic)

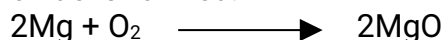
- (ii) loss of electrons and

- (iii) increase in the charge

When carbon is burned in air, carbon dioxide is formed:



Carbon has gained oxygen and is said to have been oxidised. Similarly, when Mg metal atoms are burned in air (oxygen), magnesium oxide is formed:



Magnesium has gained oxygen and has been oxidised. This oxidation reaction can be looked at through its electrical charges. Mg, as a reactant, has a charge of zero. The Mg in the product has a charge of +2 (i.e. Mg^{+2}). During the reaction, the magnesium atom has lost two electrons and the charge on the magnesium atom has been increased from 0 to +2.

OXIDATION NUMBER

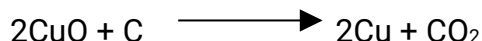
The charge on an atom, or on an ion, is called its **oxidation number** or **oxidation state**. In the case above, it means the oxidation number of Mg has increased from 0 to +2 during the reaction.

REDUCTION

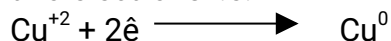
This is described as:

- (i) taking away of oxygen from a substance
- (ii) decreasing the oxidation number
- (iii) accepting (gaining) electrons

When copper oxide is heated with carbon, the following reaction takes place:



During the reaction oxygen has been taken away from the copper. In the same reduction reaction, the reactant copper starts with a +2 charge in CuO. The product formed is copper solid (Cu). The charge has changed from Cu^{+2} to Cu^0 . This means Cu^{+2} has gained two electrons i.e.



USING OXIDATION NUMBERS

1. All elements have an oxidation number of zero. This is because all elements are made of neutral atoms that always have a final charge of zero.
2. Oxygen has an oxidation number of -2, except when it is a gaseous element, then its oxidation number is zero.
3. Hydrogen has a charge of +1, except when it is an element.
4. For all neutral molecules the sum of the separate charges inside the molecule must add up to zero.
5. For a charged ion, the sum of the separate charges inside the ion, must be equal to the final charge on the ion itself.

Examples:

1. Work out the oxidation number of Na in NaOH.

Sum of the separate charges will be zero (0) since NaOH is neutral compound.

$$\text{Na} + (-2) + (+1) = 0$$

$$\text{Na} - 2 + 1 = 0$$

$$\text{Na} - 1 = 0$$

$$\text{Na} = +1$$

Therefore, the oxidation number of Na in NaOH is +1.

2. Find the oxidation number of Mn in MnO_4^- .

Sum of separate charges will be -1 since the net charge on the ion is -1.

$$\text{Mn} + 4(-2) = -1$$

$$\text{Mn} - 8 = -1$$

$$\text{Mn} = 8 - 1$$

$$\text{Mn} = +7$$

Therefore, the oxidation number of Mn in MnO_4^- is +7.

Work:

1. Work out the oxidation number of S in SO_2 .
2. Work out oxidation numbers of the elements in brackets in the following molecules and ions;
 - (a) $\text{Ca}(\text{OH})_2$ (Ca)
 - (b) NH_4^+ (N)

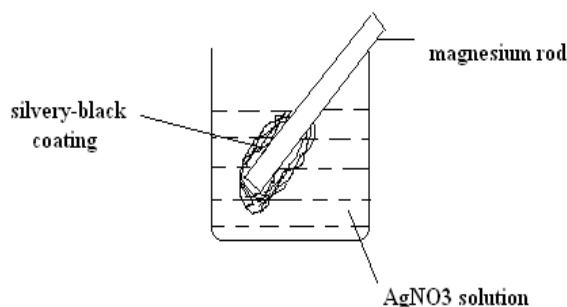
- | | |
|----------------------------------|-------------------------------------|
| (c) CO_2 | (C) |
| (d) MgCl_2 | (Mg) (Oxidation number of Cl is -1) |
| (e) $\text{Cr}_2\text{O}_4^{2-}$ | (Cr) |
| (f) CO_3^- | (C) |
| (g) SO_4^{2-} | (S) |

DISPLACEMENT REACTIONS

When a piece of magnesium metal Mg(s) is placed in an aqueous solution of silver nitrate (AgNO_3) the following things are observed:

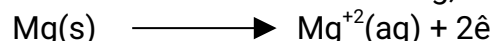
- (i) the magnesium rod begins to dissolve
- (ii) the rod becomes covered with shiny, silvery-black coating of silver metal.

This is described as displacement reaction because the magnesium seems to be dissolving and going into the solution and the silver metal is coming out of solution to deposit a solid silvery-black coating on top of the magnesium rod.

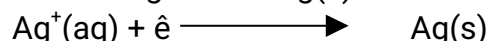


Explanation:

When a metal dissolves in an aqueous solution it ionises. For the case of Mg,



The silver metal coating on the rod is made up of silver atoms. One $\text{Ag}^+(\text{aq})$ ion needs 1 electron to change into a Ag(s) atom.



When a zinc rod is placed in aqueous solution of copper sulphate, the zinc rod dissolves and black-brown coating of copper metal forms on the zinc rod. The zinc is displacing copper from the solution.

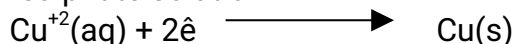
Blueness of the solution begins to get paler after sometime.

Explanation:

When the zinc rod dissolves, it ionises according to the equation:



The lost electrons are immediately accepted by the surrounding $\text{Cu}^{+2}(\text{aq})$ ions in the copper sulphate solution



This equation explains how copper metal is produced and then coated onto the zinc rod.

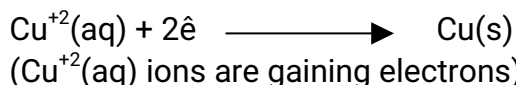
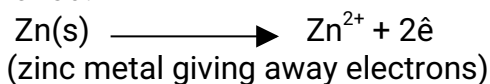
After many similar experiments a **Displacement Table of Metals** is drawn up. The metals at the top of the table would be able to displace the metals lower down the table.

Potassium (K)
Sodium (Na)
Lithium (Li)
Calcium (Ca)
Magnesium (Mg)
Aluminium (Al)
Zinc (Zn)
Iron (Fe)
Tin (Sn)
Lead (Pb)
Hydrogen (H)
Copper (Cu)
Silver (Ag)

THE ELECTROCHEMICAL SERIES

(Giving electrons to other metal ions)

We have seen that Zn rod dissolves and copper metal is deposited (precipitated) onto the zinc rod.



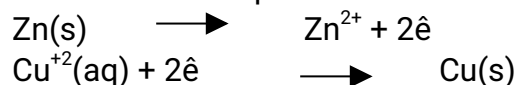
What has been shown here leads to the rule that: any metal can both gain electrons from one metal or donate electrons to another metal. This rule leads to a table of metals being set up. In this table the metals at the top are able to donate (give) electrons to

other metals below them in the table. This table is called **Electrochemical series** or **Activity series**.

K	→	$\text{K}^+ + \text{e}^-$
Na	→	$\text{Na}^+ + \text{e}^-$
Li	→	$\text{Li}^+ + \text{e}^-$
Ca	→	$\text{Ca}^{2+} + 2\text{e}^-$
Mg	→	$\text{Mg}^{2+} + 2\text{e}^-$
Al	→	$\text{Al}^{3+} + 3\text{e}^-$
Zn	→	$\text{Zn}^{2+} + 2\text{e}^-$
Fe	→	$\text{Fe}^{2+} + 2\text{e}^-$
Sn	→	$\text{Sn}^{2+} + 2\text{e}^-$
Pb	→	$\text{Pb}^{2+} + 2\text{e}^-$
H_2	→	$2\text{H}^+ + 2\text{e}^-$
Cu	→	$\text{Cu}^{2+} + 2\text{e}^-$
Ag	→	$\text{Ag}^+ + \text{e}^-$
Au	→	$\text{Au}^{3+} + 3\text{e}^-$

It is observed that the Electrochemical series has the same order of metals as the Displacement Table. In Displacement Table metals higher in the table can displace metals below them from solutions of their ions. In the Electrochemical series metals higher in the series are able to donate (give) electrons to ions of metals below them in the series. There seems to be a connection between displacement and ability to donate electrons.

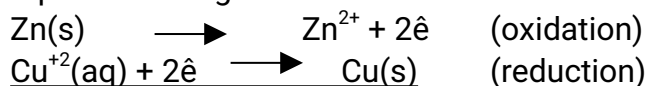
Consider the two half equations below:

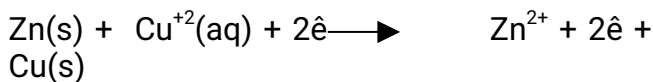


It is observed that the zinc metal atoms are giving away electrons and that the $\text{Cu}^{+2}(\text{aq})$ ions are accepting the same electrons. This would put zinc higher in the Electrochemical series. At the same time zinc metal is dissolving and displacing the copper metal from a solution of its ions.

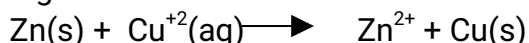
OVERALL EQUATION

The two half equations above can be combined to give the overall equation for the displacement reaction. Adding the two equations will give:





Cancelling out the electrons on both side



Such equation is known as the **overall net ionic equation**.

Work:

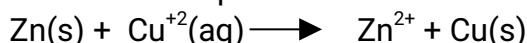
In a reaction involving magnesium rod dipped in a silver nitrate solution $[\text{AgNO}_3]$:

(a) Write the oxidation and reduction equations.

(b) Determine the overall net ionic equation for the reaction. (*Hint*: Balance the number of electrons)

OXIDISING AND REDUCING AGENTS

Consider the redox equation below:



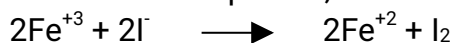
Cu^{+2} is being reduced because it is gaining two electrons and its oxidation number is reduced from +2 (in Cu^{+2}) to 0 (in Cu).

Zn is being oxidised because it is losing two electrons and its oxidation number is increasing from 0 to +2 (in Zn^{2+}). Since Zn is being oxidised, then the **oxidising agent** must be the other reactant Cu^{+2} . Cu^{+2} is being reduced, then, the other reactant Zn that has caused the reduction is the **reducing agent**.

Definitions:

- An oxidising agent is the species that has been reduced in a redox reaction.
- A reducing agent is the species that has been oxidised in a redox reaction.

Example: Identify the reducing and oxidising agents in this redox equation;



Fe^{+3} is the oxidising agent i.e. it has oxidised I^- to I_2

I^- is the reducing agent i.e. it has reduced Fe^{+3} to Fe^{+2}

Another *hint* to identify reducing and oxidising agents:

- A species whose oxidation number (state) has increased becomes a reducing agent i.e. the oxidation number in I^- is -1 and has increased to 0 in I_2 .
- A species whose oxidation number (state) has decreased or reduced becomes an oxidising agent i.e. the oxidation number of Fe^{+3} is +3 and has decreased to +2 in Fe^{+2} .

Work:

Identify oxidising and reducing agents in the following redox equations:

1. $\text{Mg} + \text{Sn}^{+2} \longrightarrow \text{Mg}^{+2} + \text{Sn}$
2. $2\text{Mg} + \text{O}_2 \longrightarrow 2\text{MgO}$
3. $5\text{Fe}^{+2} + 8\text{H}^+ + \text{MnO}_4^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{+3}$

RUSTING

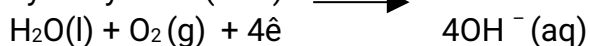
HOW RUST IS FORMED

Rusting is the name we give to the oxidation of iron or steel in damp air. It is also called **corrosion**.

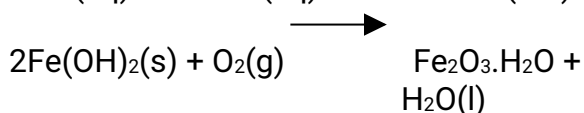
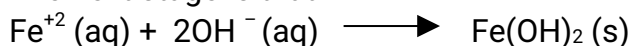
Iron ionises in water and loses electrons according to this equation:



The presence of H_2O , O_2 and electrons forms hydroxyl ions (OH^-):

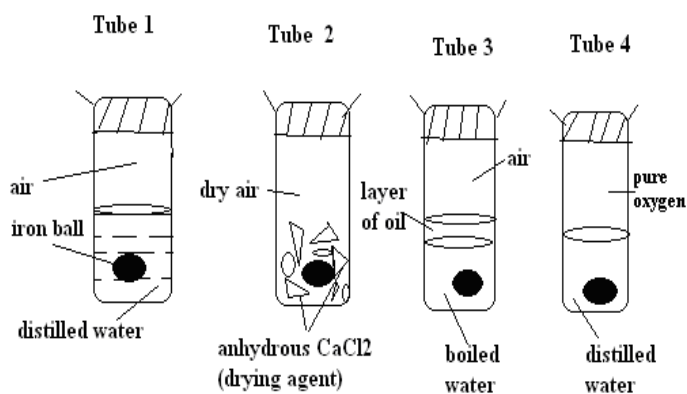


The next stage is that



$\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$ is what is known as **rust**.

From the explanation it is undeniable point to say water and oxygen are essential for iron to rust and if one of these two substances is not present then rusting will not take place. The following figures show experiments carried out to confirm that water and oxygen (from the air) are needed for iron to rust.



Results:

Tube 1: Rusting is observed on the iron ball since oxygen (from the air) and water are available.

Tube 2: No rusting, CaCl₂ being a drying agent makes the tube to have no water. There is only one essential requirement which is oxygen (from the air)

Tube 3: No rusting, in boiled water dissolved oxygen is removed and the layer of oil prevents air from dissolving in the water. As such water alone cannot make rust on the iron ball.

Tube 4: Rusting is observed, pure oxygen and water are essential for rusting.

RUST PREVENTION

To prevent iron rusting, it is necessary to stop oxygen (from the air) and water coming into contact with it. There are several ways of doing this:

- **Painting** – materials are painted to prevent rusting. However, if the paint is scratched, the iron beneath it will start to rust.
- **Oiling/greasing** – the iron and steel in the moving parts of machinery are coated with oil or grease to prevent them from coming into contact with air or moisture.

- **Coating with plastic** – most refrigerators, freezers and many other items are coated with plastic such as PVC to prevent the steel structure rusting.
- **Plating/electroplating** – cans for food can be made from steel coated with tin. Some car bumpers, as well as bicycle handlebars are electroplated with chromium to prevent rusting. The chromium gives a decorative finish as well as protecting the steel beneath.
- **Galvanising** – some steel girders, coal bunkers and steel dustbins are galvanised. This involves dipping the object into molten zinc. The thin layer of the more reactive zinc metal coating the steel object slowly corrodes and loses electrons to the iron, thereby protecting it. Zinc is higher in the electrochemical series i.e. more reactive than iron. Thus, whenever iron ionises it is forced to accept electrons from zinc and this causes iron, Fe(s) to be formed. This prevents (stops) the formation of Fe⁺²(aq) and electrons which would have been needed to allow the rusting equations (discussed earlier on) to take place. This process continues even when much of the layer of zinc has been scratched away, so the iron continues to be protected.
- **Sacrificial protection** – iron pipes under the soil, or under water, can easily corrode. It is difficult and costly to dig up a pipe and replace it. It is easier to connect an iron pipe to a piece of metal higher in the electrochemical series, such as aluminium or magnesium. It works like this: a magnesium ball is connected to an iron pipe and the iron has to accept electrons from magnesium. So the iron cannot lose electrons and it does not dissolve. In this way the iron pipe is prevented from rusting. Instead it is the magnesium that dissolves away very

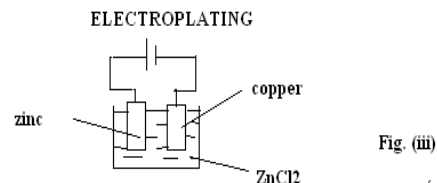
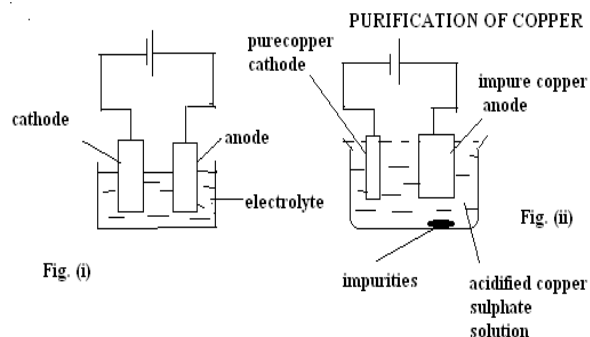
slowly in place of the iron. As it dissolves, the magnesium slowly loses electrons to the iron. A large ball of magnesium, or aluminium, can protect an iron pipe many kilometres long, and for many years, before being used up.

ELECTROLYSIS AND ITS USES

Definitions:

- Electrolysis – a process in which a chemical reaction is caused by the passage of an electric current in an electrolyte.
- Electrolyte – a substance which will carry electric current only when it is molten or dissolved. The electricity is carried through the electrolyte by ions.
- Electrodes – points where the electric current enters and leaves the electrolyte. Electrodes are usually made of unreactive metals such as platinum or of the non-metal carbon (inert electrodes).
- Cathode – the negative electrode which attracts cations (positively charged ions). Excess electrons move towards it.
- Anode – the positive electrode which attracts anions (negatively charged ions). It is positively charged because electrons are drawn away from it.

Study **Fig. (i)** below to understand these definitions very well.



USES OF ELECTROLYSIS

1. Purification of metals i.e. copper
2. Electroplating
3. Producing gases and other solutions.

PURIFICATION OF COPPER

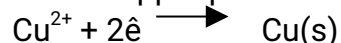
The impure copper is used as the anode. The cathode is made from very pure copper. The electrolyte is a solution of copper (II) sulphate (acidified), to help the solution conduct electricity.

Refer to **Fig. (ii)** above.

During the process the impure anode loses mass because the copper atoms lose electrons and become copper ions;



The electrons released at the anode travel around the external circuit to the cathode. There the electrons are passed on the copper ions (from CuSO_4 solution) and the copper is deposited or copper plated on the cathode.



Any impurities fall to the bottom of the cell and collect below the anode.

ELECTROPLATING

This is a process that involves electrolysis to plate or coat one metal with another. The purpose of electroplating is to give a protective coating to the metal beneath.

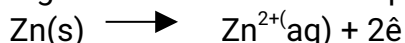
Conditions:

- The metal to be covered (plated) is made cathode.

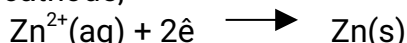
- The covering metal is made anode.
- The electrolyte should be a soluble salt of the same metal that makes up the anode.

Refer to **Fig. (iii)** above.

At the anode, zinc will dissolve and ionise according to this half oxidation equation;



Through the external circuit the electrons travel to the cathode where they are accepted by Zn^{2+} ions from the solution to form zinc atoms. The reduction half equation at the cathode;



The zinc atoms accumulate on the copper cathode thereby plating it. In this case, copper is said to be electroplated with zinc.

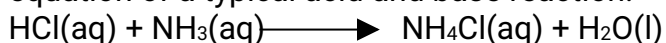
PROTON TRANSFER REACTIONS

LOWRY-BRONSTED THEORY OF ACIDS AND BASES

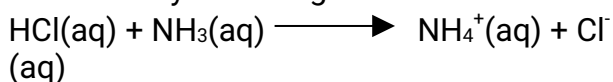
A proton is a positive hydrogen ion (H^+).

An acid is an ion or molecule that can donate (give away) protons.

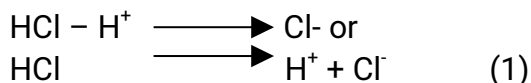
A base is an ion or molecule that can accept protons. An acid-base reaction takes place when an acid and base react with each other. In the reaction, the acid donates a proton which is accepted by the base. Consider the equation of a typical acid and base reaction.



Another way of writing this would be:



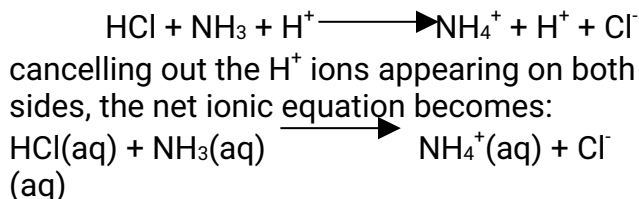
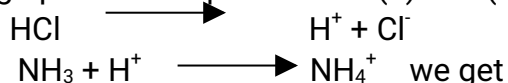
HCl is giving away a H^+ (a proton) to become a Cl^- ion. Therefore HCl is acting as a L/B acid, and this equation should be written fully as:



NH_3 has accepted a H^+ (a proton) and therefore NH_3 is acting as L/B base.

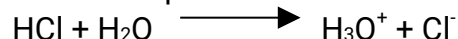


Adding up the two equations i.e. (1) and (2)

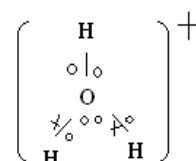


LOWRY-BRONSTED SYSTEM AND WATER

Ionisation equations involve a reaction with water, and the water molecule acts as a L/B base by accepting a proton (H^+). An example of this is the equation:

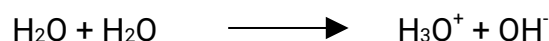


The water molecule has accepted the H^+ (proton) to form H_3O^+ ion. This ion is called the **HYDRONIUM ION**. The dot and cross diagram for the hydronium ion is:



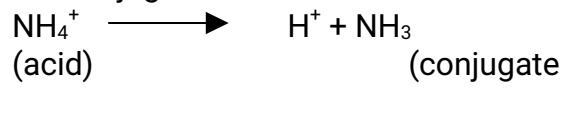
Water can act as a base or an acid. This means that water can donate protons as well as accepting them. Such substances are referred to as **amphoteric substances**.

The equation below is showing water molecule donating a proton to another water molecule:



CONJUGATE ACIDS AND BASES

When an acid loses a proton it forms what we call its conjugate base:

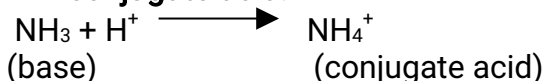


In this case the remaining species after NH_4^+ has lost a proton is NH_3 and becomes the **conjugate base**.

A conjugate base is the species that remains after an acid has given up a proton.

If an acid is strong (because it loses a proton easily), then conjugate base must be weak (because it does not accept a proton easily)

When a base accepts a proton it forms what we call its **conjugate acid**.

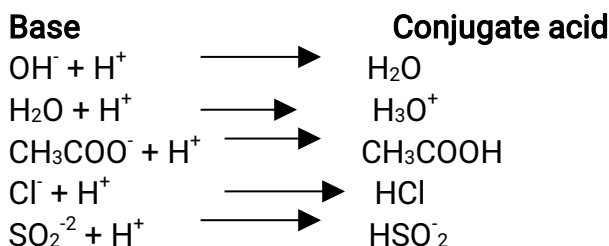
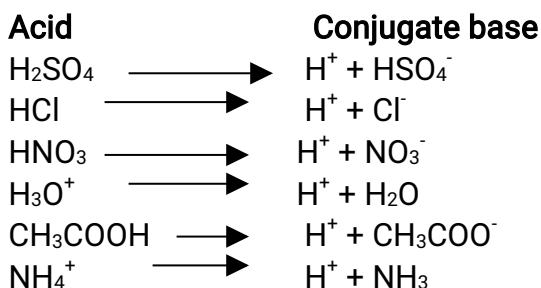


Ammonia (a base) has accepted a proton to form its conjugate acid i.e. NH_4^+

A conjugate acid is the species formed when a base takes on a proton.

If a base is strong (because it accepts a proton easily) then the conjugate acid must be weak (because it does not lose a proton easily).

The equations below show acids and their conjugate bases as well as bases with their conjugate acids.



STRENGTH OF ACIDS AND BASES

Strong acids in terms of the Ionic Theory, are those which ionise (break up) almost completely into their ions. These acids give plenty of hydrogen ions (H^+) into the solution. A weak acid donates protons a little to give a few protons (H^+) to the receiving base.

A base with strong basic strength, according to the Lowry – Bronsted Theory, is the one that accepts protons well and holds them strongly. The conjugate base in a strong acid/weak base reaction must have a very weak basic strength because it is not able to hold on the protons (H^+).

Examples of strong acids: HClO_4 (perchloric acid), HCl , HNO_3 (nitric acid), H_2SO_4

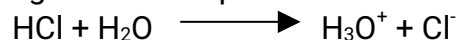
(sulphuric acid).

Examples of weak acids: CH_3COOH (ethanoic acid), H_2CO_3 (carbonic acid), H_2S (hydrogen sulphide), H_3PO_4 (phosphoric acid).

Examples of strong bases: NH_3 (ammonia), H_2O , KOH (potassium hydroxide), NaOH (sodium hydroxide).

Worked example:

Identify acid and base conjugate pairs in the following chemical equation.



Solution:

HCl/Cl^- (acid/conjugate base)

$\text{H}_2\text{O}/\text{H}_3\text{O}^+$ (base/conjugate acid)

TOPIC: FORCES AND MOTION

SCALAR AND VECTOR QUANTITIES

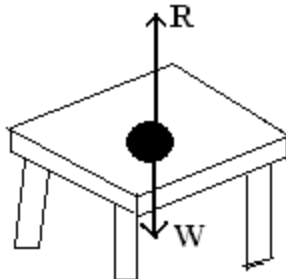
A scalar quantity is a quantity that has magnitude (size) only. Mass is a scalar quantity and is completely described when its value is known. Scalars are added by ordinary arithmetic. Other examples of scalar quantities include; speed, work done, temperature, distance, e.t.c.

A vector quantity is a quantity that has both magnitude (size) and direction. A vector quantity is one such as force which is described completely only if both its size (magnitude) and direction are stated. It is not enough to say, for example, a force of 10N, but rather a force of 10N acting vertically downwards. Other examples of vector quantities are displacement, velocity, acceleration e.t.c.

A vector can be represented by a straight line whose length represents the magnitude of the quantity and whose direction gives its line of action. An arrow on the line shows which way along the line it acts.

As already explained, force has both magnitude and direction. It is therefore represented in diagrams by a straight line with an arrow to show its direction of action.

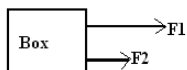
Usually more than one force acts on an object. As a simple example, an object resting on a table is pulled downwards by its weight **W** and pushed upwards by a force **R** due to the table supporting it. Since the object is at rest, the forces must balance i.e. **R = W**



When two or more forces act on an object a single force can be found which will replace these two or more forces and produces the same effect as the two or more forces do. This single force is called the **resultant force**.

FORCES ACTING IN THE SAME STRAIGHT LINE

When two given forces act in the same straight line the resultant force is relatively easy to find.



In this case the resultant force **R** can be obtained by adding together the forces **F₁** and **F₂**.

$$R = F_1 + F_2$$

Consider also a case where two teams are in a tug of war. The team on the right pull on the rope from left to right while the other team exerts its pull from right to left. If the force exerted by the team to the right is **F₁** and the other team pulls with a force of **F₂**, the resultant force **R** will be given by:

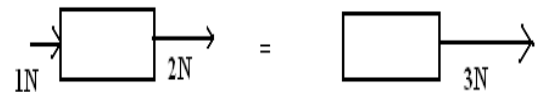
$$R = F_1 - F_2$$

If the value of the forces **F₁** and **F₂** are equal the **R = 0**, zero unbalanced force. In this case equilibrium exists i.e. a piece of rope does not move in either direction.

Examples:

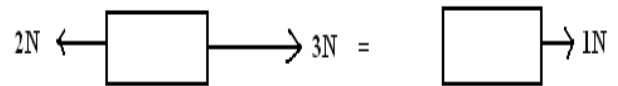
a) Determine the resultant force in the figure

below:



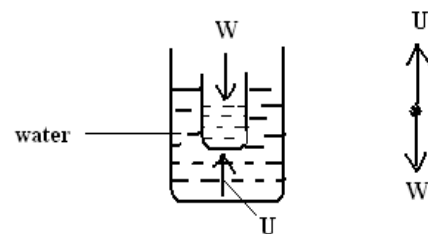
$$R = 1\text{N} + 2\text{N} = 3\text{N to the right hand.}$$

b) Work out the resultant force in the diagram below:



$$R = 3\text{N} - 2\text{N} = 1\text{N to the right hand.}$$

c) A loaded test tube of weight 0.6N floats in equilibrium in a tank of water. Calculate the upthrust **U** of the water on the test tube.



$$\text{For equilibrium, } R = U - W = 0$$

$$U - 0.6\text{N} = 0$$

$$U = + 0.6\text{N}$$

Work:

- A man is running forward at a velocity of 5Km/h on top of a moving train which is travelling 50Km/h to the east. What is the resultant velocity of the man compared to the ground?
- A lady is trying to swim upstream (in opposite direction to the river current) with a velocity of 3Km/h. If the river's velocity is 8Km/h, what is the resultant velocity of the swimmer relative to the bank?
- Jo, Dan and Helen are pulling a metal ring. Ho pulls with a force of 100N in one direction and Dan with a force of 140N in the opposite direction. If the ring does not move, what force does

Helen exert if she pulls in the same direction as Jo?

- d) A boy drags a suitcase along the ground with a force of 100N. If the frictional force opposing the motion of the suitcase is 50N, what is the resultant forward force on the suitcase?

VECTORS AT AN ANGLE TO EACH OTHER

When two give forces are not acting in the same straight line, nor parallel to each other, they must be acting at some angle to each other. If the object they act on does not break apart it can not move in two different directions at once! Therefore the object will move in some direction between the two lines of the forces.

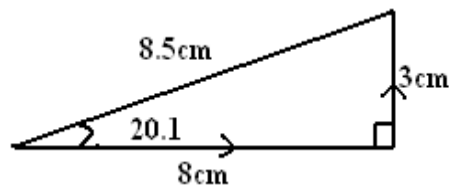
Consider a lady swimming across a river with velocity of 3Km/h at right angles. If the rivers velocity is 8Km/h, what is the swimmer's true velocity (resultant) compared to the river bank?

Solution:

By scale diagram

Let the scale be: 1cm on paper = 1Km/h on the velocity vector diagram.

Draw each velocity vector in turn as shown below:



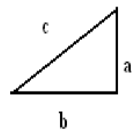
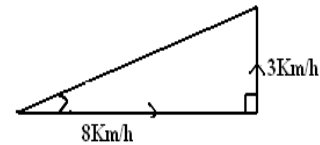
The resultant velocity vector is found from the hypotenuse of the completed right angled triangle.

$R = 8.5\text{Km/h}$

The resultant velocity is 8.5Km/h at angle of 20.1° to the river bank.

By calculation

From the right angled velocity vector diagram, the resultant velocity is worked out using Pythagoras' theorem:



$$a^2 + b^2 = c^2$$

$$R^2 = 3^2 + 8^2$$

$$R^2 = 9 + 64 = 73$$

$$R = 8.5$$

$$R = 8.5\text{Km/h}$$

$$\tan x = 8/3 = 2.67$$

$$x = 69 \quad 27^\circ$$

An angle formed between R and the bank

$$= (90^\circ - 69 \quad 27^\circ)$$

$$= 20^\circ 33'$$

Therefore, the resultant velocity R is **8.5Km/h** at an angle of **$20^\circ 33'$** with the river bank.

Work:

- cyclist travels due south from point R for 12Km to point P, then she turns to the west and cycles to point Q for another 5Km. Work out displacement RQ (i) by scale drawing (ii) by calculation.
- Using a scale of 1cm to represent 10N find the size and direction of the resultant of forces of 30N and 40N acting at right angles to each other.

PARALLELOGRAM RULE

To find the resultant force R when two forces act at a point and the direction in which it acts we use the parallelogram of forces rule. This rule states that

"if two forces F_1 and F_2 , acting at a point, are represented to scale in both magnitude and direction by the two adjacent sides of a parallelogram, then the diagonal from their point of intersection is the resultant force R in both magnitude and direction."

Example:

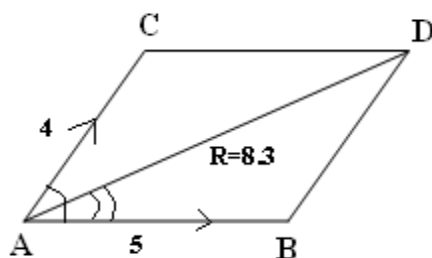
Find the resultant of two forces of 4.0N and

5.0N acting at an angle of 45° to each other.
(Use a scale of $1\text{cm} = 1\text{N}$)

Procedure:

Using a scale of $1\text{cm} = 1\text{N}$, draw a parallelogram ABCD with $AB = 5\text{N}$, $AC = 4\text{N}$ and angle $= 45^\circ$.

The diagonal AD represents the resultant in magnitude and direction. Using a protractor measure the angle the diagonal makes with 5.0N force line.



Therefore, resultant is a force of 8.3N acting at an angle of 21° to the force of 5.0N.

Work:

- Two forces of magnitude 8N and 6N act on the same body. The angle between the directions of the forces is 40° . Using the parallelogram rule, determine the magnitude and direction of the resultant of the forces.
- Using a suitable scale, find the resultant of the two forces of 4N and 3N acting at an angle of 60° to each other.
- A magnetic compass needle is subjected to a force of 0.02N acting north and force of 0.04N acting east. Calculate the resultant force on the needle and the direction in which it sets.
- Forces of 7.0N and 11.0N act at a point so that the angle between their lines of action is 35° . By means of a scale diagram, determine the magnitude of the resultant of these two forces.

RESOLVING VECTORS

Clearly if two forces can be replaced by a

single resultant force, then a single force can be 'split' into two forces acting at some appropriate angle between them. If, as is often convenient, the angle between these two component forces is 90° , the procedure is known as **resolving** a force into two components at right angles.

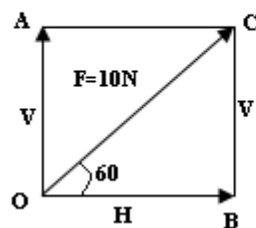
Example:

A 10N force acting at an angle of 60° to the horizontal can be resolved into two components at right angles. Calculate the vertical and horizontal forces.

By calculation method:

Procedure:

Sketch diagram of the force $F = 10\text{N}$ at 60° to the horizontal and the two components V and H resolved in the vertical and horizontal directions respectively. The triangle OBC enables the values of V and H to be calculated using trigonometry.



$$\begin{aligned}\sin \phi &= V/F \\ F \sin \phi &= V \\ V &= 10 \sin 60^\circ \\ V &= 8.66\text{N}\end{aligned}$$

$$\begin{aligned}\cos \alpha &= H/F \\ H &= 10 \cos 60^\circ \\ H &= 10 \times 0.50 \\ H &= 5.0\text{N}\end{aligned}$$

Therefore, the vertical force is 8.66N and the horizontal force is 5.0N.

By scale drawing method:

A force can be resolved into two components by using a scale drawing method. Begin by drawing the given single force and then construct the four – sided figure which has this line as a diagonal. Then, measure the vertical and horizontal lengths. These lengths will be converted to forces using the given

scale. Try to do this and you will be able to get 8.7N and 5.0N as vertical and horizontal lengths respectively.

Work:

A girl pulls a friend sitting in a trolley by means of a rope inclined at 30° to the horizontal. If the tension T in the rope is 200N, find

- the effective force pulling the trolley along
- the force tending to lift the trolley off the ground.

DISTANCE AND DISPLACEMENT

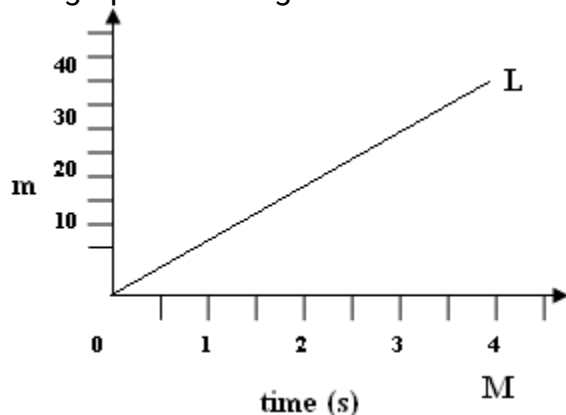
Distance is a scalar quantity whereas displacement is a vector quantity. This means distance has magnitude only and displacement has both magnitude (size) and direction.

Distance is the length between two points. It is measured in meters or kilometres.

Displacement can be defined as the distance moved in a stated direction.

Distance – time Graphs

A body travelling with uniform velocity covers equal distances in equal times. Its distance – time graph is a straight line.



The slope gradient is $\frac{LM}{OM} = \frac{40\text{m}}{4\text{s}} = 10\text{m/s}$

10 m/s is the value of the velocity. The following statement is true in general:

"The slope or gradient of a distance – time graph represents the velocity of the body."

SPEED, VELOCITY AND ACCELERATION

SPEED: The speed of an object is its distance moved per unit time.

$$\text{Speed} = \frac{\text{distance moved}}{\text{time taken}} \text{ (m/s)}$$

However, this definition applies only to an object which is moving steadily during the time 'taken' of the quotient. When the object is not moving at a constant speed then the quotient distance moved/time taken gives the average speed and therefore it is better to write:

$$\text{Average speed} = \frac{\text{distance moved}}{\text{time taken}} \text{ (m/s)}$$

VELOCITY:

This is the distance travelled in unit time in a stated direction. This means velocity of an object is its displacement per unit time.

Velocity

= $\frac{\text{displacement}}{\text{time}}$ in the direction of displacement

Time

The velocity of a body is uniform or constant if it moves with a steady speed in a straight line. It is not uniform if it moves in a curved path.

ACCELERATION:

When the velocity of a body changes we say the body accelerates.

Acceleration is the change of velocity in unit time.

$$\text{Acceleration} = \frac{\text{change of velocity}}{\text{time taken for change}}$$

For example:

For a steady increase of velocity from 20 m/s to 50 m/s in 5 seconds. Calculate the acceleration.

$$\text{Acceleration} = \frac{(50 - 20) \text{ m/s}}{5 \text{ s}} = 6 \text{ m/s}^2$$

Acceleration is also a vector and both its magnitude and direction should be stated. Note that an object moving with uniform

velocity has zero acceleration because there is no change in velocity.

Uniform acceleration

Uniform acceleration requires that the velocity changes regularly (uniformly) with time. The speeds of a car accelerating on a straight road are shown below:

Time (s)	0	1	2	3	4	5	6
Speed (m/s)	0	5	10	15	20	25	30

The speed increases by 5 m/s every second and the acceleration of 5 m/s^2 is said to be uniform. An acceleration is positive if the velocity increases and negative it decreases. A negative acceleration is also called a **deceleration** or **retardation**.

Example:

A car travelling at a velocity 25 m/s due west slows down uniformly to a velocity of 5 m/s due west in 10 seconds. Calculate the acceleration of the car during these 10 seconds.

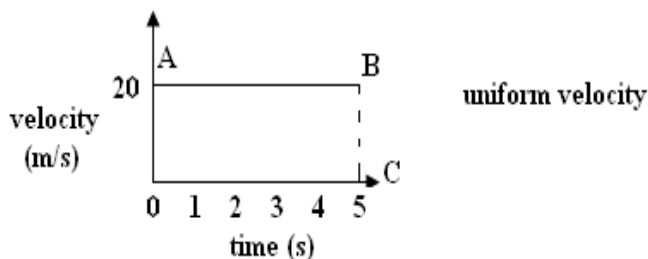
Solution:

Since there is decrease in velocity, it means there is deceleration.

$$\begin{aligned}\text{Deceleration} &= \frac{(5 - 25) \text{ m/s}}{10 \text{ s}} \\ &= -2 \text{ m/s}^2.\end{aligned}$$

Velocity – time graphs

If the velocity of a body is plotted against the time, the graph obtained is a velocity – time graph. It provides a way of solving motion problems.

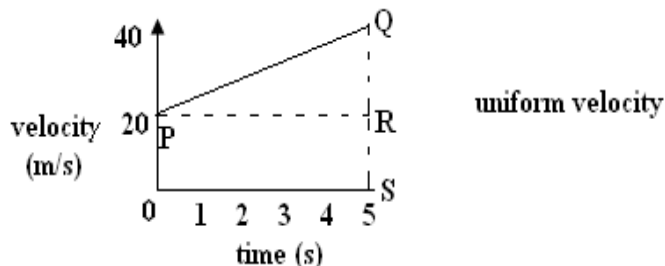


The area under a velocity – time graph measures the distance travelled.

AB is the velocity – time graph for a body moving with uniform velocity of 20 m/s.

Since distance = average velocity $\times$ time, after 5 seconds it will have moved $20 \text{ m/s} \times 5 \text{ s} = 100 \text{ m}$. This is the area under the graph i.e. rectangle OABC.

Consider also the graph below:



PQ is the velocity – time graph for a body moving with uniform acceleration. If the distance covered equals the area under PQ, then

$$\begin{aligned}\text{Distance} &= \text{area of rectangle OPRS} + \text{area of triangle PQR} \\ &= (20 \text{ m/s} \times 5 \text{ s}) + \left(\frac{1}{2} \times 5 \text{ s} \times 20 \text{ m/s}\right) \\ &= 100 \text{ m} + 50 \text{ m} \\ &= 150 \text{ m}\end{aligned}$$

The distance can be calculated by looking at the entire shape of the graph as a trapezium i.e. OP // SQ.

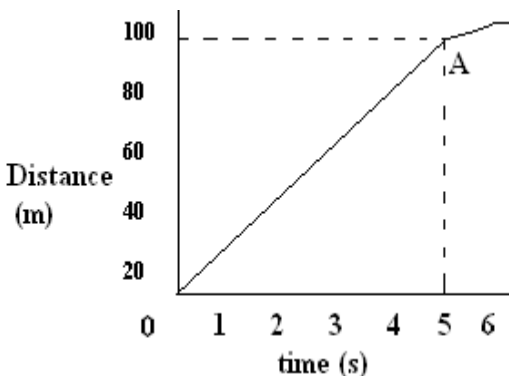
$$\begin{aligned}\text{Distance} &= \text{area of trapezium OSQP} \\ &= \frac{1}{2} (\text{sum of // sides}) h \\ &= \frac{1}{2} (20 + 40) 5 \text{ m} \\ &= 150 \text{ m}\end{aligned}$$

The slope or gradient of a velocity – time graph represents the acceleration of the body. From the graph above:

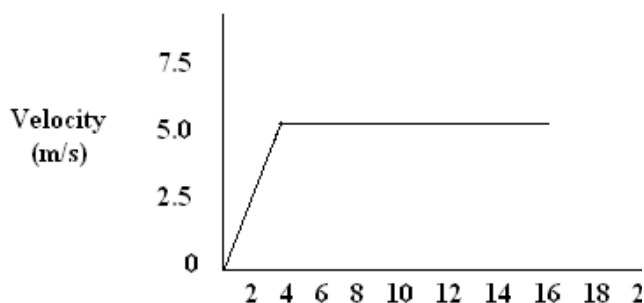
$$\begin{aligned}\text{Slope of PQ} &= \frac{\text{QR}}{\text{PR}} = \frac{20}{5} = 4 \\ &= 4 \text{ m/s}^2\end{aligned}$$

Work:

- The graph below represents the distance travelled by a car plotted against time.



- How far has the car travelled at the end of 5 seconds?
 - What is the speed of the car during the first 5 seconds?
 - What has happened to the car after A?
 - Draw a graph showing the speed of the car plotted against time during the first 5 seconds.
- b) Graph below shows an incomplete velocity – time graph for a young lady running a distance of 100m.

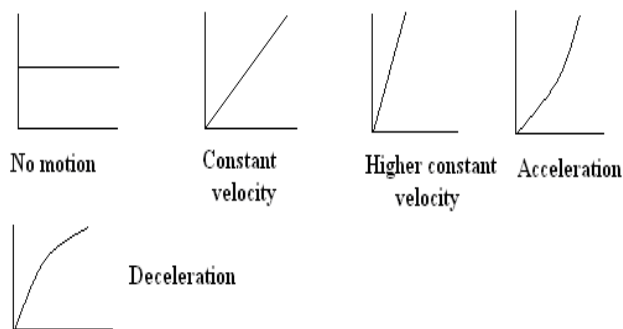


- What is the acceleration of the girl during the first 4 seconds?
 - How far does the girl travel during (1) the first 4 seconds? (2) the next 9 seconds?
 - Copy and complete the graph showing clearly at what time the girl has covered the distance of 100 m. Assume her speed remains constant at the value shown by horizontal portion of the graph.
- c) A car, initially at rest, moves with uniform acceleration for 20 seconds until it attains a velocity of 15 m/s. It then proceeds at this velocity for 60

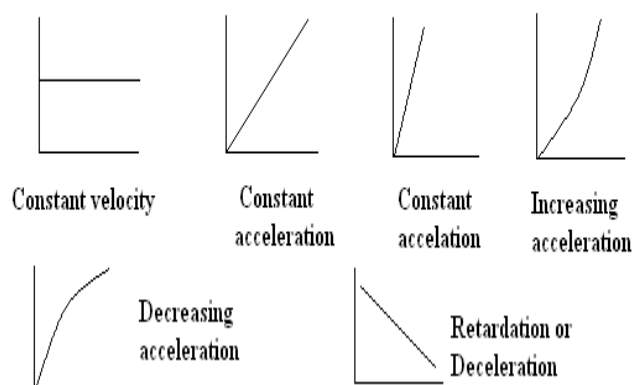
seconds and finally comes to rest after retarding uniformly for a further 10 seconds. Draw a velocity – time graph of the motion and calculate

- the retardation
- the total distance moved.

Distance - time graphs



Velocity - time graphs



NEWTON'S LAWS OF MOTION

Isaac Newton (1642 – 1727) studied the motion of objects and formulated three laws of motion.

THE FIRST LAW OF MOTION

The states that:

“Every object which is at rest will remain at rest unless an unbalanced external force is applied to it.”

It can also be stated in this way:

“An object which is moving at uniform speed in a straight line will continue to do so unless an unbalanced

external force is applied to it."

The first statement above is common experience for everyone. For example, if you leave a book on a bench then it will stay in the same position until someone moves it; i.e. they apply a force to it. If the book is not in the place where you left it then the only sensible conclusion is that someone 'pushed' or 'pulled' it into its new position.

The second statement is not quite as obvious as the first. When you see an object moving in a straight line, for along a bench, it does not continue to move at the same speed, but always slows down and eventually stops. This can be reasoned as follows: It takes a force to make a stationary object move, so it should also take a force to make a moving object slow down and stop. Hence if a moving object is observed to slow down and stop you can conclude that a force has made this happen.

The force is mainly the frictional force between the moving object and the surface over which it travels. Air also exerts a 'drag' force on an object pushing its way through it thereby slowing the object down.

THE SECOND LAW OF MOTION

Experiments have shown roughly that the acceleration a of an object is

- 1) directly proportional to the applied force F for a fixed mass i.e. $a \propto F$, and
- 2) inversely proportional to the mass m for a fixed force, i.e. $a \propto 1/m$.

Combining the results into one equation, we get

$$a \propto F/m \text{ or } F \propto ma$$

Therefore $F = kma$, where k is the constant of proportionality.

One Newton (1 N) is defined as the force which gives a mass of 1 Kg an acceleration of 1 m/s^2 i.e.

$$1 \text{ N} = 1 \text{ Kg m/s}^2$$

So if $m = 1 \text{ Kg}$ and $a = 1 \text{ m/s}^2$ then $F = 1 \text{ N}$.

Substituting in $F = kma$ we get $k = 1$ i.e. $k = F/ma = 1$.

So we can write, $F = ma$ and this is Newton's

second law of motion.

Newton's second law of motion can be stated as follows:

"Force exerted on an object is directly proportional to the product of its mass and acceleration."

Newton's second law of motion can also be stated as:

"The rate of change of momentum of an object is equal to the unbalanced force applied to it causing the change in momentum to take place in the direction of the force."

That is:

Unbalanced force (F) = rate of change of momentum.

$$= \frac{mv - mu}{t}$$

where m = mass of an object, v = final velocity, u = initial velocity and t = time taken

$$F = \frac{m(v - u)}{t}$$

but $\frac{v - u}{t}$ = acceleration (a)

Therefore, $F = ma$

Examples:

- a) A body of mass 15Kg was given a push that caused the body to accelerate by 3 m/s^2 . What is the magnitude of the push?

Solution.

The question demands us to

find

the force.

Mass $m = 15 \text{ Kg}$,

acceleration $a = 3 \text{ m/s}^2$

$$F = ma$$

$$F = (15 \times 3) \text{ N}$$

$$F = 45 \text{ N}$$

- b) A car of mass 1200Kg is travelling at a steady speed of 30m/s from Thondwe to Zomba. A goat steps out in front causing the driver to brake suddenly. It takes 4 seconds to stop the car.

i. Work out the deceleration.

$$\text{Deceleration} = \frac{v - u}{t}$$

Initial velocity $u = 30\text{m/s}$, final velocity $v = 0\text{m/s}$, $t = 4\text{s}$

$$\text{Deceleration} = \frac{0 - 30}{4} \text{ m/s}^2$$

$$= -7.5\text{m/s}^2$$

- ii. What is the average stepping force?

$$\text{Force } F = ma$$

$$\text{Mass} = 1200\text{Kg}, a = -7.5\text{m/s}^2$$

$$\text{Force} = 1200 \times (-7.5)\text{N}$$

$$\text{Force} = -9000\text{N}$$

Therefore, the average stepping force F is 9000N . (The negative sign (-) simply shows direction.

Work:

- A body receives a force of 10N . This makes the body move with an acceleration of 2.5m/s^2 . What is the mass of this body?
- A lorry has a mass of 400Kg and the force of the engine is 2400N .
 - What is the acceleration of the lorry?
 - What force is needed to decelerate the same lorry at 2.0m/s^2 ?
- A horizontal force of 20N is applied to a mass of 5Kg at rest on a smooth horizontal surface.
 - Calculate the acceleration of the mass.
 - If the same force is applied to the same mass when resting on a rough surface, a constant frictional force of 5N acts through out the motion of the mass. What is the acceleration of the mass on the rough surface?
- A car of mass 1000Kg travelling at 20m/s crashes into a brick wall and comes to rest in 0.5s . Calculate the average force exerted by the wall on the car.
- A block of mass 2Kg is pushed along a table with a constant velocity by a

force of 5N . When the push is increased to 9N what is

- the resultant force?
 - the acceleration?
- f) A car of 1200Kg travelling at 72Km/h is brought to rest in 4s . find
- the average deceleration
 - the average braking force.
- (Hint: change Km/h to m/s)

THE THIRD LAW OF MOTION

Newton's third law of motion states that:

"If a body A exerts a force on body B, then body B exerts an equal but opposite force on body A."

This law says that forces always occur in pairs as the result of the interaction between two objects. Note that two objects are involved and the two forces each act on a different object, never on the same object. The two opposing forces are sometimes called the **action** and the **reaction** forces. This is the reason why Newton's third law of motion is also stated as:

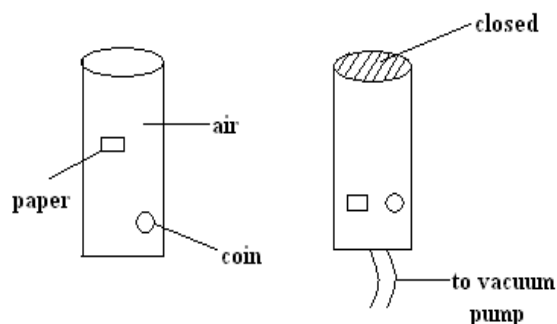
"For every force of action acting on a body, there is an equal and opposite force of reaction by the body."

Applications of Newton's third law

- An appreciation of the third law and the effect of friction is desirable when stepping from a rowing boat. You push backwards on the boat and, although the boat pushes you forwards with equal force, it is itself now moving backwards (because friction with water is slight). This reduces your forwards motion by the same amount – so you may fall in!
- When you walk the Earth pushes on you in response to your push on the Earth. Therefore a friction force between your shoe and the ground is essential for forward motion.
- Motor cars are able to move along a road because the reaction of the road pushes the car along in response to the action of the wheels pushing on the road.

FREE FALL

In air, a coin falls faster than a small piece of paper. In vacuum they fall at the same rate as may be shown with the apparatus below:



The difference in air is due to air resistance having a greater effect on light bodies than on heavy bodies. The air resistance to a light body is big when compared to with the body's weight. With a dense piece of metal the resistance is negligible at low speed.

Acceleration of free fall

All bodies falling freely under the force of gravity do so with uniform acceleration if air resistance is negligible. The acceleration, called the **acceleration of free fall**, is denoted the Italic letter *g*. its value varies over the earth. In Britain it is about 9.8m/s^2 or near enough 10m/s^2 . The velocity of a free-falling body therefore increases by 10m/s every second.

AIR RESISTANCE: TERMINAL VELOCITY

When an object falls in air, the air resistance (fluid friction) opposing its motion increases as its speed rises, so reducing its acceleration. Eventually, air resistance acting upwards equals the weight of the object acting downwards. The resistant force on the object is then zero since the gravitational force balances the frictional force. The object falls at a constant velocity, called its **terminal velocity**, whose value depends on the size, shape and weight of the object.

Terminal velocity is the constant velocity attained by a falling object and it results from the balancing of gravitational force (weight)

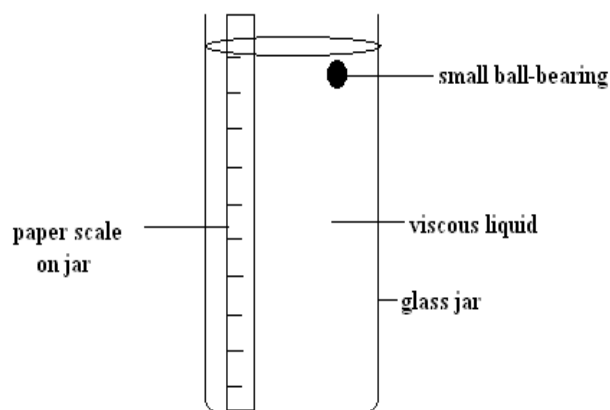
and air resistance (fluid friction).

A small dense object, e.g. steel ball-bearing, has a high terminal velocity and falls a considerable distance with a constant acceleration of 9.8m/s^2 before air resistance equals its weight.

A light object, e.g. a rain drop, or one with a large surface area e.g. a parachute, has a low terminal velocity and only accelerates over a comparatively short distance before air resistance equals its weight.

OBJECT FALLING IN A LIQUID

The time of fall of small ball-bearings dropped into a viscous liquid such as glycerine, engine oil e.t.c. can be studied using the apparatus below:



The glass jar should be as wide and as tall as possible. To reduce the risk of air bubbles sticking to the ball, it should be dipped in some of the liquid and thereby coated with it, before dropping. The investigations that can be carried out include:

- Measurement of the speed of a ball over a distance at different depths
- The effect of using balls of different diameters
- A comparison of the viscosities of different liquids.

At the start the weight W of the ball causing it to fall vertically downwards under gravity is greater than the upwards viscous force opposing its motion. As the ball speeds up, the upward force (fluid friction) increases and eventually equals weight W of the ball.

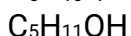
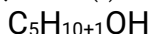
The forces acting on the ball then balance and it moves with constant speed (terminal speed).

TOPIC: ORGANIC CHEMISTRY I

ALKANOLS (ALCOHOLS)

The alcohols form another homologous series with the general formula $C_nH_{2n+1}OH$. The letter **n** in the general formula stands for number of carbon atoms. All the alcohols possess an **-OH** as the **functional group**. The functional group is the group of atoms responsible for the characteristic reactions of a compound.

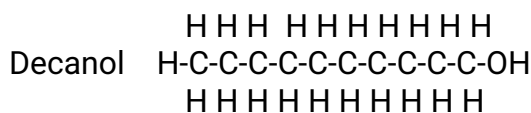
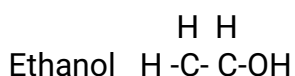
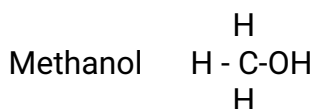
An alkanol with 5 Carbon atoms has this molecular formula; $C_5H_{2(5)+1}OH$



COMMON NOMENCLATURE

NO ₂ of Carbon atoms	Formula	Prefix	Suffix	Name
1	CH ₂ OH	Meth	-anol	Methanol
2	C ₂ H ₅ OH	Eth	-anol	Ethanol
3	C ₃ H ₇ OH	Prop	-anol	Propanol
4	C ₄ H ₉ OH	But	-anol	Butanol
5	C ₅ H ₁₁ OH	Pent	-anol	Pentanol
6	C ₆ H ₁₃ OH	Hex	-anol	Hexanol
7	C ₇ H ₁₅ OH	Hept	-anol	Heptanol
8	C ₈ H ₁₇ OH	Oct	-anol	Octanol
9	C ₉ H ₁₉ OH	Non	-anol	Nonanol
10	C ₁₀ H ₂₁ OH	Dec	-anol	Decanol

STRUCTURAL FORMULA



PHYSICAL PROPERTIES OF ALKANOLS

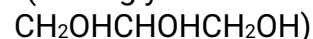
The sizes and masses of the molecules steadily increase passing along the series. This means the intermolecular forces (IMF) also increase passing along the series. Intermolecular forces decide the physical properties and the greater IMF causes alkanols to have

(i) higher boiling points and melting points

(ii) to be liquids at room temperature (i.e. methanol to pentanol)

(iii) to be more viscous (sticky) liquids when

more than one -OH group is present in a molecule (i.e. in glycerol



(iv) to be soluble in water for the small molecules of series e.g. methanol. Larger

alkanols are partially soluble in water and

others are completely insoluble.

(v) Alkanols do not conduct electricity.

The first 3 members of the alcohol family:

Alcohol	Condensed Formula	Melting point/°C	Boiling point/°C
Methanol	CH ₃ OH	-94	64
Ethanol	CH ₃ CH ₂ OH	-117	78
Propanol	CH ₃ CH ₂ CH ₂ OH	-126	97

CHEMICAL PROPERTIES

1. Combustion reaction: Alkanols burn in air and react with oxygen to form carbon dioxide and water. For example:



Along the alkanol series molecules get bigger but the relative size of the -OH group gets smaller. The relative size of -OH group

in a molecule has an effect on the flame colour.

Alkanol	Flame colour
Ethanol	blue/yellow
Butanol	yellow/little blue
Octanol	all yellow

Explanation: The colour of the flame depends upon the relative amount of oxygen in the molecule. The proportion of oxygen in a molecule decreases as alkanol molecules get larger. This means larger alkanols produce less blue (more yellow) flame and smaller alkanols produce more blue flame.

2. Reaction with sodium (Na) metal

Alkanols react with sodium metal to produce hydrogen gas and a basic solution. For example:

$$2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$$

Smaller alkanols give faster reactions with sodium metal and hydrogen bubbles rise faster in the test tube

TESTING FOR ALKANOLS

1. SOLUBILITY TEST

Alkanols dissolve in water. Adding water to an alkanol will give one layer. Only small alkanols will dissolve. This is case because as the molecule increases in size the relative size of the -OH group gets smaller as compared to the carbonyl chain. The -OH is responsible for the solubility of alkanols in water.

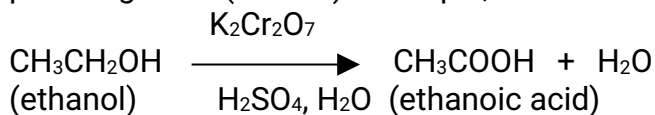
2. SODIUM TEST

When a small piece of freshly cut sodium metal is put in a test tube with an alcohol (alkanol) bubbles of a gas are formed that give a 'pop' sound when ignited (brought near a flame). The gas is hydrogen. The liquid formed turns phenolphthalein solution into a pink colour or changes litmus paper to blue colour. The liquid is a base.

OXIDATION OF ALKANOLS TO CARBOXYLIC ACIDS

Primary alcohols may be oxidised to carboxylic acids by powerful oxidising agents, such as warm acidified potassium

dichromate (VI) $\text{K}_2\text{Cr}_2\text{O}_7$ and potassium permanganate (KMnO_4). Example;



During the reaction the orange colour of potassium dichromate (VI) changes to dark green as the ethanol is oxidised to ethanoic acid.

A similar oxidation process takes place, but more slowly, if wine or beer is left open. It will eventually turn to 'vinegar' (ethanoic acid) due to bacterial oxidation of the ethanol in the alcoholic drink.

PREPARATION OF ETHANOL

Ethanol is by far the most important of the alcohols and is often just called "alcohol". Ethanol can be produced by **fermentation** as well as by the **hydration** of ethane. Ethanol is neutral, colourless, volatile liquid which does not conduct electricity. Ethanol burns quite readily with a clean and hot flame.

Some **uses** of ethanol:

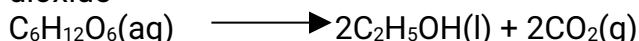
- methyated spirit is ethanol with small amount of poisonous substances added to stop people from drinking it.
- some food flavourings are made from ethanol.
- ethanol is a very good solvent and used extensively as a solvent for paints, glue, aftershave and many other everyday products.
- ethanol is also used as a source of energy i.e. for heating, cooking and lighting.
- It is also used as a raw material for making other substances, such as synthetic rubbers and flavourings
- Ethanol is well known for another reason too – it makes people drunk. Beer, wine and all alcoholic drinks contain some ethanol.

(a) FERMENTATION

Biotechnology involves making use of micro-organisms or their components, such as

enzymes, for the benefit of humans to produce, for example, foods such as yoghurt and bread. One of the oldest biotechnologies is that of fermentation. It involves a series of biochemical reactions brought about by micro-organisms or enzymes. Fermentation is the basic process behind the baking, wine and beer making industries. Fermentation in the laboratory can be carried out using sugar solution. A micro-organism called yeast is added to the solution. The yeast uses sugar for energy during anaerobic respiration (respiration without oxygen), and so the sugar is broken down to give carbon dioxide and ethanol. The best temperature for this process to be carried out is at 37°C.

Glucose $\xrightarrow{\text{yeast}}$ ethanol + carbon dioxide



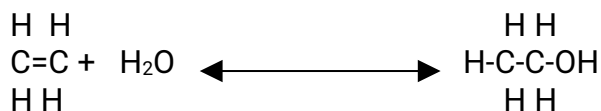
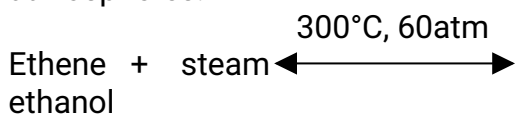
Alcoholic drinks such as beer and wine are made on a large scale in vast quantities. Beer is made from barley whereas wine is made by fermenting grape juice, which contains glucose. The micro-organism in yeast will carry out fermentation quite successfully in both cases. Beer normally contains only about 4% by volume of ethanol, whereas wine contains about 11%.

Definition:

Fermentation is a process in which sugars (i.e. glucose) are broken down (decomposed) in presence of enzymes (yeast) to form ethanol and carbon dioxide.

(b) HYDRATION OF ETHENE

One other method of producing ethanol is by addition reaction of ethane. Ethanol is produced when water (as steam) is added across the double bond in ethene. For this reaction to take place, the reactants have to be passed over a catalyst of phosphoric (V) acid (absorbed on silica pellets) at a temperature of 300°C and a pressure of 60 atmospheres.



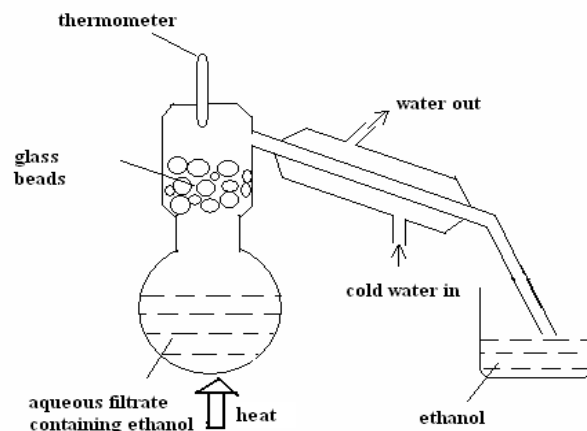
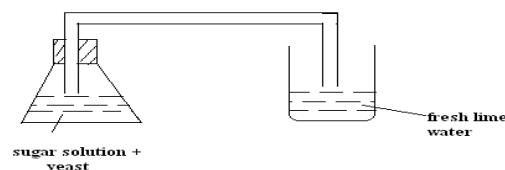
The reaction is reversible and the conditions have been chosen to ensure the highest possible yield of ethanol.

PRACTICAL WORK

Fermentation

Procedure:

1. Add about a teaspoonful of sugar to water in a flask with a little yeast as shown in a diagram below and allow any gas produced to bubble out through fresh limewater.



2. After a few days, when the bubbles have stopped, filter the solution to get rid of much of yeast. Put the filtrate in the distillation flask together with a few pot chips to help smooth boiling.

Results:

The lime water in the initial fermentation step will turn milky showing that the process of fermentation produces carbon dioxide.

From the second step, a small amount of alcohol (ethanol) will be collected after the

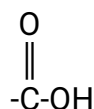
distillation process.

CARBOXYLIC ACIDS/ALKANOIC ACIDS

Carboxylic acids are oxycarbons just like alkanols.

Functional group:

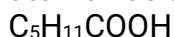
The functional group of the carboxylic acid is **-COOH, the carboxylate group**. This group decides the chemical properties and this is its structure:



The general formula:

Carboxylic acids have a general formula of $\text{C}_n\text{H}_{2n+1}\text{COOH}$ where **n** is the number of carbon atoms.

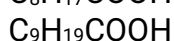
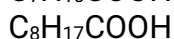
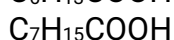
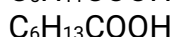
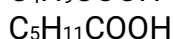
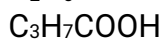
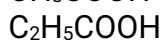
For example, a carboxylic acid with 6 carbon atoms has the formula:



Note: The sixth carbon is the one on the functional group.

Common nomenclature

Molecular Formula



Name

Methanoic acid

Ethanoic acid

Propanoic acid

Butanoic acid

Pentanoic acid

Hexanoic acid

Heptanoic acid

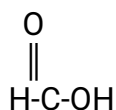
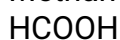
Octanoic acid

Nonanoic acid

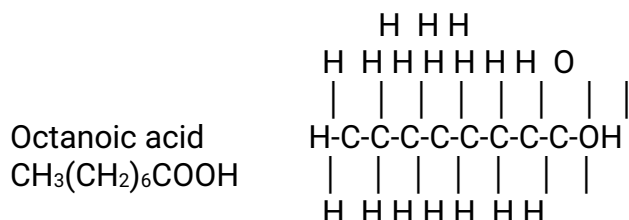
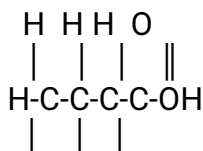
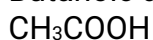
Decanoic acid

Names, Condensed formulae and Structural formulae of some carboxylic acids are given below;

Methanoic acid



Butanoic acid



Octanoic acid
 $\text{CH}_3(\text{CH}_2)_6\text{COOH}$

Physical properties of carboxylic acids

1. Along the homologous series, the sizes and masses of the molecules increase. In turn this causes the intermolecular forces (IMF) between carboxylic acid molecules to increase also. As expected there are general increases in melting and boiling points along the series.

Carboxylic acid	Condensed formula	Melting point/ $^{\circ}\text{C}$	Boiling point/ $^{\circ}\text{C}$
Methanoic acid	HCOOH	9	101
Ethanoic acid	CH_3COOH	17	118
Propanoic acid	$\text{CH}_3\text{CH}_2\text{COOH}$	-21	141

The IMF of the carboxylic acids are stronger compared to those of the hydrocarbons and alkanols. This is shown in the higher melting and boiling points of carboxylic acids.

2. Small molecule carboxylic acids are soluble in water. Large molecules are less soluble because of the size factor, as explained for alkanols. Both carboxylic acids and water contain -OH groups, and therefore acids dissolve in water. "Like dissolves like."
3. Carboxylic acids can show high viscosity (stickiness) as liquids. Both viscosity and density increase on going along the series.

Chemical properties

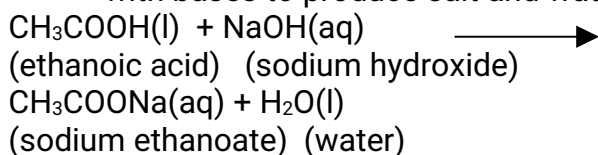
1. **Carboxylic acids change colours of indicators.** Indicators are often natural dyes. Local flowers like hibiscus or jacaranda flowers are examples. Indicators show different colours when placed in liquids of different pH. Often the same indicator will show one colour in acid solution and

another colour in a basic solution. Some common indicators used in the laboratory are shown in the table below together with their colours in acidic and basic solution.

Name of indicator	Colour in acid solution	Colour in basic solution
Blue litmus	Red	Blue
Methyl orange	Pink	Yellow
Methyl red	Red	Yellow
Phenolphthalein	Colourless	Pink
Red litmus	Red	Blue
Universal indicator	Red	Purple/blue

2. Neutralisation reaction

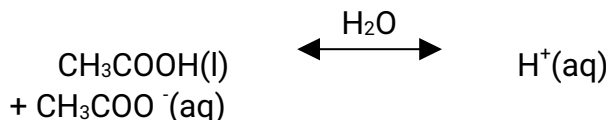
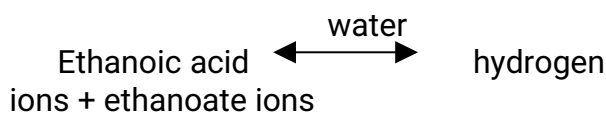
Carboxylic acids take part in neutralisation reaction by reacting with bases to produce salt and water.



The salt sodium ethanoate is formed when ethanoic acid reacts with dilute sodium hydroxide.

3. Ionisation in water

carboxylic acids are weak acids and therefore releases few hydrogen ions (H^+) when they dissolve in water compared with a strong acid of the same concentration.



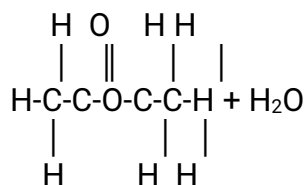
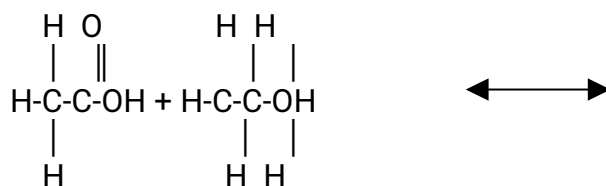
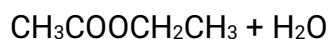
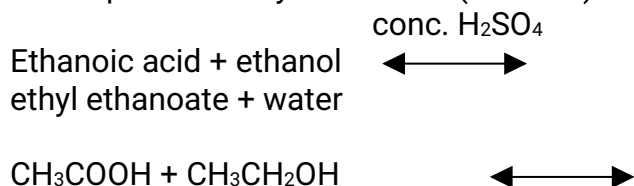
The $\rightleftharpoons$ sign means that the reaction is reversible. This means that if the ethanoic acid molecule breaks down to give hydrogen ions and ethanoate ions then they will react together to re-form the ethanoic acid molecule.

4. Carboxylic acids will also react with

metals such as magnesium.

5. Esterification

Carboxylic acids react with alkanols to form esters and the reaction is called **esterification**. For example, ethanoic acid will react with ethanol, in the presence of a few drops of concentrated sulphuric acid, to produce ethyl ethanoate (an ester).



Esterification is an example of **condensation reaction**.

Members of the 'ester' family have strong and pleasant smells. Many of them occur naturally and are responsible for the flavours in fruits and the smells of flowers.

USES OF ESTERS

1. They are used in some **food flavourings** and in **perfumes**.
2. **Fats** and **oils** are naturally occurring esters which are used as energy storage compounds by plants and animals (i.e. olive oil is obtained from a plant).
3. An ester known as amyl acetate is used as the **solvent** for removing nail varnish.

Some examples of natural sources of carboxylic acids and their names

<u>Name of acid</u>	<u>source</u>
Formic acid	the 'sting' of ants
Ethanoic acid	vinegar
Citric acid	in citrus fruits like lemons and oranges
Tartaric acid	used in cooking
Malic acid	in apples
Lactic acid	in sour milk
Ascorbic acid (vitamin C)	from foods

TESTING FOR CARBOXYLIC ACIDS

1. Solubility test: carboxylic acids dissolve in water and therefore they will give clear and one layer solution when mixed with.
2. Acid test: it is based on the ability of carboxylic acids to change the colour of an indicator during neutralisation reaction.

Procedure:

To a solution of about 15 drops of sodium hydroxide (NaOH) solution is added 2 drops of phenolphthalein indicator. This gives a pink colour (phenolphthalein in basic solution). To this is added 6 drops of the test liquid.

Results:

No change in colour means that the added liquid is not an acid. A change to colourless shows the test liquid contains the carboxylate group, -COOH and therefore an acid.

Work:

1. Write the molecular, condensed and structural formulae for Propanoic acid.
2. Write word and balanced chemical equations for the esterification of Propanoic acid with ethanol. Give the name of the ester formed.

FLOW DIAGRAMS

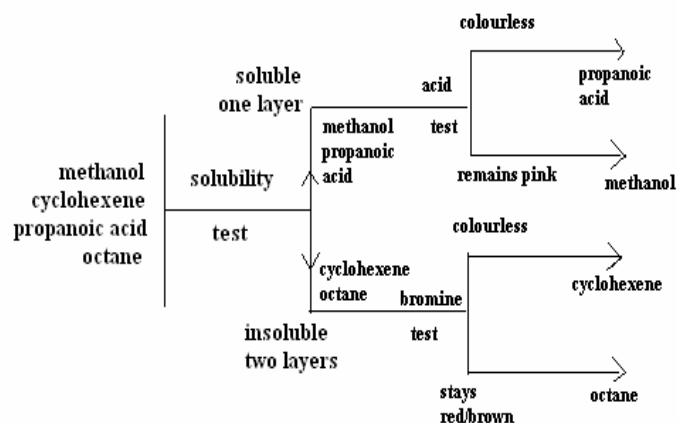
A flow diagram is a kind of a 'map' a chemist

needs to follow in order to analyse unknown organic compounds. A flow diagram helps to carry out an analysis of unknown organic substances by first separating them and then identifying each one. The identification uses the special functional group tests.

Example:

Construct a flow diagram that can be used to identify the following organic substances; methanol, cyclohexene, propanoic acid and octane. Include relevant tests and expected results.

Solution:



Note: The solubility, acid and bromine tests in the flow diagram should be briefly explained as illustrated in the notes. This is recommended when writing exams.

Work:

Prepare a flow diagram that can be used to separate and identify the following organic compounds;

- (a) Propanol, cyclobutane, decene and Methanoic acid
- (b) Hexane, hexene, ethanol and ethanoic acid
- (c) Hexane, Propanol, pentane and oxalic acid.

TOPIC: ORGANIC CHEMISTRY II

ISOMERISM

Isomerism takes place when two or more

molecules with the same molecular have different structural formulae. Each molecule having a different structure, but the same molecular formula, is called an isomer.

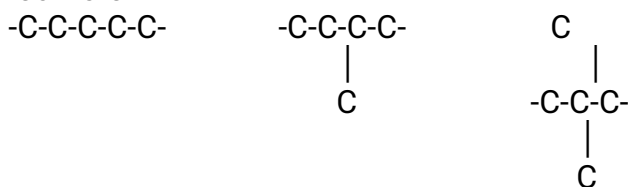
Definition:

Isomers are molecules having the same molecular formula but different structural formulae.

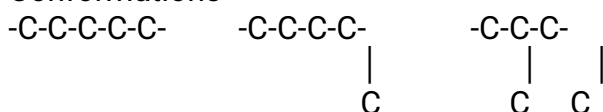
Molecules that have the same molecular formula and the same structural formula are called **conformations**.

Example: Write three isomers and three conformations of pentane (C_5H_{12}).

Isomers



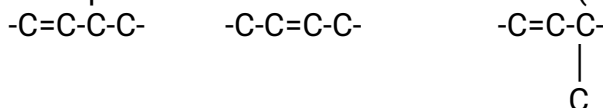
Conformations



Note: A bend is structurally the same as straight chain of carbon atoms for the same molecule.

ISOMERS OF ALKENES

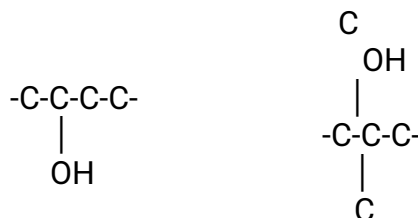
Example: Write three isomers of butene (C_4H_8)



Note: In the first and second molecules the positions of double bonds are different thereby making them isomers.

ISOMERS OF ALKANOLS

Example: Write four isomers of butanol.



Note: The different positions of the functional group, -OH makes two molecules to be isomers. Look at the first and third molecules.

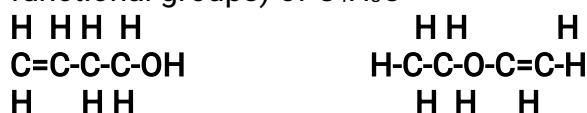
ISOMERS WITH DIFFERENT FUNCTIONAL GROUPS

Example: Write two isomers with a molecular formula C_4H_8



But-1-ene cyclobutane

Example: Two isomers (with different functional groups) of C_4H_8O



3-buten-1-ol

ethoxyethene

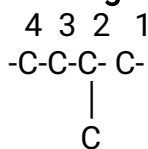
Nomenclature

Rules in naming isomers

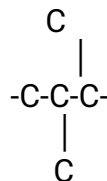
1. Write down the carbon skeleton of the molecule together with the functional group.
2. Number the longest chain. The numbering starts at the end of the chain nearest to the first junction in the 'parent chain' (if it is an alkane) and if it is any other organic compound, the numbering should start with the carbon nearest the functional group.
3. Count up all the carbon atoms in the parent chain and use the number to give a basic name to the compound.
4. State the number of junctions using prefix numbers and number the

carbon atoms where the junctions are placed.

Naming of alkanes

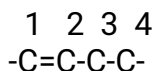


2-methylbutane
dimethylpropane

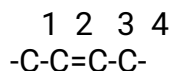


2,2-

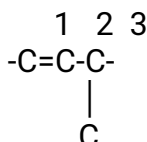
Naming of alkenes



But-1-ene
ene



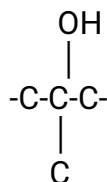
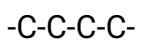
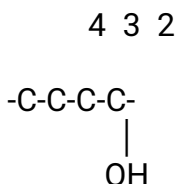
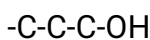
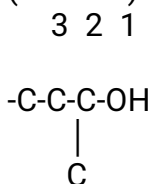
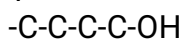
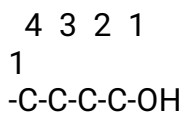
but-2-ene



2-methylprop-1-

Naming of alkanols

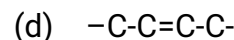
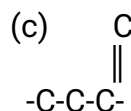
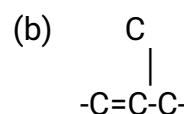
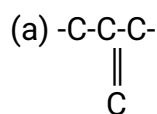
Isomers of butanol (C₄H₉OH)



The names are as follows with respect to the above order: butan-1-ol or 1-butanol or n-butanol, 2-methylpropan-1-ol, butan-2-ol 2-methylpropan-2-ol

Work:

1. Draw the structural formulae of three isomers whose molecular formula is C₅H₁₀. Give the names of the isomers.
2. Draw all isomers of pentane and give their names.
3. Study the organic compounds below and answer the questions that follow;



(i) Write down the molecular formula for each

compound.

(ii) Which molecular are conformations with

(a)?

(iii) Which molecules are isomers with (a)?

POLYMERISATION

Small organic compounds of alkenes such as ethene can actually react with itself to form very long chain molecules. When small molecules like ethene join together to form long chains of atoms called **polymers**, the process is called **polymerisation**. The small molecules, like ethene, which join together in this way, are called **monomers**. A polymer chain often consists of many thousands of monomer units and in any piece of plastic there will be many millions of polymer chains. Examples of polymers include plastics, proteins, DNA, nylon, terylene, carbohydrates e.t.c.

Definitions:

Monomers are small organic units that join/combine/add to form long chain molecules.

Polymers are long chain molecules formed when monomers join together.

Polymerisation is the process in which monomers join together to form polymers.

Polymers can be grouped as **natural** and **synthetic (artificial)**.

Natural polymers occur naturally and synthetic polymers are made by man and do not occur naturally.

Examples of natural include proteins, DNA and carbohydrates.

Examples of synthetic polymers include

polythene, terylene and nylon.

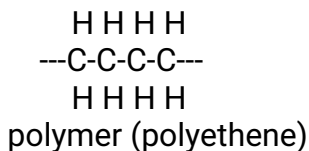
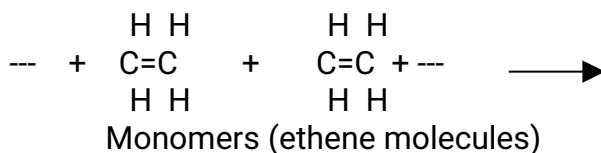
TYPES OF POLYMERISATION

1. ADDITION POLYMERISATION

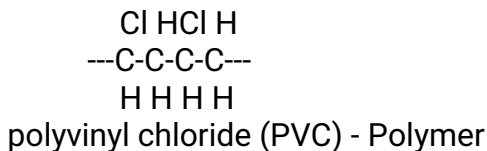
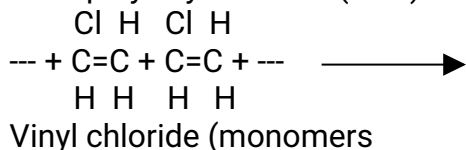
Polythene is a very common material. Its correct name is polyethene because the monomer from which it is made is ethene. The ethene molecules are able to form chains like polyethene because they possess carbon-carbon double bonds. In this process of polymerisation, ethene molecules undergo relatively high temperature and a high pressure in the presence of a catalyst. Since in this polymerisation process the monomer units add together to form only one product, the polymer, the process is called **addition polymerisation**.

Examples:

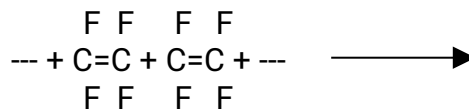
(a) polymerisation of ethene to form polyethene



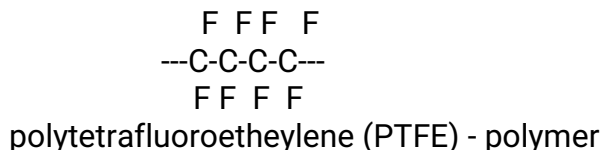
(b) polymerisation of vinyl chloride to form polyvinyl chloride (PVC)



(c) polymerisation of tetrafluoroethene to form polytetrafluoroethylene (PTFE).



Tetrafluoroethene (monomers)

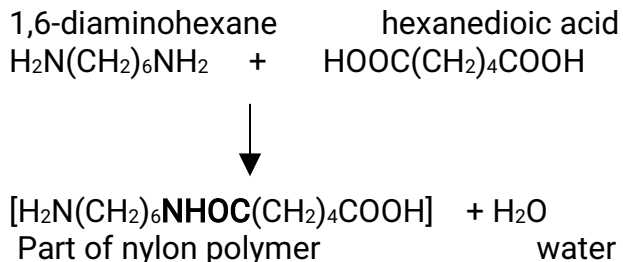


2. CONDENSATION POLYMERISATION

This is a type of polymerisation in which a small molecule (frequently water) is split off as each monomer unit is attached to the growing polymer. The term condensation comes in because two molecules combine to form a larger one while liberating a small molecule of water. The polymer chain is made up from the two starting molecules arranged alternately as these molecules react and therefore link up. Each time a reaction takes place a molecule of water is lost, and because of this it is called **condensation polymerisation**.

Examples:

(a) Nylon is made by reacting two different chemicals i.e. 1,6-diaminohexane and hexanedioic acid. Nylon is also known as **polyamide**.

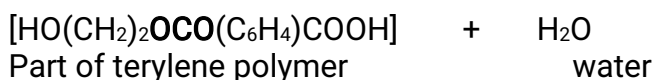


-NHOC- is referred to as the amide link.

Nylon fibres are used to make shirts, ties, sheets and parachutes or turned into ropes or racket strings for tennis.

- (b) Terylene is also produced through condensation polymerisation reactions between ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

ethane-1,2-diol benzene-1,4-dicarboxylic acid.



-OCO- is known as an ester link and because an ester link is formed during the polymerisation, terylene is known as **polyester**.

Terylene is also used to make fibres which are then woven into fabrics (clothes), ropes, bristle of brushes, tights.

PROPERTIES AND USES OF SOME POLYMERS

1. Plastics

Polythene

- it is easily moulded
- it is an excellent electrical insulator
- it does not corrode
- it is tough
- it is not affected by weather
- it is durable

It is therefore used as carrier bags, bowls, buckets e.t.c.

- Polypropylene- harder and tougher than polythene. It is used as water pipes, ropes, packaging e.t.c.
- PVC – slightly stronger and harder than polythene. Polyvinyl chloride (PVC) is used in electrical insulation and pipes for plumbing.

- PTFE - non –stick surface (slippery), withstands high temperature and resistant to chemical attack. Polytetrafluoroethylene is used as a ‘non-stick’ coating for frying pans and saucepans.
- Polystyrene – light, poor conductor of heat.
- Perspex – transparent.
- Melamine – used to make ‘unbreakable’ dishes and mugs, ashtrays.

2. Proteins

- Made from amino acids
- Insoluble in water
- Easily affected by acids, alkalis and temperature increase, when they are said to be denatured.

Uses: They control chemical processes that keep our bodies going.

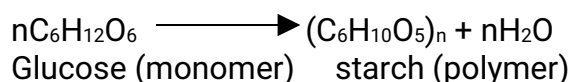
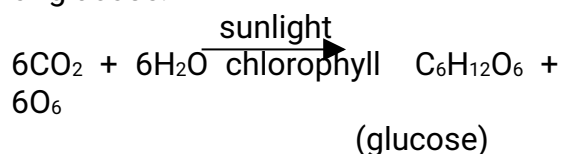
3. DNA (Deoxyribonucleic acid) – they are biopolymers

Uses:

- Controls the protein synthesis within body cells.
- DNA “fingerprinting” is a very powerful tool in the investigation of crime.

4. Carbohydrates

Starch is a biopolymer or natural polymer. It is a condensation polymer of glucose.



Both starch and glucose are carbohydrates, naturally occurring organic compounds. Starch occurs in potatoes, rice and wheat.

Properties of starch

- Does not form a true solution

- Does not have a sweet taste
- With iodine it gives an intense blue colour (nearly black)

Use: Often produced as a way of storing energy.

THERMOPLASTIC AND THERMOSETTING POLYMERS

Plastics can be put into one of two categories, thermoplastics (thermosoftening plastics) or thermosetting plastics (thermosets).

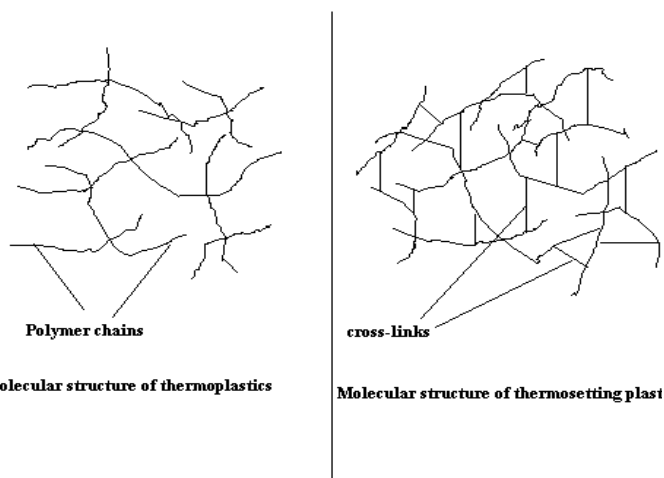
Thermoplastics like polythene, PVC and polystyrene melt or soften when heated.

Thermosetting plastics do not melt or soften on heating but only char and decompose on further heating. To char simply means to partially burn so as to blacken the surface. Examples of thermosets include melamine (for table ware) and phenolic (for formica and saucepan handles).

Thermoplastics are easily moulded or formed into useful articles. Once they are molten they can be injected or blown into moulds, a variety of different-shaped items can be produced.

Thermosetting plastics can be heated and moulded only once, usually by compression moulding.

The diagram below shows the molecular structures of thermoplastics and thermosets.



Thermosoftening (thermoplastics) do not have polymer chains that give a cross-linked structure, so when they are subjected to heat

their polymer chains flow over one another and the plastic softens and later melts.

Thermosetting plastics (thermosets) have polymer chains which are linked or bonded to each other to give a cross-linked structure, and so the chains are held firmly in place and no softening takes place on heating.

ARE PLASTICS GOOD OR BAD?

Plastics are widely used in different areas; schools, homes, industries e.t.c. Much as these plastics are widely used they have their own disadvantages.

Good points (advantages)	Bad points (disadvantages)
1. Quite cheap, and easily made	1. Difficult to dispose of. Plastic bags and cartons do not rot when they are thrown away, so they pollute the countryside, but biodegradable plastics rot away.
2. Lighter than wood, stone or metal	2. Some plastics catch fire very easily
3. Unreactive. They do not corrode in air or water. Many are not affected by acids or alkalis	3. When they burn, they often produce harmful gases. For example, PVC gives off fumes of hydrogen chloride when it burns. This would form hydrochloric acid in your eyes and throat.
4. Do not conduct heat or electricity, so can be used as insulators	4. They usually do not look as good as wood or stone.
5. Can be moulded into any shape	
6. Can be made very strong	

7. Can be coloured, by adding pigments	
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DISPOSAL OF PLASTICS

Plastics have replaced materials for metals, glass, paper and wood as well as for natural fibres such as cotton and wool. This is because of the numerous advantages discussed in the table above. However, this situation has contributed significantly to the household waste problem.

Ways of disposing plastics

1. Land-filling disused quarries (although it is becoming harder to find such sites).
2. Incineration – heating used plastics and the heat generated can be used for heating purposes.
3. Recycling – large quantities of plastic bags and sheeting are reproduced or reprocessed for resale.
4. Biodegradable plastics and photodegradable plastics have been developed. Biodegradable plastics are plastics which are broken down by bacteria (they decompose) as well as plastics which dissolve in water. Photodegradable plastics degrade (decompose) in sunlight. Therefore, the use of biodegradable and photodegradable plastics can reduce waste problem of plastics.

TOPIC: ELECTRICITY, MAGNETISM AND ELECTROMAGNETIC INDUCTION I

ELECTROSTATICS – CHARGING BY INDUCTION

They are two types of electricity and these are static and current electricity. Static electricity is often formed on insulators e.g. polythene, cellulose acetate and rubber. Static electricity, also known as electrostatics, has so many uses in our everyday life.

Activity:

Materials: balloons, woollen cloth or cotton wool, thread or rubber bands and a smooth

wall.

Procedure:

- Inflate two balloons to a reasonable size and tie the mouth with a rubber band or thread.
- Rub the surface of one inflated balloon using the woollen cloth or cotton wool.
- Bring the rubbed and unrubbed surfaces of the balloons without touching them close to the wall and leave them.
- Record the results in a table.

Material	Results
1. Unrubbed balloon against the wall	
2. Rubbed balloon against the wall	

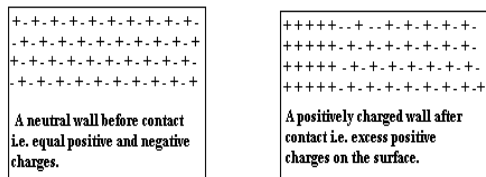
Food for thought: Discuss why the rubbed balloon remained close or attached to the wall for a long time.

WAYS OF CHARGING OBJECTS

- **By friction/rubbing:** It is making objects to become charged by rubbing them. Under normal circumstances an object has equal positive and negative charges i.e. it is neutral. If an object is rubbed by cotton wool and acquires a positive charge, it would mean that it has lost electrons upon rubbing. The rubbing material i.e. cotton wool becomes negatively charged because it has excess electrons and the object becomes positively charged because it has excess positive charges.
- **By contact:** Suppose we have a neutral object and the other one is charged by friction and are brought in contact for a short time the following might happen; if the charged object is strongly negatively charges then the neutral object will gain electrons from

the negatively charged body. The neutral body now becomes negatively charged.

- **By induction:** Suppose we have two neutral objects such as a smooth wall and a balloon. If the balloon is rubbed it might become negatively charged and when brought close to the neutral wall it gets attracted to the wall. This shows that the electrons in the wall are repelled and the wall becomes positively charged. The balloon then gets stuck to the wall.



Work:

Write a procedure you would use to identify sign of the charges of different objects given the following materials: polythene strips, cellulose acetate, a balloon, pen, and cloth/cotton wool.

USES OF ELECTROSTATICS

1. **ELECTROSTATICS PRECIPITATION OR FLUE PRECIPITATION:** In industries which produce a lot of smoke there are some charged plates in the chimney. Rising smoke particles are charged and as they pass through the charged plates in the chimney they are attracted thereby reducing environmental pollution.
2. **PHOTOCOPIERS:** These contain a charged drum and when the paper to be copied is laid on the glass plate, the light reflected from the white parts of the paper causes the charge to disappear from the corresponding parts of the drum opposite. The charge pattern remaining on the drum

corresponds to the dark-coloured printing on the original. Special toner powder is then dusted over the drum the drum and sticks to those parts which are still charged. When a sheet of paper passes over the drum, the particles of toner are attracted to it and fused into place by a short burst of heat.

3. CAPACITOR IN A CAMERA WITH A FLUSH:

If one operates a camera with a flush it must have a capacitor. Battery circuit charges the capacitor when on the flush mode. One side of the capacitor accumulates negative charges while the other plate positive charges. Pressing the shooting button it discharges the capacitor through the flush. By charging the capacitor, it means there is an accumulation of charges. Discharging means the capacitor is used and it makes the charges to flow.

4. **SPRAY PAINTING:** Body of a car is neutral and the paint is negatively charged. The paint being spread induces positive charges by repelling the electrons in the car, hence the paint is attracted to the car body.

5. INKJET PRINTERS

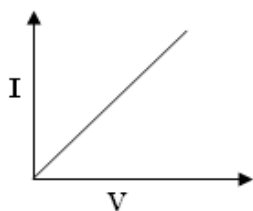
ELECTRICAL RESISTANCE

Electrons move more easily through some conductors than others when a potential difference (p.d.) is applied. The opposition of a conductor to current is called its **resistance**. A good conductor has a low resistance and a poor conductor has a high resistance.

The resistance of a wire of a certain material

- (a) increases as its length increases
- (b) increases as its cross-section area (thickness) decreases
- (c) depends on the material
- (d) increases as its temperature increases.

This means a long thin wire has more resistance than a short thick one of the same material. Silver is the best conductor but the most common used metal is copper. Copper



is cheaper and is used for connecting wire and domestic electric cables.

THE OHM

If the current through a conductor is I when the voltage across it is V , its resistance R is defined by

$$R = \frac{V}{I}$$

Resistance R is measured in ohms (Ω).

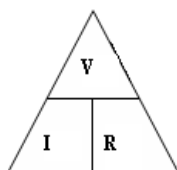
Alternatively, if R and I are known, V can be found from

$$V = IR$$

Also knowing V and R , I can be calculated from

$$I = \frac{V}{R}$$

The triangle below is an aid to remembering the three equations



Metallic conductors

Metals and some alloys give $I - V$ graphs which are a straight line through the origin, so long as their temperature is constant. Current I is directly proportional to voltage V . Doubling V doubles I . Such conductors obey

Ohm's law, stated as follows:

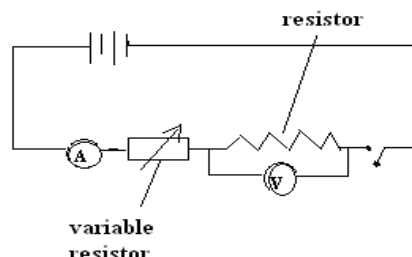
'The current through a metallic conductor is directly proportional to the voltage across its ends if the temperature and other conditions are constant.'

PRACTICAL WORK ON MEASURING RESISTANCE

The resistance R of a conductor can be found by measuring the current I through it

when a p.d. V is applied across it and then using $R = \frac{V}{I}$.

This is called the ammeter – voltmeter method. Set up the circuit as below with a resistor of an unknown resistance.



Altering the rheostat (variable resistor) changes both the p.d. (V) and the current (I). Record in a table, with three columns, five values of I (e.g. 0.10, 0.15, 0.20, 0.25 and 0.30A) and the corresponding values of V . Work out R for each pair of readings.

Current (I)	Voltage (V)	Resistance (R) $=V/I$
0.10 A		
0.15 A		
0.20 A		
0.25 A		
0.30 A		

It is observed that $I \propto V$, and it follows that $V/I =$ a constant (obtained from the slope of the $I - V$ graph).

Example:

Calculate the resistance of a conductor, if the current passing through it is 2 A and voltage across it is 12 V.

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{12V}{2A} \\ &= 6 \Omega \end{aligned}$$

Work:

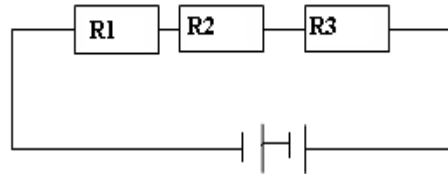
1. What is the resistance of a lamp when a voltage of 12V across it causes a current of 4A?
2. Calculate the p.d. across a 10Ω

resistor carrying a current of 2A.

3. Calculate the current flowing through a $4\ \Omega$ resistor when a potential difference of 6 V is applied across it.

RESISTOR IN SERIES

The resistors in the figure below are in series.

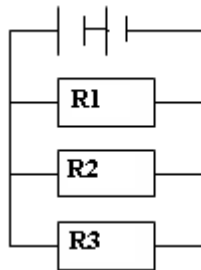


The total resistance R is equal to the sum of the resistances R_1 , R_2 , and R_3 of the separate resistors, i.e.

$$R = R_1 + R_2 + R_3$$

RESISTORS IN PARALLEL

The resistors in the figure below are in parallel.



Experiments have shown that the reciprocal of the total resistance is equal to the sum of the reciprocals of the separate resistances, i.e.

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

For the simpler case of two resistors in parallel

$$\begin{aligned} \frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2} \\ &= \frac{R_2 + R_1}{R_1 R_2} \end{aligned}$$

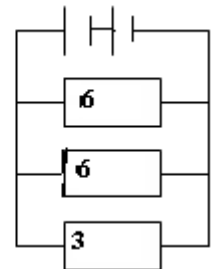
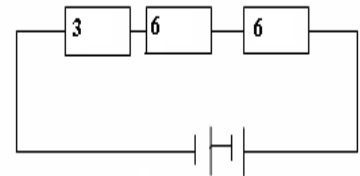
Inverting both sides,

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{\text{product of resistance}}{\text{sum of the resistances}}$$

NOTE: This formula applies only if there are **two** resistors connected in **parallel**.

Examples:

1. Calculate the effective resistance in the following circuits;



Solutions:

For the series circuit:

$$\begin{aligned} \text{Total resistance (R)} &= 3 + 6 + 6 \\ &= 15\ \Omega \end{aligned}$$

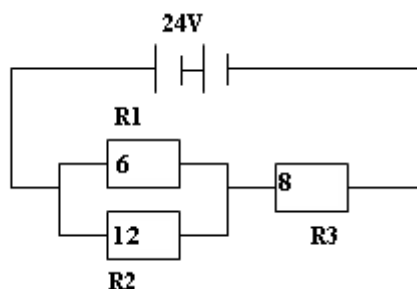
For parallel circuit;

$$\begin{aligned} \frac{1}{R} &= \frac{1}{6} + \frac{1}{6} + \frac{1}{3} \\ &= \frac{1 + 1 + 2}{6} \\ &= \frac{4}{6} \end{aligned}$$

Therefore, $R = 6/4 = 1.5\ \Omega$

From the above results it is observed that the combined or effective resistance is greater in series circuit than in parallel circuit.

2. A p.d. of 24V from a battery is applied to the network of resistors in the figure below:



$$I = \frac{V}{R}$$

$$= \frac{8V}{6\Omega}$$

$$= 4/3A$$

Work:

1. (a) The student obtains the following results (filament lamp).

Voltage (V)	0	3.0	5.0	7.0	9.0	11.0
Current (A)	0	1.0	1.4	1.7	1.9	2.1

- i. Plot a graph of current against voltage.
- ii. Use your graph to find the current when the voltage is 10V.
- iii. Use your answer in (ii) to calculate the resistance of the lamp when the voltage is 10V.

- (b) i. What happens to the resistance of the lamp as the current through it increases?

- ii. Explain your answer.

2. The 6Ω and 3Ω resistors are connected in parallel and an 8Ω resistor is connected in series with them. A current of 2A passes through the 8Ω resistor.

- (a) Sketch the circuit diagram.
- (b) Find the combined resistance in the circuit.
- (c) Find the potential difference across the combined resistor.
- (d) Calculate the p.d. across the parallel network.
- (e) Calculate the current through the 3Ω resistor.

3. A 4Ω coil and a 2Ω coil are connected in parallel. What is their combined resistance?

A total current of 3A passes through the coils. What current flows through the 2Ω coil?

4. The diagram below shows a 2V battery connected to an arrangement of resistors, part in series and part in parallel.

- (a) What is the combined resistance of the 6 Ω and 12 Ω resistors in parallel?

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$= \frac{1}{6} + \frac{1}{12}$$

$$= \frac{2}{12} + \frac{1}{12}$$

$$= \frac{3}{12}$$

$$\therefore R = 12/3 = 4 \Omega \quad \text{Or}$$

$$R = \frac{6 \times 12}{6 + 12} = \frac{72}{18} = 4 \Omega$$

- (b) What is current in the 8Ω resistor?

$$\text{Total resistance of the circuit} = 4 + 8 = 12\Omega$$

$$\text{Total voltage} = 24V$$

Current passing through the 8Ω resistor is the same the total current in the circuit,

$$I = \frac{V}{R}$$

$$R = \frac{24V}{12\Omega}$$

$$= 2A$$

- (c) What is the voltage across the parallel network?

$$\text{Total current in the circuit} = 2A$$

$$\text{Combined resistance of parallel network} = 4 \Omega$$

$$V = I \times R$$

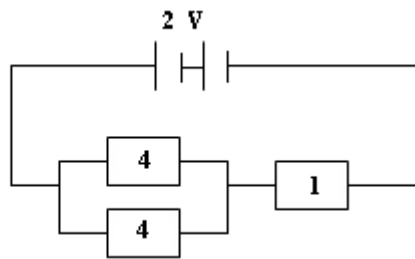
$$= 2A \times 4\Omega$$

$$= 8V$$

- (d) What is the current in the 6Ω resistor?

$$\text{Voltage across the } 6\Omega \text{ resistor} = 8V$$

$$\text{Resistance} = 6\Omega$$



- i. What is the total resistance of the two 4Ω resistors in parallel?
- ii. What is the current flowing in the 1Ω resistor?
- iii. What is the current in one of the 4Ω resistors?

RESISTANCE OF RESISTORS

COLOUR CODE FOR RESISTORS

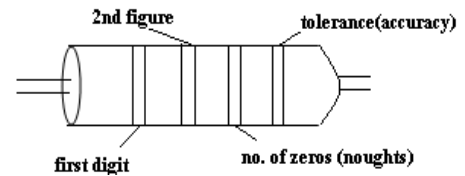
It is useful to know the value of a resistor; this is indicated by the four coloured bands on the resistor.

The first three colours give the first digit, the second digit and the number of zeros (noughts), the fourth band, if there is one, gives the tolerance (accuracy) as follows;

None	= ±20%
Silver	= ±10%
Gold	= ±5%
Red	= ±2%
Brown	= ±1%

Gave in Willingly

Black Brown Red Orange Yellow Green Blue
Violet Grey White



Examples:

1. Find the value of a resistor with the following colour bands in that order:

(a) Brown, green, brown, (**no** fourth band)

Solution:

-First digit is 1 (brown corresponds to 1 in the table)

-Second digit is 5 (green corresponds to 5 in the table)

-there is only one zero (nought) i.e. brown corresponds to 1 and the third band gives the number of zeros (noughts)

-tolerance is ±20% (there is no fourth band i.e. none)

Therefore, the resistor has a value of **150Ω ±20%**

(b) red, violet, orange, gold

Solution:

-first digit is 2

-second digit is 7

-the number of zeros is 3 i.e. 000

-the tolerance (accuracy) is ±5%

Therefore, the resistor has a value of **27000Ω ±5%**

2. Determine the colour codes of a resistor whose resistance value is **3500Ω ±10%**.

Solution:

- first colour is orange (3 corresponds to

orange in the table)

- second colour is green (5 corresponds to green)

- third colour is red (the number of

Colour	Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Grey	White
Figure	0	1	2	3	4	5	6	7	8	9

Hint for remembering the order of the colours:

Bad Boys Raped Our Young Girls But Violet

©TemboJPPCDSS

zeros

is 2 and 2 corresponds to red)

- fourth colour is silver (silver colour means an accuracy of $\pm 10\%$)

Therefore, the colour code is in this order: **orange, green, red, silver.**

THE RESISTANCE CODE

Resistance keep currents and voltages at the levels needed for other components to work properly. The resistance in ohms is marked on by using either the **colour code** (the one discussed above) or the **resistance code**. The value is only approximate. Resistance values are printed on the resistor including the tolerance (accuracy). **Tolerance of a resistor tells you by how much the resistance may differ from the marked value.**

R27 means 0.27Ω

2R7 means 2.7Ω

3K0 means $3.0\text{ k}\Omega$

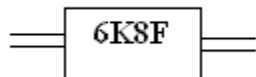
5K6 means $5.6\text{ k}\Omega$

47K means $47\text{ k}\Omega$

Note: Where there is R it means resistance will expressed in ohms (Ω) and where there is K it means resistance is calculated in kilo-ohms ($\text{k}\Omega$).

The extra letter at the end gives the tolerance (accuracy): F = 1%, G = 2%, J = 5%, K = 10% and M = 20%

Example: Work out the value of resistor below;



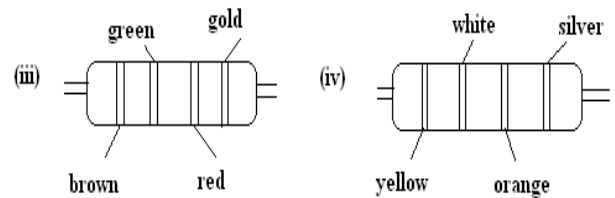
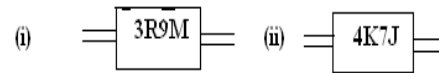
Solution:

The resistance value is **$6.8\text{ k}\Omega \pm 1\%$**

Work:

What is the resistance of each of the resistors in the figures below? What is the tolerance of each? Explain what is meant by

term 'tolerance' or 'accuracy'.



ELECTRICAL POWER

Power is defined as work done per unit time.

$$\text{Power} = \frac{\text{work done}}{\text{time taken}} = \frac{\text{energy transferred}}{\text{time taken}}$$

In symbols, $P = W/t$

where W is in joules (J) and t in seconds (s).

Then, P is in J/s or watts (W).

From the definition of potential difference p.d. (voltage), work done (W) is the electrical energy transferred when a steady current I (in amperes) passes for time t (in seconds) through a device with a p.d. V (in volts) across it.

Then, $W = ItV$

Substituting for W in the first equation we get

$$P = W/t$$

$$= ItV/t$$

$$P = IV$$

Therefore to calculate the power P of an electrical appliance we multiply the current I through it by the p.d. V across it.

Example:

If a lamp on a 240V supply has a current of 0.25A through it, calculate the power of the lamp.

Solution:

$$P = IV$$

$$P = 0.25\text{ A} \times 240\text{ V}$$

$$P = 60\text{ watts}$$

$$P = 60\text{ W}$$

Note: 1 kW = 1000 W and 1 MW = 1 000 000 W

The power equation may be written in three forms:

$$P = IV$$

$$P = I^2R$$

$$P = \frac{V^2}{R}$$

The power of normal electrical appliances is usually stated in watt or kilowatt. The potential difference p.d. which should be applied to it is also stated, because it is meaningless to state the wattage without also stating the voltage. For example, an electric light bulb for use in the house is marked 100W 240V, a car headlamp bulb is marked 48W 12V. The resistance of the bulbs may be obtained from the equation: $P = V^2/R$ or $R = V^2/P$

Example: Find the resistance of 100W 240V bulb.

$$R = V^2/P$$

$$R = \frac{240 \times 240}{100} \Omega$$

$$R = 576 \Omega$$

This is the working resistance of the light bulb, i.e. the resistance when the bulb is hot. When the bulb is cold the resistance is much less, but putting a p.d. of 12V across it will not make it light and the power produced will be very low.

It is observed that if much lower voltages are applied to electrical appliances the wattage is much less than that required, and if a voltage greater than the stipulated voltage is applied it is likely that the element or filament will **burn out**.

Electrical appliances should **not** be used with voltages higher than those marked upon them.

Example:

A 3kW electric fire is designed to be run on a 250V supply. Assuming it to be operating at the correct voltage,

(a) What current will it draw from the supply?

$$P = IV$$

$$I = P/V$$

$$P = 3kW = 3000W, V = 250V$$

$$\begin{aligned} \text{Therefore, current (I)} &= 3000W/250V \\ &= 12A \end{aligned}$$

(b) What is the resistance of the element of the fire?

$$R = V^2/P$$

$$R = \frac{250 \times 250}{3000} \Omega$$

$$R = 20.83 \Omega$$

Work:

- Find the resistance of 48W 12V car headlamp bulb.
- What current is taken by a 3kW electric fire on the 240V mains? What size fuse should be used in the plug?
- What is the maximum power in kilowatts of the appliance(s) that can be connected safely to a 13A 230V mains socket?
- A bulb is labelled 12V 36W, when used on a 12V supply, (a) what current will it take? (b) What is its resistance?
- A 3kW heater is fitted with a 35W indicator lamp and a 15W fan. These three components of the appliance are connected directly to the 250V mains and switch on and off together. When the appliance is operating, calculate
 - the total power
 - the total current
 - the energy used in 4 hours.
- A car battery sends a current of 8A through its headlamp for 1 minute. The lamp resistance is 4Ω.
 - What is the power of the headlamp?
 - How much heat

- energy is given out?
7. An electric kettle gives out 1250 joules of heat every second working on 250V mains,
- What is the power dissipated by the kettle element?
 - What current would normally flow in the element?
 - What is the resistance of the element?

ELECTRICAL ENERGY

Electrical energy is calculated by multiplying power by time.

$$\text{Electrical energy} = \text{power} \times \text{time}$$

(watt or J/s) (s)

Electrical energy that is used in homes and industries has to be paid for and that is why electricity supply companies (i.e. ESCOM) charge for the electrical energy they supply. A joule is a very small amount of energy and a large unit, the kilowatt – hour (kWh) is used. This means electricity is sold in units of kilowatt – hours (kWh). As the name suggests, **a kilowatt-hour is the electrical energy used by a 1kW appliance in 1 hour.**

$$1\text{kWh} = 1000\text{J/s} \times 3600\text{s}$$

$$= 3600000\text{J} = 3.6\text{MJ}$$

$$1\text{ kWh of electrical energy} = 1\text{ unit}$$

This means a 3KWh electric fire working for 2 hours uses 6KWh of electrical energy usually called 6 units.

Examples:

- If 1 unit (1KWh) costs 15t, work out the cost of a 5KWh oven running for 90 minutes.

Solution:

$$\text{Power} = 5\text{KW}, \text{ time} = 90/60 \text{ hours} = 1.5\text{h}$$

$$\begin{aligned} \text{Electrical energy} &= \text{power} \times \text{time} \\ &= 5\text{KW} \times 1.5\text{h} \\ &= 7.5\text{KWh} \end{aligned}$$

$$= 7.5 \text{ units}$$

$$1 \text{ unit costs } 15\text{t}$$

$$\text{Therefore, } 7.5 \text{ units will cost} = 7.5 \times 15\text{t}$$

$$= 112.5\text{t}$$

$$= \text{K}1.125$$

$$= \text{K}1.13$$

(to the nearest

tambala)

- If a 1 unit of electrical energy costs 75t, work out the cost of a 150W refrigerator working for 48 hours.

Solution:

$$\begin{aligned} \text{Electrical energy} &= \text{power} \times \text{time} \\ &= 150\text{KW} \times 48\text{h} \\ &= 7.2\text{KWh} \end{aligned}$$

$$= 7.2 \text{ units}$$

$$1 \text{ unit costs } 75\text{t}$$

$$7.2 \text{ units will cost} = 7.2 \times 75\text{t}$$

$$= 540\text{t}$$

$$= \text{K}5.40$$

Work:

- Each day in the school kitchen, the 10KW cooker runs for 6 hours, the two 1.5KW deep freezers are run for 20 hours and five 100W light bulbs run for 15 hours. If the cost of 1 unit of electrical energy is 250 tambala, then work out the cost to the school for a whole month of 31 days?
- What is the cost of heating a tank of water with a 3000W immersion heater for 80 minutes if electricity costs K2.10 per unit?
- Mr. Thomas uses a kettle to boil water. It is a 2KW kettle. It takes 6 minutes to boil the water.
 - Calculate the energy transferred in kilowatt-hours (KWh).
 - Electricity costs K2.70 per KWh. How much does it cost Mr Thomas to boil the water?

- An electric cooker has the following specification:

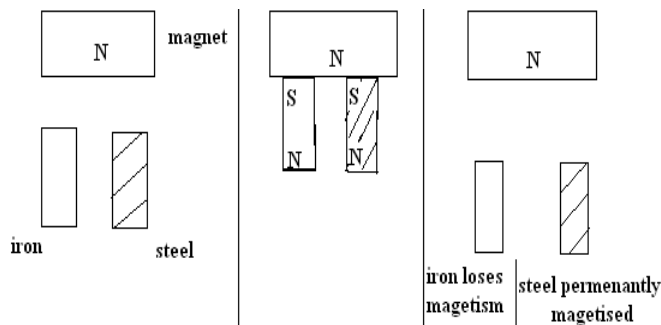
Item	No. of items	Power of
item		

Ceramic hob (small)	2	1250W
Ceramic hob (large)	2	1500W
Grill	1	2000W
Oven	1	2500W

- Calculate how much energy (in KWh) the cooker will use in 30 minutes when all the items are used simultaneously.
- How much will it cost to run the cooker during this time if electrical energy costs K2.75 for 1 unit?
- Calculate the maximum current that will be carried by the cable connecting this cooker to a 250V mains supply.

MAGNETISATION

Materials such as iron and steel are attracted to magnets because they themselves become magnetised in the presence of a magnet.



The magnet is said to induce magnetism in both metals (iron and steel) and the induced pole nearer the magnet is the opposite of the pole at the end of the magnet i.e. in the above diagram the induced pole is S-pole and is opposite to N-pole of the magnet. It is the attraction between these unlike poles that holds each piece of metal firmly to the magnet.

If the steel is pulled away from the magnet, it keeps some of the induced magnetism, and itself becomes a **permanent magnet**.

Magnetism induced in the iron is only **temporary** and is virtually all lost when the iron is pulled away from the magnet.

SOFT AND HARD MAGNETIC MATERIALS

Magnetic materials are known as

ferromagnets and they are classified as magnetically hard or soft.

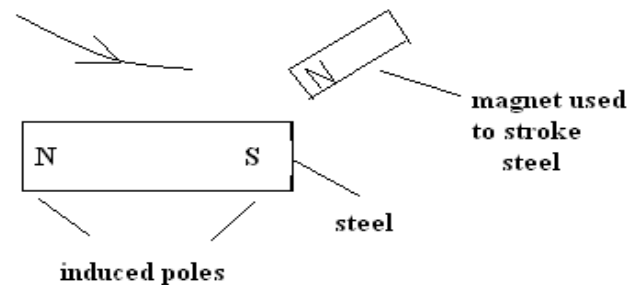
Soft magnetic materials such as iron are relatively easy to magnetise but their magnetism is only temporary.

Hard magnetic materials such as steel are the most difficult to magnetise but do not readily lose their magnetism. They are used to make permanent magnets.

MAKING MAGNETS

1. STROKING METHOD

A piece of metal (steel) becomes permanently magnetised when placed near a magnet, but its magnetism is usually very weak. The steel can be magnetised more strongly by stroking it with a bar magnet. The magnet must be moved along the steel over and over again in one direction only. The pole produced at the end of the stroke is always the opposite of the stroking pole.



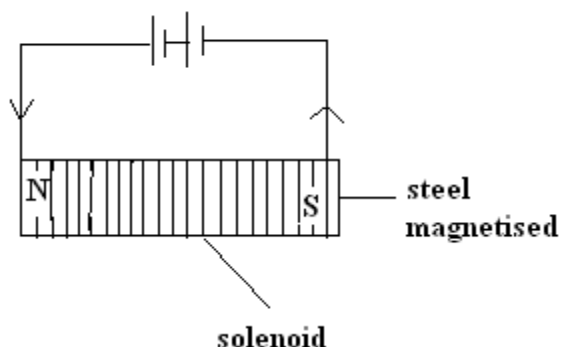
Magnetisation by stroking can be **single** or **divided**.

Single – when one magnet is used just like the case above.

Divided – when two magnets are used in the stroking process.

2. ELECTRICAL METHOD

The most effective method of magnetising a piece of steel is to place it in a long coil or solenoid made up of several hundred turns of conducting wire, and then pass a large direct current through the coil for a second or so. This produces a very much stronger magnet than the stroking method.



To identify the N and S poles, the right-hand grip rule is used.

Imagine your right-hand gripping the coil such that your fingers point the same way as the conventional current arrows. Your thumb then points towards the N-pole.

To be exact, the right-hand grip rule states that if the fingers of the right hand grip the solenoid in the direction of the current (conventional), the thumb points to the N-pole.

DEMAGNETISATION

The process of removing magnetism from a magnet is known as demagnetisation. The following are some of the ways that can be used to demagnetise a magnet.

- Heating – a magnet will become demagnetised if it is heated strongly because the vibrating molecular magnets eventually gain enough energy to overcome the forces holding them in a common direction.
- Hammering – hammering the magnet or dropping it repeatedly makes the magnet to be demagnetised.
- Alternating current method – the best method to demagnetise a magnet is to connect an already magnetised solenoid to an alternating current (a.c.) supply. Hold the magnet in the solenoid and switch on the current. With the current still switched on, pull out the rod (steel or iron) along the axis of the solenoid in an east – west direction until it is a good distance away. The magnet becomes demagnetised.
- Stopping the flow of electric current –

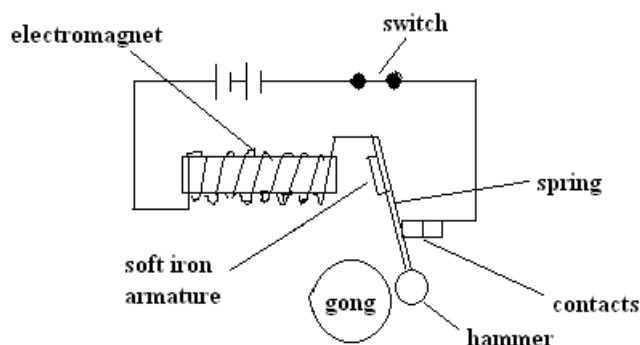
the same effect as in c) is produced when a steel rod is left in the solenoid while the current is reduced to zero. An a.c. current is essential for both methods i.e. in c) and d).

ELECTROMAGNETISM

The term 'electromagnetism' combines both the production of magnetic fields by means of electric current and the production of electric current by means of magnetic fields.

USES OF ELECTROMAGNETISM

1. Electric bell



An electric bell contains an electromagnet that switches itself on and off very rapidly, moving the bell hammer as it does so.

When the bell switch is pressed, current flows through the electromagnet and it becomes magnetised. The soft iron bar (armature) is attracted to the electromagnet. The hammer hits the gong but the circuit is now broken at the point of contact. The electromagnet loses its magnetism and no longer attracts the armature. The springy metal strip is then able to pull the armature back, remaking the contact and so completing the circuit again. This cycle is repeated so long as the bell switch is on and continuous ringing occurs.

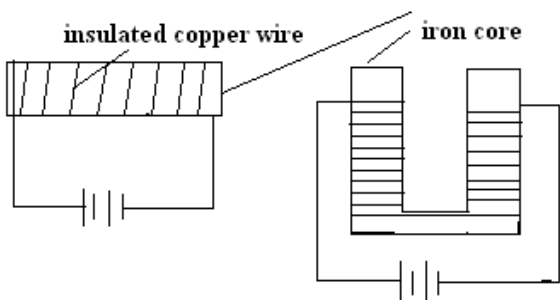
- Electromagnetism is also helpful in lifting of scrap metal.
- Electromagnetism is used in the

earpiece of a telephone.

4. Relay and circuit breaker make use of electromagnetism.
5. Microphones also apply the principle of magnetism.
6. Electromagnetism is helpful in removing steel splinters from a patient's eye.

ELECTROMAGNETS

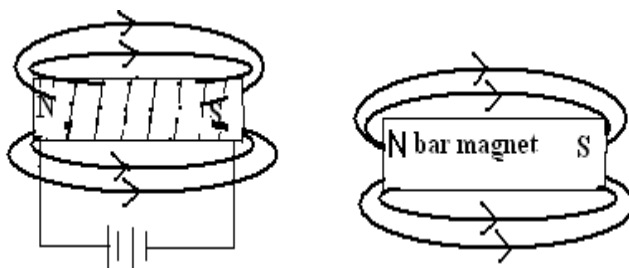
A simple electromagnet is made of a coil which consists of several hundreds turns of insulated copper wire, wound round a core made of a soft ferromagnetic material such as iron. A magnetic field is produced when a current is passed through the coil. This magnetises the iron core, which in turn produces a magnetic field about a thousand times stronger than that from the coil alone. As the core is magnetically soft, its magnetism is only temporary, and is virtually all lost when the current through the coil is switched off. Electromagnets are made in a variety of shapes and sizes. The U-shaped electromagnet produces a very concentrated magnetic field in the gap between its pole pieces.



U - shaped electromagnet

Magnetic fields of a bar magnet and a solenoid

A magnetic field of a bar magnet is invisible and spreads out a bit like a spider's web, but it can be detected and mapped out. To do this, iron filings are sprinkled onto a piece of white paper placed on top of a bar magnet. The magnetic field reaches out from the bar magnet and attracts the pieces of iron filings at a distance.



Note: The direction of the field lines is from the N - pole to the S - pole.

A copper wire coiled into a helical shape is called a **solenoid**.

Increasing the strength of an electromagnet

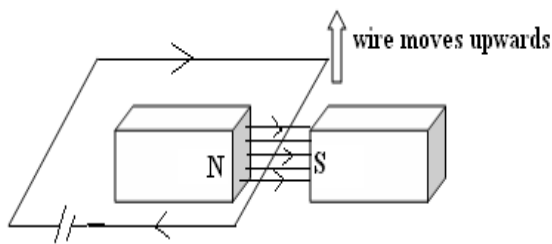
The strength of an electromagnet increases if

- a) The current in the coil increases i.e. increasing number of cells.
- b) The number of turns on the coil increases.
- c) The poles are closer together i.e. like in U - shaped electromagnet.

FORCE ON A CONDUCTOR IN A MAGNETIC FIELD

When a current is passed through a conductor, a magnetic field is produced. When a current-carrying conductor is placed in a magnetic field, a force is produced. The figure below shows an experiment to demonstrate the force produced when a current is passed through a conductor in a magnetic field. The conductor in this case is a length of stiff copper wire, and it is at right angles to the field provided by the magnet. When the switch is closed, current flows through the wire. The wire moves upwards, indicating that there is upward force acting on it. If the direction of either the current or the field is reversed, the wire moves downwards.

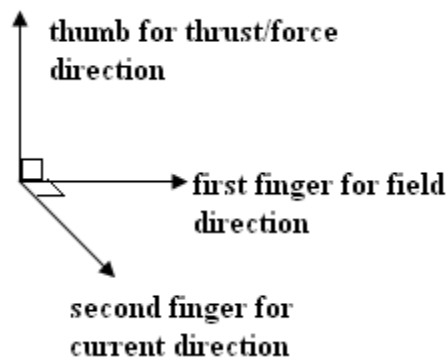
It should be noted in each case that the force on the wire acts at right angles to both the current direction and the field direction. The wire is moved across the field; it is not attracted to either pole of the magnets.



FLEMING'S LEFT-HAND RULE

Knowing the direction of the current and the field in the diagram, the direction of the force can be found using Fleming's left-hand rule:

*"If the thumb and the first two fingers of the left hand are held at right angles to one another, the **THumb** gives the direction of the **THrust** (the force) if the **First** finger points in the same direction as the **Field** and the **seCond** finger points in the same direction as the **Current**."*



In applying the rule, remember the following;

- The direction of the field is from the N-pole to the S-pole.
- The direction of the current is from the positive (+) terminal of the battery to the negative (-) i.e. conventional current direction.
- The rule applies only where the current and field directions are at right angles. A force still acts if the current and field directions are at some other angle, but its direction is more difficult to predict.

Magnitude (size) of the force on a current-carrying conductor

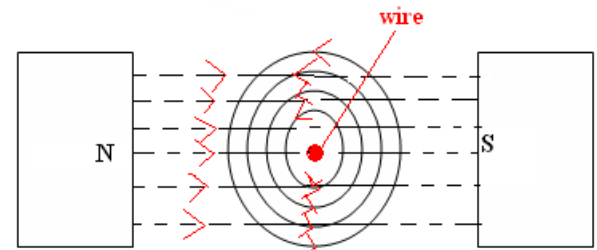
Experiments with current-carrying conductors placed at right angles to a magnetic field show that the force acting depends on:

- The strength of the magnetic field
- The size of the current
- The length of the conductor.

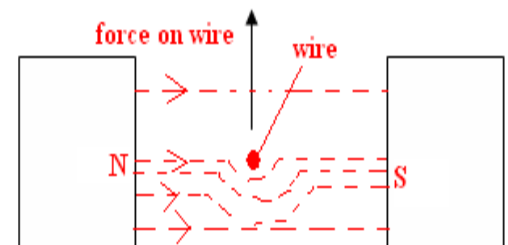
An increase in any of these three produces an increase in the magnitude (size) of the force acting.

FLUX (FIELD) PATTERNS

The figure below is a side view of the magnetic field lines due to the wire and the magnet:



The magnetic field lines due to the wire are circles and we will assume their direction is as shown. The dotted lines represent the field lines of the magnet and their direction to the right. The resultant field obtained by combining both fields is shown in the figure below:



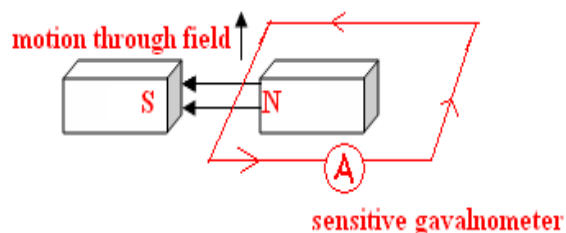
There are more lines below than above the wire since both fields act in the same direction below but in opposition above. If we suppose the lines are like stretched elastic, those below will try to straighten out and in so doing will exert an upward force on the wire.

ELECTROMAGNETIC INDUCTION

By 1821, scientists had discovered that motion could be produced by passing current through a conductor in a magnetic field. Ten years later, Michael Faraday found that the reverse was possible: a current could be produced by moving a conductor through a

magnetic field.

When a wire is moved across a magnetic field as in the next figure, a small e.m.f. is produced in the wire. The effect is called **electromagnetic induction**; an e.m.f. has been induced. If the wire forms part of a circuit as in the diagram, the induced e.m.f. causes the current to flow. The current can be detected using a sensitive galvanometer. The induced e.m.f. is present only while the wire is moving, and cutting through magnetic field lines. There is **NO** induced e.m.f. if the wire is held still, nor if it is moved parallel to the field.



The size of the induced e.m.f. can be increased by:

- Moving the wire at a higher speed
- Using a stronger magnet
- Increasing the length of the wire in the magnetic field – for example, by looping the wire through the field several times.

The above results are summed up by Faraday's law of electromagnetic induction:

"The e.m.f. induced in a conductor is directly proportional to the rate at which the conductor cuts through the magnetic field lines."

Induced e.m.f. and current in a coil

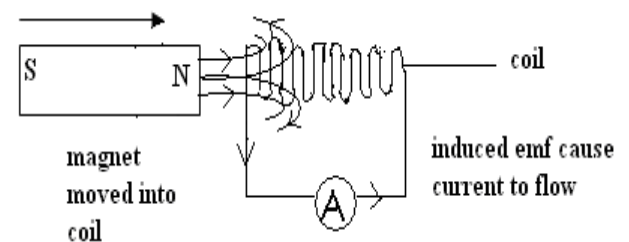
If a bar magnet is moved into a coil an e.m.f. is induced in the coil. In this case it is the magnet that is being moved, but the motion still causes field lines to be cut by the wire. The coil forms part of a complete circuit, so a current is also induced when the magnet is moved. The size of the induced e.m.f. and therefore of the induced current, can be increased by:

- Moving the magnet at a higher speed

b) Using a stronger magnet

c) Increasing the number of turns in the coil, as this increases the length of the wire cutting through the magnetic field.

All these factors will make more "cuts per second". It must be noted that **NO** e.m.f. or current is induced if the magnet is held still because no magnetic field lines are then being cut.



It should be learnt that it is the 'cutting' effect that produces the electricity. It is observed that the current is reversed when the magnet is pulled out of the coil.

Electric current produced by magnetic lines of force cutting the coil is called **an induced current**. The name of the effect is called **electromagnetic induction**.

Direction of induced current: The direction of induced current can be determined by using

a) Lenz's law:

"An induced current always flows in a direction such that it opposes the change producing it."

b) Fleming's right-hand rule:

"If the thumb and the first two fingers of the right hand are held at right angles to one another then the second finger gives the direction of the induced Current if the First finger points in the same direction as the Field and the thumb points in the direction of the Motion."

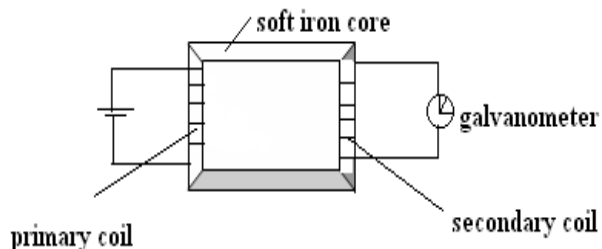
To avoid confusion on how to use Fleming's right-hand and left-hand rules there is need to know that:

- The left-hand rule applies when a current causes motion.
- The right-hand rule applies when motion causes a current.

Applications of electromagnetic induction

- a) Moving-coil microphone
- b) Magnetic recording (magnetic tapes and disks)
- c) Hydroelectric power stations (HEPs)
- d) Alternators in most cars to give a greater output than dynamos.
- e) Bicycle dynamos.

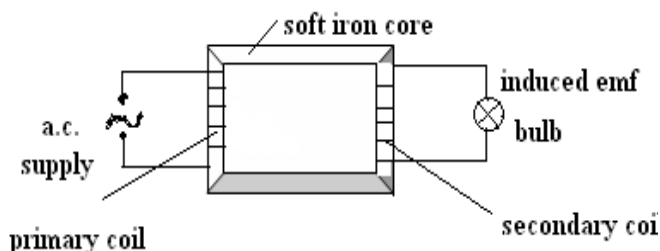
TRANSFORMERS



When the switch is closed, current flows in the primary circuit and the primary coil acts as the electromagnet and the core guides the magnetic field so that field lines pass through the secondary coil thereby inducing current.

So it is a change in the current through one coil which brings about an induced e.m.f. in the other. The effect is called **mutual induction**. Mutual induction has many applications, including the heart pacemaker, car ignition system and transformers.

When the current is steady and the magnetic field unchanging, no e.m.f. is induced in the coil. This means there is need to switch on and off the electromagnet continuously so that there is cutting and re-cutting of magnetic fields on the secondary coil thereby inducing an e.m.f. much the same effect can be produced more simply by passing alternating current (a.c.) through the primary. Unlike direct current (d.c.), a.c. is changing all the time.



The primary is connected to a low voltage a.c. supply. The secondary is connected to a small bulb. As current flows backwards and forwards through the primary, it sets up an alternating magnetic field in the core. This induces an alternating e.m.f. and current in the secondary and so the bulb lights up.

The primary and secondary coils and core together form a transformer. When a transformer is connected to an a.c. supply, electrical energy is continuously transferred from the primary to the secondary circuit.

STEP-DOWN AND STEP-UP TRANSFORMERS

By choosing suitable numbers of turns for the coils of a transformer, it is possible to make the alternating e.m.f. (a.c. voltage) in the secondary different from the voltage across primary. In other words, a transformer can be used to change or transform an a.c. voltage from one value to another. A simple equation links the a.c. voltages across primary and secondary coils and the number of turns on each.

$$\frac{\text{Secondary voltage}}{\text{Primary voltage}} = \frac{\text{No. of secondary turns}}{\text{Number of primary turns}}$$

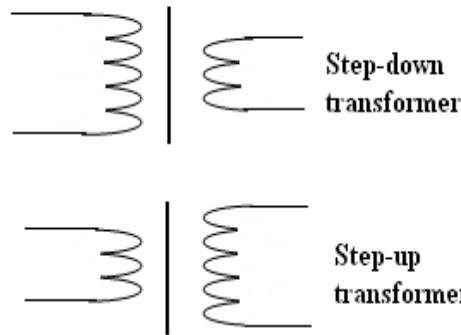
In symbols,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

A transformer is said to be a **step-down** if it has few turns in secondary than in primary, and the secondary or output voltage is less than the primary or input voltage.

A **step-up** transformer has more turns in the secondary than in the primary, and the output voltage is greater than the input voltage.

The circuit symbols for step-down and step-up transformers are given below;



If no energy is wasted in a transformer, all the power (energy per second) supplied to the primary will be delivered by the secondary. As power is calculated by multiplying voltage by current,

$$\text{Primary voltage} \times \text{primary current} = \text{secondary voltage} \times \text{secondary current}$$

In symbols: $V_P \times I_P = V_S \times I_S$

Transformers therefore transform current as well as voltage.

Another relationship is between the currents and number of turns:

$$\frac{N_S}{N_P} = \frac{I_P}{I_S}$$

N_S = number of turns in the secondary, I_P = current in primary, N_P = number of turns in primary and I_S = current in secondary.

TRANSFORMER CALCULATIONS

Example 1:

A transformer steps down the mains supply from 230V to 10V to operate an answering machine.

- a) What is the turns ratio of the transformer windings?

Primary voltage (V_P) = 230V

Secondary voltage (V_S) = 10V

$$\begin{aligned} \text{Turns ratio} &= \frac{V_S}{V_P} = \frac{N_S}{N_P} \\ \frac{N_S}{N_P} &= \frac{10V}{230V} \\ \frac{N_S}{N_P} &= \frac{1}{23} \end{aligned}$$

- b) How many turns are on the primary if the secondary has 100 turns?

$$\begin{aligned} \frac{N_S}{N_P} &= \frac{1}{23} \\ \frac{100}{N_P} &= \frac{1}{23} \\ N_P &= 23 \times 100 \\ N_P &= 2300 \text{ turns} \end{aligned}$$

- c) What is the current in the primary if the transformer is 100% efficient and the current in the answering machine is 2A?

$$\begin{aligned} V_P \times I_P &= V_S \times I_S \\ 230V \times I_P &= 10V \times 2A \\ I_P &= \frac{10V \times 2A}{230V} \\ I_P &= 0.09A \end{aligned}$$

Example 2:

The primary of the transformer is connected to an a.c. supply, and the secondary to a bulb rated at 12V 36W. There are 4000 turns on the primary and 200 on the secondary. If the bulb is delivering its rated power, find

- a) The supply voltage

$$\begin{aligned} \frac{V_S}{V_P} &= \frac{N_S}{N_P} \\ \frac{12V}{V_P} &= \frac{200}{4000} \\ V_P \times 200 &= 12V \times 4000 \\ V_P &= \frac{12V \times 4000}{200} \\ V_P &= 240V \end{aligned}$$

- b) the current through the primary

$$\begin{aligned} V_P \times I_P &= V_S \times I_S \\ 240V \times I_P &= 36W \end{aligned}$$

$$\begin{aligned} I_P &= \frac{36W}{240V} \\ &= 0.15A \end{aligned}$$

- c) the power taken from the supply, if the transformer is 100% efficient.

Power taken from the supply is the same as the power delivered to the bulb.

Therefore, Power = 36W

Or

$$\begin{aligned} P &= IV \\ P &= 0.15A \times 240V \\ P &= 36W \end{aligned}$$

WORK:

- a) The step-up transformer has an input supply of 20V and delivers an output of 250V. This transformer is 80% efficient and the secondary coil is connected to a 250V, 100W bulb. Work out the following:
- the input power in the primary circuit
 - the secondary current
 - the turns ratio
 - if there are 50 turns in the primary coil, how many turns are there in the secondary coil?
- b) A transformer has an input of voltage of 240V a.c. and a current of 10/3A. The voltage is stepped down to light an 80V bulb normally, using a current of 9A. Work out the following:
- The output power in the secondary
 - The turns ratio
 - The input power in the primary
 - The efficiency of the transformer.
- c) i. Calculate the number of turns on the secondary of a step-down transformer, which would enable a 12V bulb to be used with a 230V a.c. mains power, if there are 460 turns on the primary.
ii. What current will flow in the secondary when the primary current is 0.10A?
Assume there are no energy losses.
- d) A transformer has a primary of 6000 turns and a secondary of 150 turns. The primary is connected to an a.c. supply of 240V. A 3Ω resistor is connected across the secondary. Draw a circuit diagram to show the above arrangement. Assuming that the transformer has an efficiency of 100%, calculate
- The voltage across the resistor
 - The current through the resistor
 - The current in the primary circuit
 - The power taken from the supply
 - The power dissipated (wasted as heat) in the resistor.
- e) A 6V, 12W bulb operates from a transformer connected to a 240V a.c. supply. If there are 8000 turns on the primary, how many turns are there on the secondary? What is the current in the primary? (Assume that the transformer is 100% efficient.)

EFFECIENCY OF A TRANSFORMER

A transformer as a machine, the input power of the transformer is always greater than the output power. A measure of the losses of any machine is given by the efficiency, which is defined as:

$$\% \text{ Efficiency} = \frac{\text{output energy}}{\text{input energy}} \times 100$$

$$\% \text{ Efficiency} = \frac{\text{output power}}{\text{input power}} \times 100$$

ENERGY LOSSES IN A TRANSFORMER (CAUSES OF LOSS OF EFFICIENCY IN A TRANSFORMER)

- a) **The resistance of the windings (coils):**
As the coils have some resistance, they give off heat when a current flows through. **Solution:** Coil resistance is kept as low as possible by making the coils from thick wire. Large transformers have to be oil-cooled to prevent overheating.
- b) **Eddy currents:** The iron core is in the changing magnetic field of the primary, and currents, called **eddy currents**, are induced in it (since it is a conductor) which cause heating. **Solution:** Eddy currents are reduced by making the core from thin sheets of soft iron which are insulated from each other so that their total resistance is high. A core made in this way is said to be **laminated (layered)**.
- c) **Leakage of field lines (flux leakage):**
All the field lines produced by the

primary may not cut the secondary, especially if the core has an air gap or is badly designed. This results into loss of energy. **Solution:** For all the flux generated by the primary to pass through (cut) the secondary there is need for a good design for the transformer. This is achieved by winding the primary on top of the secondary and having a 'closed' iron core.

- d) **Hysteresis losses (magnetisation and demagnetisation):** The magnetic fielding the core is in one direction 50 times/second and in the opposite direction 50 times/second. Consequently the domains are continually changing their alignment. The energy needed to effect this is taken from the input energy (primary). **Solution:** these energy losses are reduced by making core from a soft magnetic material such as iron which magnetises and demagnetises easily.

TOPIC: ELECTRICITY AND ELECTROMAGNETIC INDUCTION II

THE BAND THEORY

Materials behave differently when it comes to conducting electricity.

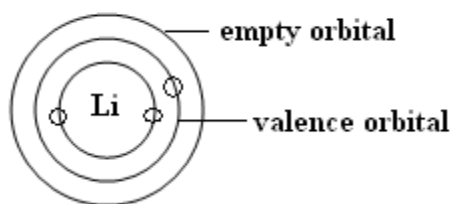
- a) Metallic substances are very good conductors of electricity e.g. copper metal.
- b) Insulators e.g. wood or glasses do not conduct electricity.
- c) However, materials called semiconductors show properties

intermediate those of metals and insulators. They conduct electricity but they are not very good conductors.

Band theory explains why the substances in such a way.

In atoms, some shells contain electrons and they are called **valence shells or orbitals**.

However, there are some shells that are empty and these are called **empty shells or orbitals**. For example the diagram of lithium atom below shows both empty and valence orbitals.

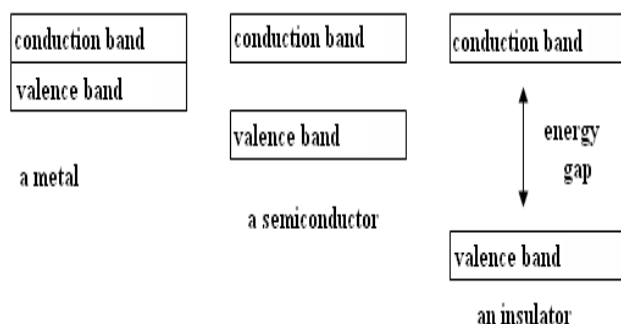


When atoms become bonded, the valence orbitals of all atoms interact to form new valence orbitals. However, these still remain to each atom and they form the **valence band** (since they are many).

The empty orbitals interact to form delocalised orbitals. These orbitals are available to all atoms and they are called the **conduction band**.

In terms of energy, the conduction band is higher than the valence band.

Note: For an electron to move or conduct it must be in the conduction band. Study the following figures.



- a) Metals are good conductors of electricity because it is easy (needs less energy) to promote an electron from the valence band to the conduction band there is no gap between the two bands.

- b) Semiconductors conduct when some energy (voltage) is supplied so that electrons are promoted from the valence band to the conduction band. The energy gap between the valence band and the conduction band is not too large.
- c) Insulators do not conduct electricity because of a large energy gap between the valence band and the conduction band. It is very difficult to promote electrons from the valence band to the conduction band.

All this explanation is what **band theory** is about.

ELECTRONICS

Electronics is being used more and more in our homes, factories, offices, schools, banks e.t.c. The development of semiconductor devices such as transistors has given us among other things, automatic banking machines, laptop computers, computer games e.t.c.

SEMICONDUCTORS

Semiconductors, of which silica and germanium are the two best known, are insulators if they are very pure, especially at low temperatures. Their conductivity increases at higher temperatures. This is the opposite to the case with metals. In metals the number of electrons available for conduction i.e. to carry current, is fixed. As the temperature increases the resistance increases due to increased vibration of the atoms making electron flow more difficult. In semi conductors, raising the temperature causes electrons to shake loose and thus the ability of a semi conductor to conduct increases with temperature. Consider the bonding in silicon in the figure below:



The stability of an element depends on the number of electrons in the outer shell; the most stable structures have eight outer electrons. Silicon has four outer electrons in its valence shell. This atom should gain (or more correctly share) four electrons in its outer shell to reach the next most stable structure. Each of the four surrounding atoms shares one electron with the central atom, which thus appears to have a stable outer shell of eight electrons. Each of the four original electrons forms what is called **covalent bond** with one electron from a surrounding atom.

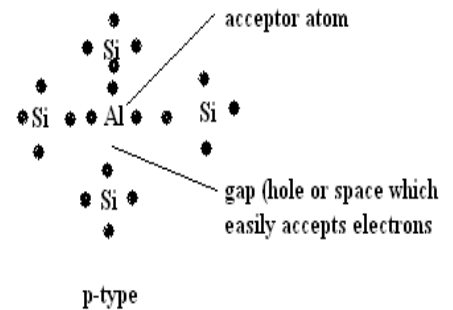
Therefore, pure silicon has all the outer, valence electrons in its atoms linked by covalent bonds; there are no free electrons and so pure silicon is a good insulator. However, a slight rise in temperature 'breaks' some of the bonds, by increasing the kinetic energy of the electrons until they have sufficient energy to break loose. These electrons are free to move and hence constitute an electric current.

The conductivity of semi conductors can be greatly increased by adding tiny but controlled amounts of certain other substances (called 'impurities') by a process known as **doping**. They can then be used to make diodes, transistors and integrated circuits.

P- type and n-type material (Semiconductors)

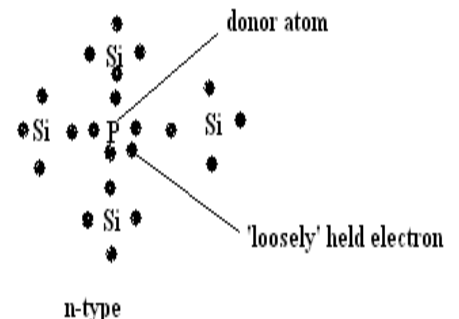
Nature of the doping element determines whether the semiconductor becomes p-type (positive type) or n-type (negative). Elements which have three valence electrons in outer shells of their atoms e.g. aluminium, boron, give p-type properties to the semiconductors.

These atoms, which do not give an electron to a covalent bond are known as **acceptor atoms**.



The acceptor atom (Al) in the figure above has seven electrons in its outer shells. If the gap or hole is filled by one electron the structure of the shell becomes stable. This positive 'hole' readily accepts electrons which may be moving in the silicon. However, the electrons move from atom to atom, and an electron which moves into the 'hole' must leave a 'hole' in the atom it has come from. Therefore the positive 'hole' appears to move through the material in the opposite direction to the electron movement. This improves the conductivity of the silicon since aluminium is accepting electrons and it is known as an **acceptor impurity**.

Elements which have five electrons in the outer shell of their atoms e.g. phosphorous give n-type properties to the silicon. These atoms are called **donor atoms**, and each forms covalent bonds with four surrounding atoms of silicon. One of the donor atom's outer electrons is held very loosely by the protons in the nucleus. This electron can easily be detached by thermal energy or by a small potential difference applied to the silicon.



The atoms of the impurity increase the conductivity of the silicon by providing a majority of positive charge carriers ('holes') in the p-type silicon and a majority of negative charge carriers (electrons) in the n-type silicon.

It must be stressed that both p-type and n-type silicon are electrically neutral. Each atom of silicon or impurity has an equal number of electrons and protons and hence is electrically neutral. However, the 'doped' material has the ability to accept electrons easily from certain surrounding atoms or to release electrons easily to the atoms of the semiconductor.

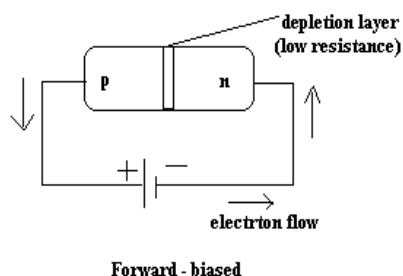
SEMICONDUCTOR DIODE

A diode is a two-terminal, one-way device which lets current pass through it in one direction only.

The p-n junction diode

A diode can be produced by doping a crystal of pure silicon so that a junction is formed between p-type and n-type regions. P-type material meets n-type material across a narrow layer called the **depletion layer**. Material in this layer conducts very poorly.

If a potential difference (p.d.) is applied across the crystal as in the figure below, current flows. Despite the poor conducting ability of the material in the depletion layer, the layer is so narrow that its resistance is low. The crystal in this case is said to be **forward biased**.

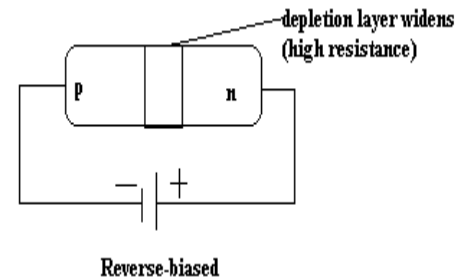


Note that:

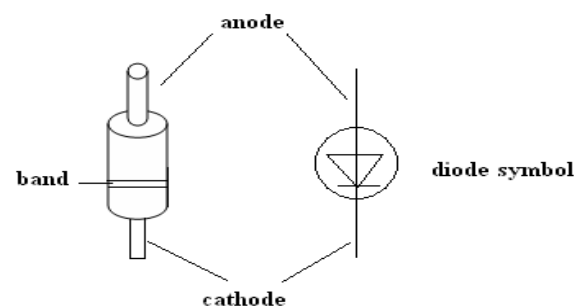
- i) the p-type material has been made positive
- ii) the p.d. is assisting the natural tendency of electrons in the n-type region to move across and fill holes in

the p-type

If the p.d. is applied in the opposite direction as in figure below, the current through the crystal is virtually zero. This is because electrons and holes are pulled away from the depletion layer, making it much wider and greatly increasing its resistance. The crystal in this case is said to be **reverse biased**.

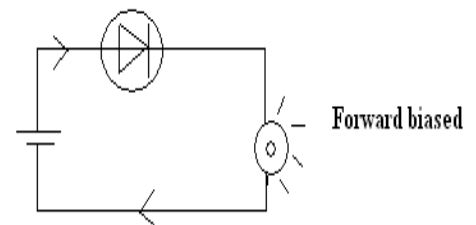


A diode and its symbol

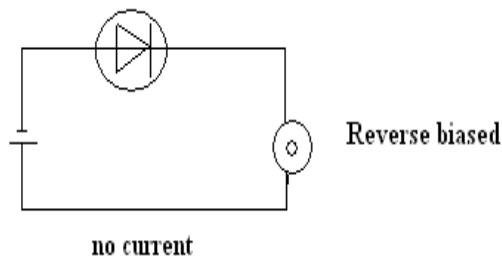


The wire nearest the band is the cathode and the one at the other end is the anode.

The diode conducts when the anode goes to the positive (+) terminal of the voltage supply and the cathode to the negative (-) terminal. It is then **forward biased**; its resistance is small and conventional current passes in the direction of the arrow on its symbol.



If the connections are the other way round, it does not conduct; its resistance is large and it is **reversed biased**.



A diode changes alternating current (a.c.) to direct current (d.c). In this way diodes are used as **Rectifiers**.

Rectification is the changing of alternating current (a.c) to direct current (d.c).

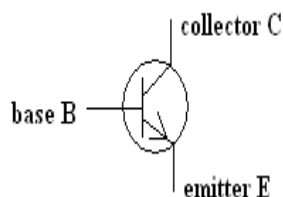
Nowadays, solid state diodes are preferred to vacuum diodes. This is because solid states diodes are

- a) Cheaper to make
- b) More rugged and less easily broken
- c) Much smaller
- d) Use up less electrical power

TRANSISTORS

Transistors are the small semiconductor devices which have revolutionised electronics. They are made both as separate components and also as parts of integrated circuits (I.Cs).

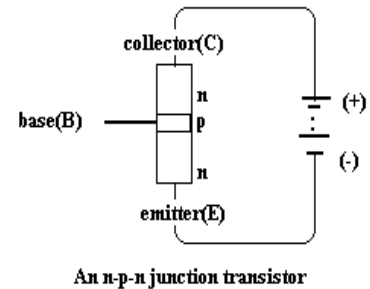
Transistors have three connections called the **base** (B) the **collector** (C) and the **emitter** (E). In the transistor symbol shown below, the arrow indicates the direction in which conventional current flows through it when C and B are connected to battery +, and E to battery -. Below is the diagram of a transistor's symbol:



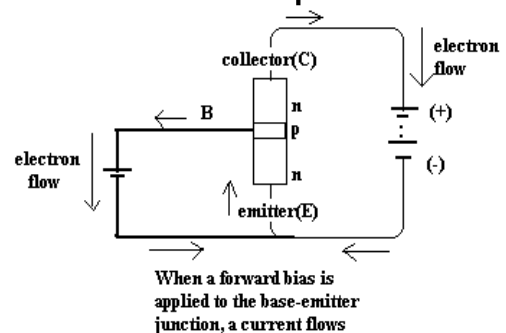
The n-p-n junction transistor

Basically, this consists of of a semiconductor chip, usually silicon, doped so that two p-n junction diodes are formed back to back. The n-type junction regions are the **emitter** and **collector** of the transistor; the thin p-type region is the **base**.

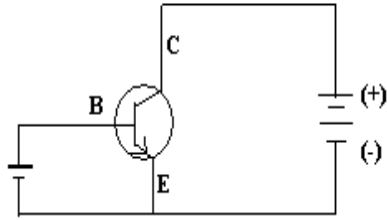
If the battery in the figure below is connected across the transistor as shown, the upper junction becomes reverse biased, so no current flows.



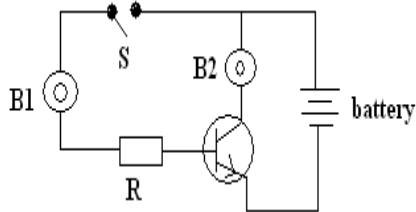
The following figure shows what happens if a small p.d. is applied across the lower junction so that it is forward biased. Not only does a current flow through the base, this current greatly reduces the 'current blocking' effect of the upper junction. The junction starts to conduct and current flows between emitter and collector. Typically, a small current in the base or **input** circuit can cause a current fifty to a hundred times larger to flow in the collector or **output** circuit.



In the following figure, the above figure has been redrawn using the circuit symbol for an n-p-n junction transistor. The arrow in the symbol indicates the conventional current direction, though the general movement of electrons is from emitter to base to collector. With the output and input circuits joined as shown, the transistor is said to be in **common-emitter mode**.



Below is another circuit diagram showing how a transistor is connected.



There are two current paths through a transistor. One is the base-emitter path and the other is the collector-emitter (via base) path.

The transistor's usefulness arises from the fact that it can link circuits connected to each path so that the current in one controls that in the other. When switch S is open the base current is zero and neither B1 nor B2 lights up, showing that the collector current is also zero even though the battery is correctly connected across the collector-emitter path. When switch S is closed, base current flows through resistor R to the positive terminal of the battery, B2 lights up but not B1. This shows there is now collector current (which passes through B2) and that it is much greater than the base current (which passes through B1, but it is too small to light it).

Therefore, in a transistor the base current switches on and controls the much greater collector current. Resistor R must be in the circuit to limit the base current which would otherwise create so large a collector current as to destroy the transistor by overheating.

In conclusion, with a small current in the circuit, a large current is produced in the main collector-emitter circuit.

USES OF TRANSISTORS

- An electronic switch:** Transistors have many advantages over other electrically operated switches. They are small, cheap, reliable, have no moving parts, their life is also indefinite and they can switch on and off millions of times a second.
- An amplifier of electrical current:** this means they increase current.

Therefore, a transistor is a semiconductor device that controls current as an electronic switch and can also amplify current as well.

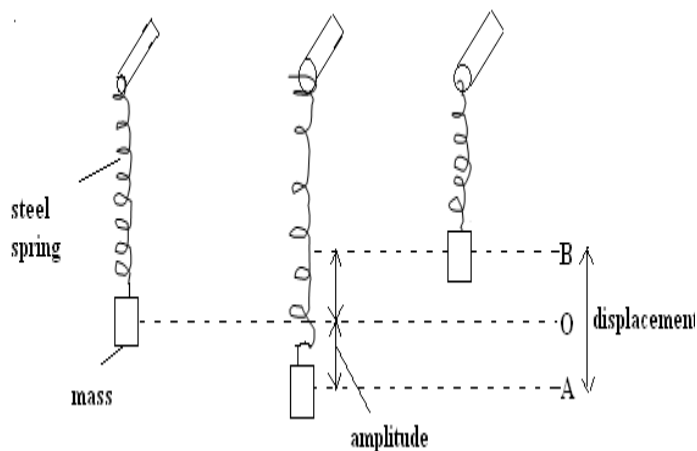
TOPIC: OSCILLATIONS AND WAVES

OSCILLATIONS: Repeating movements are called oscillations or vibrations. Oscillation can also be defined as a to and fro or an upward and downward movement of an object from its rest position.

Examples of oscillating materials include a spring, pendulum, cantilever e.t.c. Each oscillating system is an example of the continuous interchange of potential energy (PE) and kinetic energy (KE).

CHARACTERISTICS OF OSCILLATING SYSTEMS

Support a spiral steel spring as below and hang a mass from its lower end so that it is stretched several centimetres. Pull the mass vertically downwards a few centimetres and release it so that it oscillates up and down above and below its rest position.



One oscillation is from A to O to B to O to A.

Period: The time taken for a system to perform one oscillation. It is normally expressed in seconds.

Period = $\frac{\text{total time taken}}{\text{number of complete oscillations}}$

Frequency: The number of complete oscillations per second.

Frequency = $\frac{\text{number of oscillations}}{\text{total time taken}}$

Frequency is measured in Hertz (Hz). The relationship between frequency and period is:

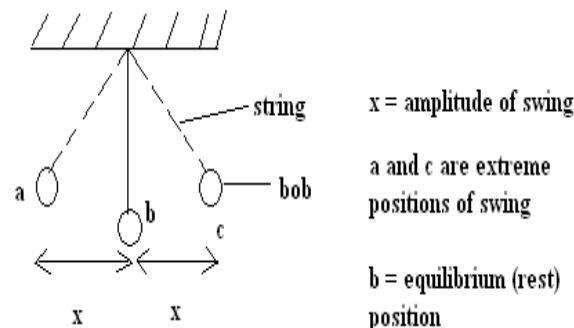
$$f = 1/T \quad \text{or} \quad T = 1/f$$

f is frequency and T is the period.

Amplitude: This is the maximum displacement of an oscillating from the equilibrium position (rest position). In the above diagram, the amplitude is from point O to A point O to B. Amplitude is measured in mm, cm or m.

Displacement: Is the distance between the extreme positions of an oscillating body i.e. from A to B.

A vibrating pendulum



One complete oscillation (cycle):

a to b to c to b to a.

Measuring the frequency f of a pendulum

Procedure:

A pendulum is set up just as in the diagram above. The stop-clock is set to zero reading and the pendulum bob is pulled aside about 10cm to the amplitude position **a** and released. Start the stop-clock as soon as the bob is released. Count 10 complete oscillation and stop the clock and record the time taken. The experiment can be repeated for several times and calculate the average frequency at the end. In all experiments the mass (bob), length of the string and the amplitude should not be altered i.e. should remain the same. A table of results can be drawn as below:

Time for 10 oscillations (s)	Time for 1 oscillation (s)	Frequency (Hz)
Average frequency		

Factors affecting frequencies of vibrating (oscillating) systems

Oscillating system	Factors affecting its frequency	Factors not affecting its frequency
Spiral spring	Mass on spring	Length of spring
	Material of spring	Changes in amplitude
Pendulum	Length of pendulum	Small amplitude changes
	Material of	Mass of bob

	pendulum	
	Amplitude changes	

Activity 1:

Aim: To find out the effect of mass on the frequency of string (pendulum)

Materials:

- a string
- different masses i.e. 20g, 25g, 30g and 35g
- a stand
- a wrist watch or stop watch

Procedure:

- Set the apparatus as shown in the diagram of pendulum above.
- Hang a mass of 20g and let it oscillate and record time taken for 10 oscillations.
- Repeat step b) without changing length of string for the 25g, 30g and 35g masses respectively.
- Record your findings in the table below:

Mass (g)	Time taken for 10 oscillations (s)	Time (T) for 1 oscillation (s)	Frequency (f), $f = 1/T$ (Hz)

From your findings **discuss** what is happening to frequency when mass is increased.

Activity 2:

Aim: To find out the effect of length on the frequency (f) of the pendulum.

Materials:

- a string of length 100cm
- a mass
- a stand
- 1 metre ruler
- Stop watch

Procedure:

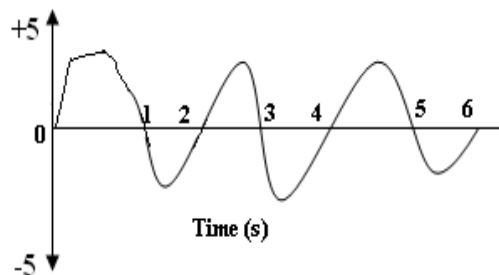
- Set the apparatus as shown in diagram of pendulum above with the length of string 75cm long.
- Set the pendulum to oscillate and record time taken for 10 oscillations for 75cm length of the pendulum.
- Repeat step b) with lengths of pendulum 60cm, 45cm, 30cm and 15cm respectively and record the results in the table below:

Length of pendulum (cm)	Time taken for 10 oscillations (s)	Time (T) taken for 1 oscillation (s)	Frequency (f) $f = 1/T$ (Hz)
75			
60			
45			
30			
15			

From your findings **discuss** what is happening to the frequency (f) when the length of the pendulum is decreasing.

Discussion question

Use the graph below to answer the questions that follow:



- How many complete oscillations are in the figure above?
There are **three** complete oscillations.
- What is the frequency (f) of these

oscillations?

Frequency = No. of complete oscillations

total time taken

$$\text{Frequency} = \frac{3}{6} \text{ Hz}$$

$$\text{Frequency} = 0.5 \text{ Hz}$$

c) What is the Periodic time (T)?

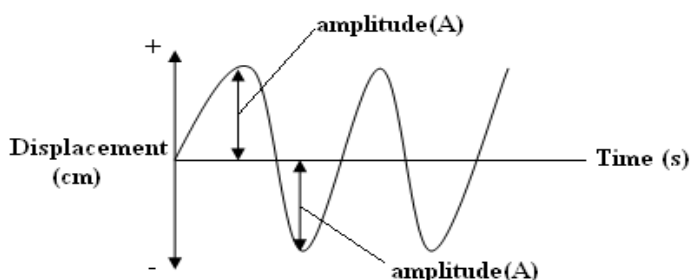
$$\begin{aligned}\text{Period (T)} &= 1/f \\ &= 1/0.5 \text{ s} \\ &= 2 \text{ s}\end{aligned}$$

d) What is the meaning of the (+) and (-) respectively above and below the 0 sign?

The positive (+) sign simply means the upward direction and the negative (-) sign simply tells us the downward direction of the oscillating body i.e. the oscillating body was moving in opposite directions.

Ideal and real oscillations

These are perfect oscillations whereby both potential energy (PE) and kinetic energy (KE) are conserved. No energy is lost. The amplitude of the oscillating body remains constant as shown in the figure below:



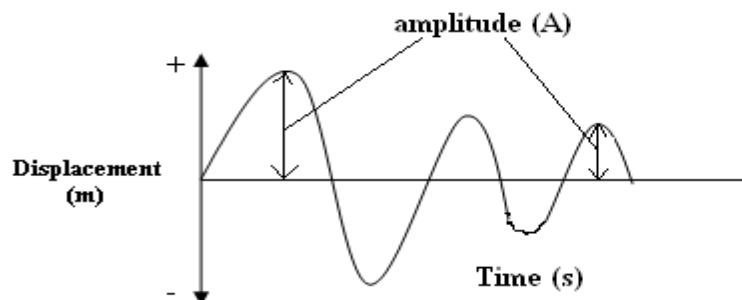
However, it is **not** practical to achieve such a situation due to:

- air resistance
- friction at the pivot and where the bob is connected to the string
- gravitational force on the bob

Damped oscillations

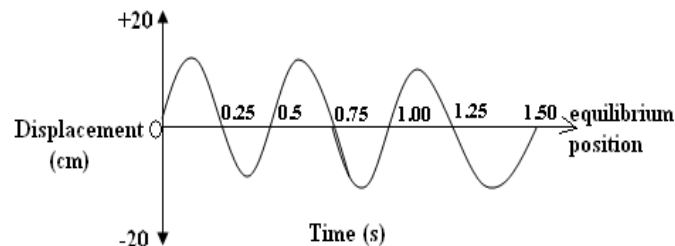
These are oscillations whereby both potential and kinetic energies are not conserved. There is energy loss due to friction. The amplitude of the oscillating system

decreases with time as shown in the figure below:



Work:

The graph below is that of a swinging pendulum. You are required to answer the questions that follow:



- What name is given to such oscillating system pattern?
- What is the value of the amplitude?
- What is the periodic time (T)?
- What is the frequency of the pendulum?
- Apart from a swinging pendulum, name any other vibrating source that would give a similar shape of the graph.
- Comment on the PE and KE in the system.

WAVES

Energy can be sent from one place to another by means of waves. In every case, the waves travel from a source to detector or receiver. Some of the examples of waves include sound, light, radio waves e.t.c.

Definition:

A wave can be defined as a disturbance which carries energy from one place to

another without transferring of matter.

A wave can also be defined as a transport of energy which does not permanently displace a medium through which it might pass and is often carried by a repeated and regular motion.

TYPES OF WAVES

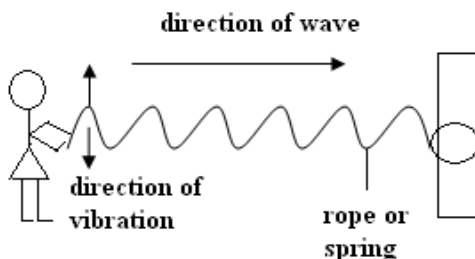
A disturbance which carries energy from one place to another without transferring matter is said to be a progressive (travelling) wave.

There are two types: **transverse** and **longitudinal**.

Transverse waves

In a transverse wave, the direction of the disturbance is at right angles to the direction of travel of the wave. In other words, the direction of motion of the particles is perpendicular to the direction of motion of the wave, and the wave is 'progressive' because the wave crest moves forward.

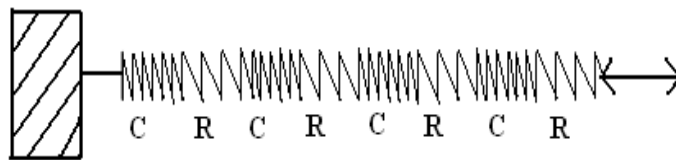
When transverse wave is propagated (i.e. energy is transmitted as a wave) the particles just vibrate in a direction perpendicular to the direction of motion of the wave. Water waves are transverse waves. So are all types of electromagnetic waves e.g. radio, TV, laser, infrared, visible light, X-rays and gamma rays. Below is a diagram showing a transverse wave generated in a rope or spring by a young girl.



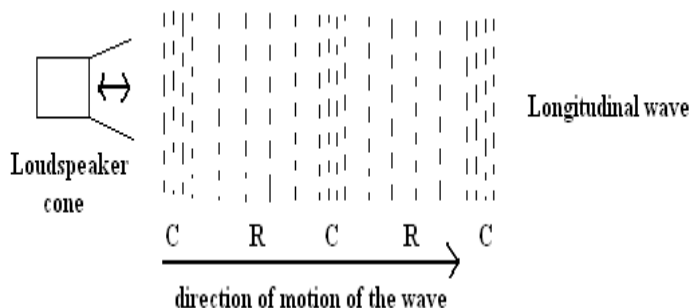
Longitudinal waves

Longitudinal waves are formed when the particles vibrate along the direction of motion of the wave. In other words, in a progressive longitudinal wave the particles of the transmitting medium vibrate to and fro along the same line as that in which the wave is travelling and not at right angles to it as in a transverse wave.

A longitudinal wave can be sent along a spring, stretched out on the bench and fixed at one end, if the free end is repeatedly pushed and pulled sharply. Compressions C (where the coils are closer together) and rarefactions R (where the coils are farther apart) travel along the spring.



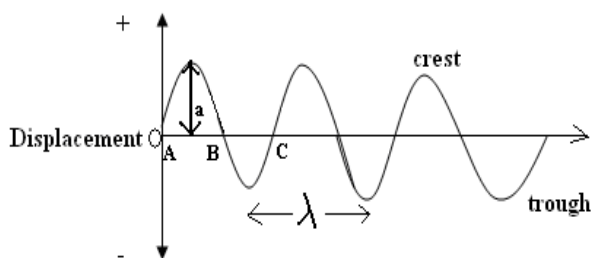
A sound wave, produced for example by a loudspeaker, consists of a train of compressions ('squashes') and rarefactions ('stretches') in the air. The speaker has a cone which is made to vibrate in and out by an electric current. When the cone moves out the air in front is compressed; when it moves in the air is rarefied (goes 'thinner'). The wave progresses through the air but the air as a whole does not move. The air particles (molecules) vibrate backwards and forward a little as the wave passes.



Wave characteristics

a) Wavelength (λ)

The wavelength of a wave, represented by the Greek letter λ (lambda), is the distance between successive crests or troughs. In a longitudinal wave, the distance between successive compressions is the wavelength.



b) Frequency (f)

The frequency f is the number of complete waves generated per second. A frequency of a wave is also the number of crests passing a chosen point per second. Frequency is measured in Hertz (Hz) i.e. number of complete cycles (waves)/second or vibrations/second. In a longitudinal wave, the number of compressions produced per second is the frequency f . It must be noted that the frequencies of the wave and its source are equal.

Frequency = $1/\text{period}$

$$f = 1/T$$

c) Speed (v)

The speed v of the wave is the distance moved by a crest or any point on the wave in 1 second.

d) Amplitude

The amplitude is the height of a crest or the depth of a trough measured from the undisturbed (equilibrium) position of what is carrying the wave. In general, amplitude is the maximum distance a point moves away from its rest position when a wave passes.

e) Displacement

This is the distance moved by a vibrating object or oscillating object in a stated direction.

f) Period (T)

A period is the time taken for one complete wave.

$$\text{Period } T = 1/f$$

In the above diagram, one complete wave is from point A through B to C.

wave

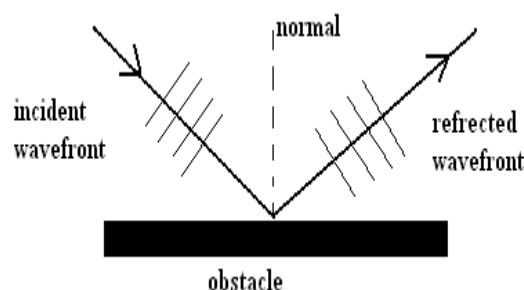
g) Crests and troughs

In the transverse wave diagram, the uphill is known as the crest and well below the equilibrium is known as the trough.

WAVE PROPERTIES

a) Reflection

This is the sending back or bouncing back of a wave. Like light, water waves are reflected when an obstacle is placed in their path, and the same laws of reflection apply i.e. angle of incidence (i) is equal to angle of reflection (r). The figure below shows the reflection of water wavefronts upon falling on a metal strip which is placed at an angle.

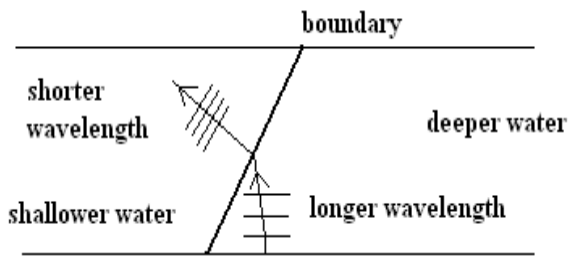


b) Refraction

This is the bending of a wave. Like light, water waves undergo refraction (bending) when they are slowed down. Water waves travel more slowly if they enter shallower water, so they can be refracted in a ripple tank by placing a piece of flat plastic in the bottom of the tray to reduce the depth.

When the waves enter the shallower water, there is no change in their frequency (as this depends only on the frequency of the source/vibrator). The reduction in speed does, however, cause a reduction in wavelength λ , a fact which emerges if you consider the equation: speed (v) = frequency (f) $\times$ wavelength (λ) with frequency kept constant. As the wavefronts close up on each other, the direction of travel

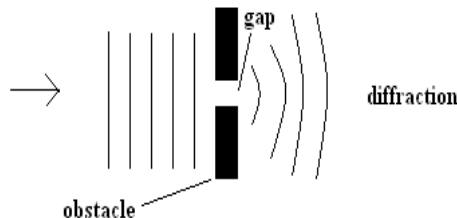
of the waves changes. Like light, water waves bend in towards the normal when they lose speed.



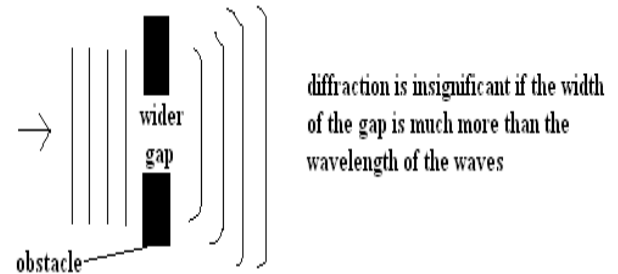
c) Diffraction

This is the bending of waves as they pass through a gap.

Something is observed when two obstacles with a narrow gap between them in the path of plane waves are placed in a ripple tank. Waves passing through the gap spread out in all directions and the wavefronts produced are circular. This is an example of an effect known as **diffraction** – the bending of waves as they pass around obstacles.



Diffraction is only significant if the size of the gap is comparable with the wavelength of the waves. The figure below shows what happens when plane waves pass through a much wider opening. The waves continue in their original direction and the wavefronts remain straight. Some diffraction does occur at the edges of the wave 'beam', but the effect is slight. In other words, some spreading occurs but it is less obvious.

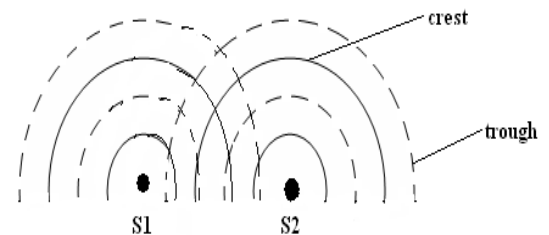


d) Interference

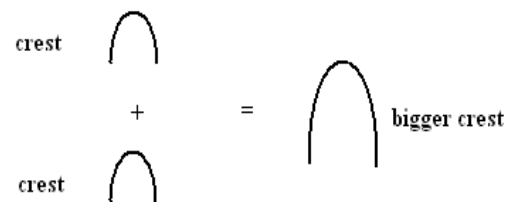
This is the overlapping of sets of waves.

If two identical sets of waves travel through the same region of water, they may, depending on their phase, reinforce each other or cancel each other out. The effect is known as **interference**.

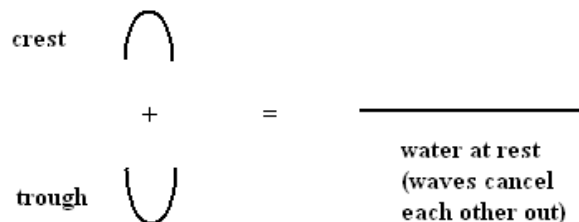
When two sets of continuous circular waves cross in a ripple tank, a pattern like the one below is obtained.



At point where a crest from one source, **S1**, arrives at the same time as a crest from the other source, **S2**, a bigger crest is formed and the waves are said to be in phase. At such points waves are said to be in phase. When waves are in phase it means there is **constructive interference**.



At points where a crest and a trough arrive together, the waves are exactly out of phase and they cancel out (if their amplitudes are equal). When waves are out of phase there is **destructive interference**.



WAVE EQUATION

The speed of a wave and its frequency and wavelength are linked by simple equation known as a **wave equation**.

Speed of a wave = frequency $\times$ wavelength

$$v = f \times \lambda$$

The equation implies that the higher the frequency of a wave the smaller its wavelength. This equation applies to any wave. The wave speed is measured in m/s, if frequency is measured in Hz and wavelength in m.

Examples:

- a) Calculate the speed of a wave if 4 complete waves are produced every second and the wavelength is 2m.

Solution:

$$\begin{aligned} \text{Frequency } f &= \frac{\text{No. of complete waves}}{\text{time taken}} = \frac{4}{1 \text{ s}} \\ &= 4 \text{ Hz} \end{aligned}$$

$$\text{Wavelength } \lambda = 2 \text{ m}$$

$$\begin{aligned} \text{Speed of a wave} &= f \times \lambda \\ &= 4 \text{ Hz} \times 2 \text{ m} \\ &= 8 \text{ m/s} \end{aligned}$$

- b) A vibrating tuning fork produces 500 waves every two seconds. If the speed of sound is 350 m/s, then work out the wavelength of the sound wave.

Solution:

$$\begin{aligned} \text{Frequency } f &= \frac{\text{No. of complete waves}}{\text{time taken}} = \frac{500}{2 \text{ s}} \\ &= 250 \text{ Hz} \end{aligned}$$

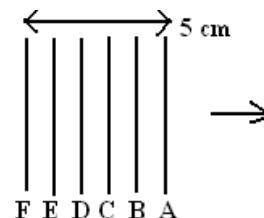
$$\begin{aligned} v &= f \times \lambda \\ \lambda &= v/f \end{aligned}$$

$$\begin{aligned} &= \frac{350 \text{ m/s}}{250 \text{ Hz}} \\ &= 1.4 \text{ m} \end{aligned}$$

Wavelength is 1.4 m.

Work:

- a) If some water waves are found to have a wavelength of 2cm and a frequency of 10 per second, work out the speed of the waves.
- b) The lines below are crests of straight ripples:



- What is the wavelength of the ripples?
 - If ripple A occupied 5 seconds ago the position now occupied by ripple F, what is the frequency of the ripples?
 - What is the speed of the ripples?
- c) The wave crests seen in a ripple tank are 5mm apart, and the frequency of the vibrator is 10Hz, what is the wave speed? If the frequency of the vibrator is doubled, what then is the spacing between the crests?

LIGHT

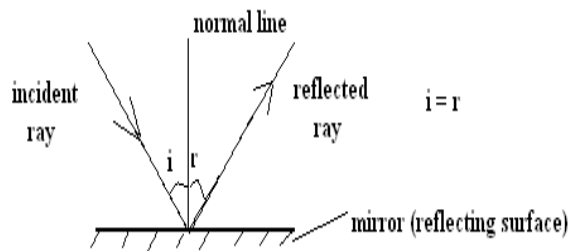
Definition:

Light is a form energy.

Properties of light

- a) **Reflection:**

This is the sending back of light. This is done when a ray of light strikes a plane surface. The light is reflected back at the same angle as the incident ray.



Incident ray is the incoming ray of light.

Reflected ray is the outgoing ray of light.

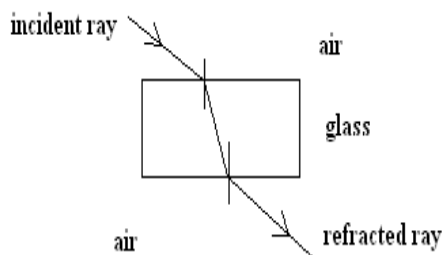
Normal line is the imaginary line that is perpendicular to the front surface of the mirror.

Angle of incidence (i) is the angle made between the incident ray and the normal line.

Angle of reflection (r) is the angle made between the reflected ray and the normal line.

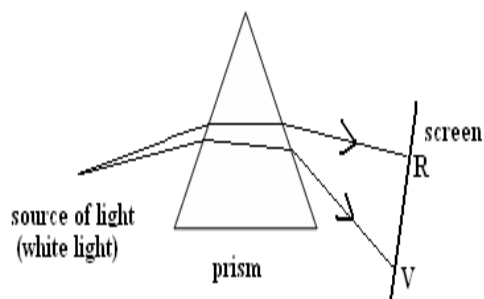
b) Refraction:

This is the bending of light when it changes the medium of travel i.e. from air to water. The ray bends towards the normal line when entering a denser medium and away from normal line when going into a less dense medium.



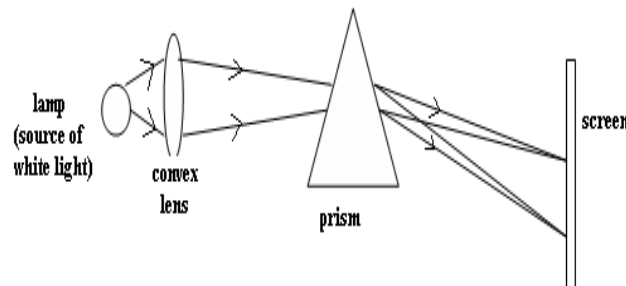
c) Dispersion:

This is the splitting up of white light into different colours of the spectrum after passing through a prism.



The colours are in this order: red, orange, yellow, green, blue, indigo and violet (**ROYGBIV**).

A pure spectrum is the one in which the colours do not overlap, as they do when a prism alone is used. It needs a lens to focus each colour as shown in the figure below:

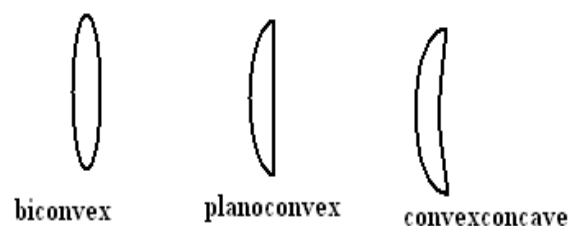


LENSES

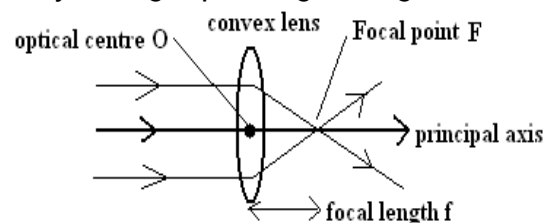
These are pieces of glass or clear plastic with curved surfaces. Lenses are used in many optical instruments. These lenses can either be curved outwards or inwards and there are two types; **convex** and **concave**.

Convex (converging) lens

A convex lens is the one which is thicker in the middle than at its edges. The three diagrams below show three shapes which act as converging lenses.

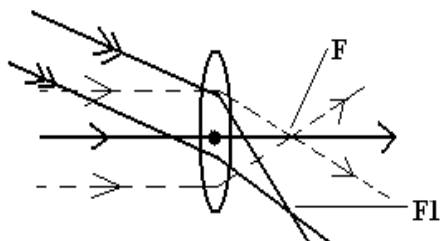


A convex lens converges (brings together) parallel rays of light passing through.



The line which passes symmetrically through the optical centre O of the lens is called the **principal axis** of the lens. All rays close to and parallel to the principal axis pass through a common point F on the axis called

the **focal point**. The distance from the optical centre O of the lens to the focal point F is known as the **focal length f** of the lens. If the parallel rays which are not parallel to the principal axis are incident on the lens these rays will also focus at one point F1 which lies directly below F. A plane passing through F and F2 is called the **focal plane** of the lens. Any parallel beam of light rays incident on the lens will be refracted so as to form a point which lies on the focal plane. This can be easily understood in the figure that follows:

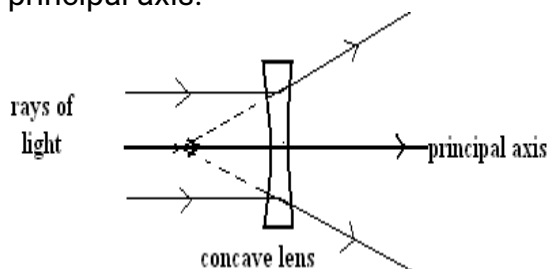


Concave (diverging) lens

A diverging lens is the one which is thinner in the middle than it is at its edge. The figures below show diverging lenses.



Concave lenses diverge (spread out) parallel rays of light that pass through the lens. In other words, light rays passing through a concave or diverging lens are bent away from the principal axis.



Lenses are used in many different kinds of optical instruments – cameras, telescopes, microscopes, binoculars e.t.c. and the focal length f of the lens is an important property

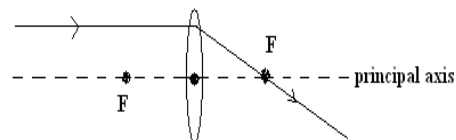
of a lens. The focal length helps when deciding which lens to use in which instrument.

It should also be noted that the more curved the lens faces are, the smaller the focal length f and the more powerful is the lens.

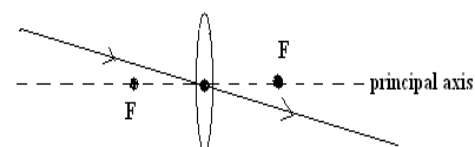
IMAGE FORMATION (RAY DIAGRAMS)

An image of any object may be located by use of ray diagrams. Ray diagrams are always drawn to scale using lines to represent rays. The position of the object decides what kind of light ray diagram will be drawn and the image position. Information about the images formed by a lens can be obtained by drawing any two of the following rays:

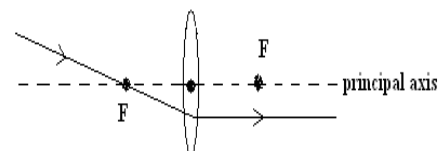
- A ray parallel to the principal axis which is refracted (bent) through the principal focus F on the other side of converging lens.



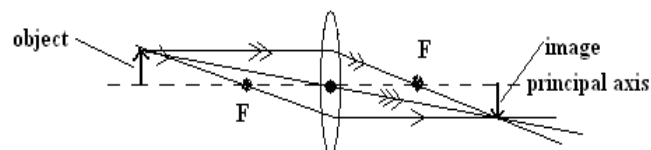
- A ray through the optical centre C which is undeviated (not refracted or bent).



- A ray through the principal focus F which refracts (bent) parallel to the principal axis.



All rays always start from the same point and a combination of **any two** of three will give us the position of the image.



The images formed from objects which are placed at different object distances have different characteristics. The following three characteristics show the nature of the image produced depending on the object distance.

- Upright or upside down
- Real or virtual
- Magnified (enlarged) or diminished.

MAGNIFICATION (m)

This is the comparison of the image formed to the object and can be calculated using the following relationship:

Magnification = $\frac{\text{image distance (v)}}{\text{object distance (u)}}$ or

Magnification = $\frac{\text{image height (h}_i\text{)}}{\text{object height (h}_o\text{)}}$

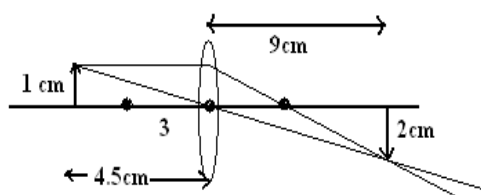
It should be noted that magnification has **no units** since the units of denominator and numerator cancel each other out.

Example:

An object 2cm high stands on the principal axis at a distance of 9cm from a convex lens. If the focal length of the lens is 6 cm, what is the position, height, magnification and nature of the image? (Scale: 1 cm on paper represents 2 cm)

Solution:

According to the scale, on paper height of object = 1 cm, object distance = 4.5 cm and focal length = 3 cm.



The position of the image = $9 \times 2 = 18$ cm from the lens.

Height of the image = $2 \times 2 = 4$ cm from the principal axis.

Magnification = $v/u = 18/9 = 2$ or $h_i/h_o = 4/2 = 2$

The image formed is real, upside down (inverted) and magnified (enlarged).

After a series of experiments, positions of images and the nature of the images (with objects placed at different distances from the lens) can be determined. The table below summarises the relationship between object distance, image distance and nature of the image formed.

Object distance (u)	Image distance (v)	Nature of image
1. Infinity	At F	Real, upside down, diminished
2. After 2F	Between F and 2F	Real, upside down, diminished
3. At 2F	At 2F	Real, upside down, same size as the object
4. Between F and 2F	After 2F	Real, upside down, magnified
5. At F	Infinity	Image not formed
6. Between F and lens	On the same side of the object	Virtual, upright, magnified

Note: All distances are measured from the lens.

THE LENS FORMULA

Converging lenses can use a lens formula in order to solve light ray diagram problems by calculation. The formula is:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Where **u** is the distance from the

object to the lens, **v** is the distance from the image to the lens, and **f** is the focal length. It should be noted that **u**, **v** and **f** can be measured using any length unit, provided the same unit is used in each case.

Example 1:

An object 2 cm high stands on the principal

axis at a distance of 9 cm from a convex lens. If the focal length of the lens is 6 cm, what is the position and nature of the image?

Solutions:

$u = 9 \text{ cm}$, $f = 6 \text{ cm}$, $v = ?$

Using lens formula: $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$$\frac{1}{9} + \frac{1}{v} = \frac{1}{6}$$

$$\frac{1}{v} = \frac{1}{6} - \frac{1}{9}$$

$$\frac{1}{v} = \frac{1}{18}$$

$$v = 18 \text{ cm}$$

The image is real and 18 cm away from the lens.

Example 2:

An upright object 6 cm long is placed 30 cm in front of a converging lens of focal length 10 cm. by using lens formula, find the position, size and magnification of the image formed.

Solution:

$f = 10 \text{ cm}$, $u = 30 \text{ cm}$, height of object (h_o) = 6 cm

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{10} - \frac{1}{30}$$

$$\frac{1}{v} = \frac{1}{15}$$

$$v = 15 \text{ cm}$$

Magnification (m) = $\frac{\text{image distance}}{\text{object distance}}$

$$m = \frac{15 \text{ cm}}{30 \text{ cm}}$$

$$m = 0.5$$

But $m = \frac{\text{image height } (h_i)}{\text{object height } (h_o)}$

$$0.5 = \frac{h_i}{6 \text{ cm}}$$

$$h_i = 0.5 \times 6 \text{ cm}$$

$$h_i = 3.0 \text{ cm}$$

Therefore, the image is found at a distance of 15 cm, with a height of 3.0 cm and a

magnification (m) of 0.5.

Work:

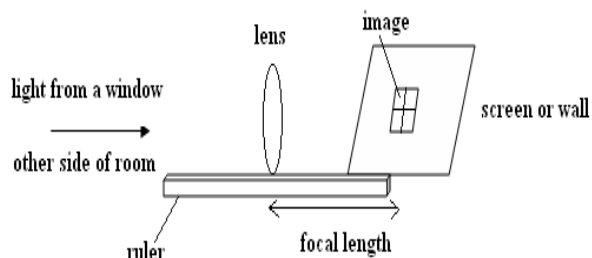
By means of **ray diagrams** (using suitable scales) or **lens formula**, work out the following problems:

- An object 6 cm high placed 18 cm in front of a lens of focal length 6 cm. Find the position and size of the image formed.
- An object 3 cm high is placed 24 cm away from a convex lens of focal length 8 cm. Find the position, height and nature of the image formed. If the object is moved to a point 3 cm away from the lens, what are the new position, height and nature of the image?
- An object 1.0 cm high stands on the principal axis of a converging lens of focal length 6 cm. What are the position, height, magnification and nature of the image formed given that the object distance is 9 cm?

DETERMINING FOCAL LENGTH OF A CONVEX (CONVERGING) LENS

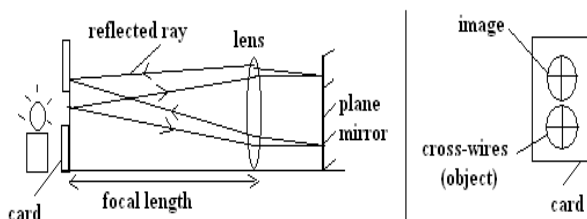
- Rough determination of the focal length of a converging lens.**

We use the fact that rays from a point on a very distant object i.e. at infinity, are nearly parallel. Sunlight rays arrive in parallel beams. Using the window as an object, you can form a sharp, inverted image on a sheet of white card (acting as screen) by adjusting the converging lens between the window and the screen.



b) **Determining the focal length f by the plane mirror method.**

The experimental arrangement is shown in the figure below. A plane mirror is placed behind the lens and a bright object in front. For instance, a set of illuminated cross-wires can be used as an object and its position is adjusted until its image appears alongside on the card. In this position, rays of light from the object are parallel when they strike the mirror and are reflected back to the lens as a parallel beam. The cross-wires and card mark the position of the principal focus of the lens.

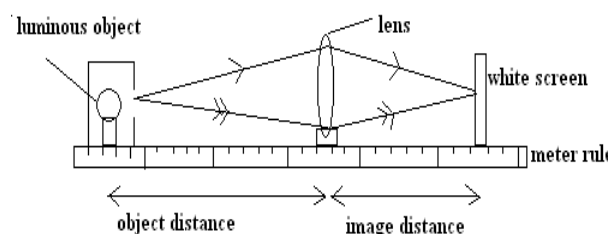


c) **Determining the focal length f of a convex lens by the object and image distance method.**

When a real, inverted image is focussed onto a screen using a convex lens, its focal length can be determined using the lens formula:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

An illuminated object, a convex lens in a holder and a white screen are set as below:



A real image can be produced only if the object distance is greater than the focal length. Therefore, set the object distance at some suitable (large)

value to be reasonably sure of focussing a real image. It may be necessary to adjust the lens in its holder until a circular patch of light is seen on the screen. Then move the screen until a sharp, inverted image is focussed on the screen. An illuminated wire grid or a flame of a burning candle is suitable object which can be focussed to give a sharp image.

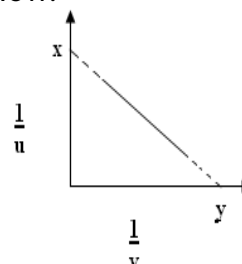
Using a metre ruler, measure the object distance u from the centre of the lens to the object and the image distance v from the centre of the lens to the sharp image formed on the screen. Record the values of u and v in a table of results like the one below. Repeat the experiment for more values of u . taking each pair of values of u and v , calculate the focal length f using lens formula:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Hence, calculate the average value of the focal length f

Object distance (u)/cm	Image distance (v)/cm	Focal length (f)/cm
70		
60		
50		
40		
Average		

Another way of treating the results in the above table is by plotting $1/u$ against $1/v$. a straight line graph is obtained and extrapolated as shown below:



At intercept x , $1/u = x$

$$u = 1/x$$

But $1/v = 0$, $v = 1/0 \rightarrow \infty$ (infinity: very very large)

$$v = \text{infinity}$$

For rays emerging from the convex lens meet at infinity, then these rays must be travelling parallel to each other. For this to happen the object must be placed at the focal point F of the lens and therefore, $u = f$

$$f = 1/x \text{ (since } u = 1/x \text{)}$$

At intercept y,

$$1/v = y$$

$$v = 1/y$$

But $1/u = 0$, $u = 1/0 \rightarrow \infty$, so $u = \text{infinity}$.

This means the object must be at infinity (so far away) like a star. The rays which are incident (arriving) on the convex lens are travelling parallel to each other and this applies that the image must be formed at the focal point F.

$$v = f$$

$$v = 1/y$$

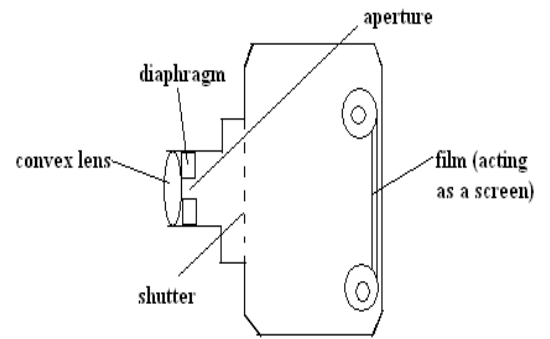
$$f = 1/y.$$

OPTICAL INSTRUMENTS

These are instruments that use light in their operations. The instruments to be studied are a camera and a projector. The human eye will also be looked at to compare it with a camera.

CAMERA

A conventional camera is a light-tight box in which a convex lens forms a real image on a film. The image is smaller than the object and nearer to the lens. The film contains chemicals that change on exposure to light; it is developed to give a negative. From the negative a photograph is made by printing. The following diagram shows some parts of a simple camera:



How a camera works

As already discussed, a **convex lens** is used to form a small, inverted, real image on a piece of photographic film. The film, which is normally kept in total darkness, contains a light-sensitive chemical i.e. silver bromide. When you press the camera button, a **shutter** in front of the film opens then shuts again, exposing the film to light for a brief moment only.

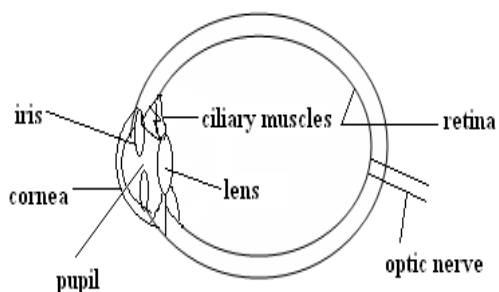
Focussing

In simple cameras the lens is fixed and all distant objects, i.e. beyond about 2 metres, are in reasonable focus. In other cameras exact focussing of an object at a certain distance is done by altering the lens position. For distant objects, the film must lie at the principal focus of the lens if the image is to be in sharp focus. For closer or nearer objects, the distance between the lens and film must be increased. This is achieved by screwing the lens backwards or forwards in its holder to suit the particular object distance.

The brightness of the image on the film depends on the amount of light passing through the lens when the shutter is opened and is controlled by the size of the hole (**aperture**) in the **diaphragm**. In some cameras this is fixed but others it can be made larger for a dull scene and smaller for a bright one.

THE HUMAN EYE

Below is diagram showing some of the parts of a human eye:



How the human eye functions

Like a camera, the eye uses a **convex lens** system to produce a real inverted image of an object in front of it. An image is formed on the **retina** of the eye by successive refraction at the **cornea**, **aqueous humour**, the **lens** and the **vitreous humour**. Electrical signals then travel along the **optic nerve** to the brain to be interpreted.

Objects at different distances are focussed by the **ciliary muscles** changing the shape and size of the lens, hence so the focal length of the lens – a process called **accommodation**.

Accommodation is the ability of the eye to focus light from objects, placed at different distances, onto the retina where they can be seen clearly and sharply.

The lens fattens (thickens) to view near objects, and the eye lens has a short focal length. The **iris** has a central hole, the **pupil**, whose size decreases in bright light and increases in dim light. Excessive brightness can damage the eye. The table below summarises the functions of some the parts of the human eye.

Part of the eye	Its use (function)
Iris	The iris controls the amount of light entering the eye. In a dark room, the pupil automatically grows larger to let in more light.
Eye lens	Used by the eye to focus a sharp and inverted image on a

	screen called retina.
Pupil	A black hole used to admit light.
Cornea	Forms a tough, transparent and protective outer cover to the eye. Some refraction takes place as light passes through it.
Retina	Found on the black inner surface of the eye and it is covered with light sensitive cells that make up a screen where images can be formed.
Ciliary muscles	Their job is to change the shape of the eye lens, so that light rays entering the eye are made to cross on the retina.
Optic nerve	The image formed on the retina is passed on to the brain by the optic nerve.

SIMILARITIES AND DIFFERENCES BETWEEN A CAMERA AND AN EYE

SIMILARITIES

- An eye uses a light sensitive retina as its screen and a camera uses a light sensitive film as its screen.
- An eye has a converging lens and a camera has a converging lens also.
- Inside surface of an eye is black just the same as in a camera.
- The iris controls light entering the eye and the shutter/diaphragm.
- Both use light in their operations.

DIFFERENCES

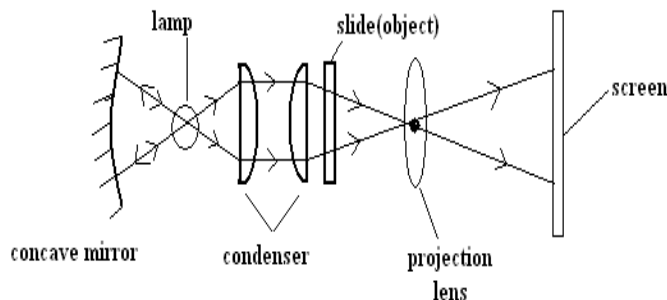
- The focal length of the eye lens changes while that of a camera is fixed.
- The pupil (hole) of the eye normally

opens while the aperture of the camera is closed except when picture is being taken.

- c) The eye has a fixed image distance while the camera has an image distance that changes.

PROJECTOR

A projector forms a real image on a screen of a slide in a slide projector, or of a film in a cine-projector. The image is larger than the slide or frame of film and is further away from the lens. It is usually so highly magnified that very strong but even illumination of the slide or film is needed if the image is also to be bright. This is achieved by directing light from a small but powerful lamp on to the 'object' by means of a concave mirror and a condenser lens system.



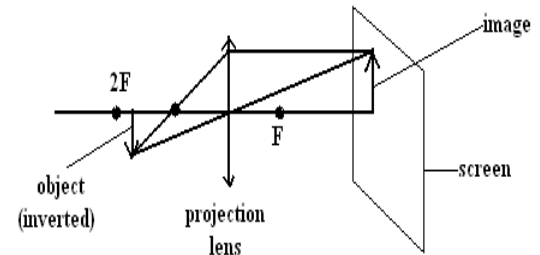
In a projector the object (**slide** or **film**) must be inverted to give an upright image and must be between $2F$ and F from the projection lens.

The **projection lens** is a lens of long focal length and its function is to produce a focussed, magnified, real and upright image (when object is inverted) on the screen.

The **lamp** (light bulb) is used to illuminate the object (slide or film). But much light from the lamp is lost because some rays do not strike and light up the object. The light rays that are directed away from the object are therefore reflected back towards the object by the **concave mirror**.

In order to concentrate the light from the lamp upon the object, a pair of converging lenses of short focal length called **condenser** is used. The function of the **condenser** is to converge and condense (concentrate) as

much light as possible onto the object (slide) so as to illuminate it very well. A ray diagram can be drawn to show how the image is formed.



The magnification (m) of the image on the screen is

$$m = \frac{\text{image height}}{\text{object height}} ; \quad m = \frac{v}{u}$$

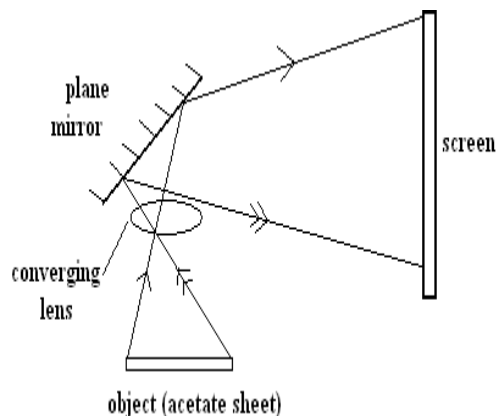
in order to increase the magnification of the image on the screen, there must be:

- a decrease in object height
- an increase in image height

To do this it is necessary to move the projection lens nearer to the object (slide or film) or the object nearer to the projection lens. Another way is by moving the screen away from the projection lens. Each of these actions will increase v and decrease u , and so increase m .

Another type of projector in fairly common use in education is the overhead projector (OHP). This consists of a quartz-halogen line-filament lamp sited at the centre of curvature of a converging mirror such that light goes directly from the lamp and by reflection from the mirror to a special type of lens called a Fresnel lens. The figure below shows the main components of an OHP in use and a ray diagram of the formation of the image on the screen.

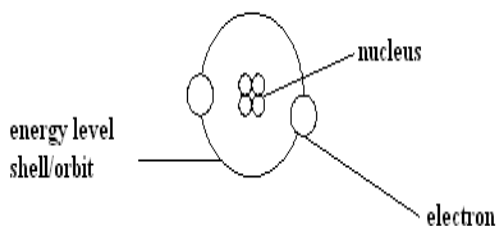
In using an OHP the teacher or lecturer can write or draw on the acetate sheet without turning away from the audience and the image is projected onto a white screen which is also facing the audience.



TOPIC: NUCLEAR PHYSICS

STRUCTURE OF AN ATOM

Everything is made of atoms. An atom is referred to as uncuttable (indivisible) basic unit of matter. The atom consists of the central part called the **nucleus** and negatively charged particles called **electrons** that are found in orbits around the nucleus. Atoms are far too small to be seen with any ordinary microscope. However, by using very advanced microscope, scientists have been able to develop models of structure of an atom and it appears as shown below:



Composition of a nucleus

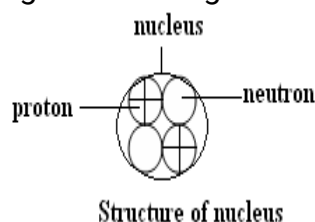
The central nucleus is made up of the protons and neutrons. Protons have a positive (+) charge while neutrons have no charge. Electrons are negatively (-) charged. Normally, an atom has the same number of electrons as protons, so its overall charge is zero. As an example, consider a sodium atom (Na): it has atomic number of 11 (number of protons) and around the nucleus there are 11 electrons.

$$\begin{aligned}\text{Total charge of protons} &= +11 \\ \text{Total charge of electrons} &= -11 \\ \text{Overall charge} &= +11 - 11 = 0\end{aligned}$$

Protons and neutrons in the central nucleus are collectively called **nucleons**. Each proton and neutron is about 1800 times heavier than an electron, so virtually all of an atom mass is in its nucleus or in other words, an electron has negligible mass.

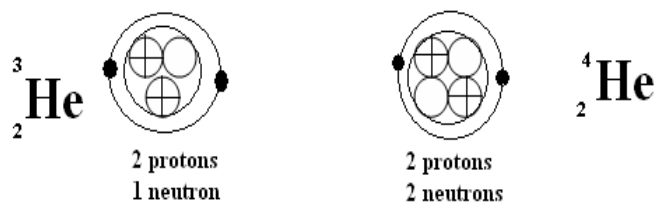
Electrons are held in orbits by the force of attraction between opposite charges while protons and neutrons are bonded tightly together in the nucleus by a different kind of force called **strong nuclear force**.

Each element has a different number of protons in its atoms; it has atomic number (Z), sometimes called **proton number**. Below is diagram showing the structure of nucleus:



Isotopes and Mass number

The atoms of any one element are not all exactly alike. Some may have more neutrons than others. For example, look at the structures of two helium atoms in the following diagram:



The difference in the number of neutrons in the two helium atoms above results to two different atoms, one with a mass number of 3 amu while the other with a mass number of 4 amu and are called **helium-3** and **helium-4** respectively. These different types of atoms of the element are called **isotopes** i.e. atoms of the same element with different mass numbers due to the difference in the number of neutrons in their nuclei.

Most elements are a mixture of two or more isotopes. Isotopes have the **same atomic number** (number of protons) but different **mass numbers** (total number of protons and

neutrons in the nucleus).

RADIOACTIVITY

Some substances contain atoms with unstable nuclei. In time, each unstable nucleus disintegrates (breaks up). As it does so, it shoots out tiny particles and, in some cases, a burst of wave energy as well. The particles and waves 'radiate' from the nucleus, so these are called **Nuclear Radiations**.

The materials whose unstable nuclei disintegrate and emit nuclear radiation are known as **Radioactive Materials**. The process whereby the nuclei disintegrate and emit nuclear radiation is called **Radioactive Decay**.

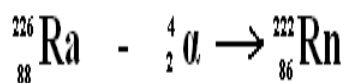
TYPES OF NUCLEAR RADIATION

There are three main types of nuclear radiation and these are:

a) ALPHA PARTICLES (α)

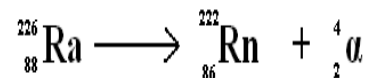
A nucleus of an atom of a particular substance is said to have undergone alpha decay when it loses two (2) protons and two (2) neutrons leaving the nucleus with 2 protons and 2 neutrons less than before. For example, Radium-226 (with atomic number 88), when the nucleus undergoes alpha decay, it loses 2 protons and 2 neutrons making the atomic number to drop by 2 and the mass number by 4 i.e.

$$\begin{aligned}\text{Atomic number} &= 88 - 2 = 86 \\ \text{Mass number} &= 226 - 4 = 222\end{aligned}$$



The equation above is showing that the Radium-226 nucleus, on the left hand side, is losing 2 neutrons and 2 protons forming a new nucleus on the right hand side with the atomic

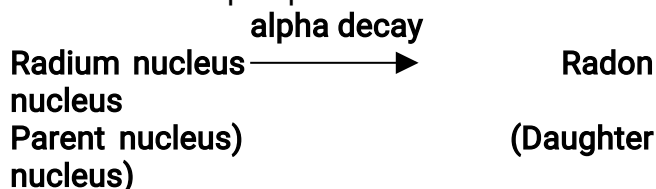
number 2 less than before. The equation can be rewritten as shown:



The equations above are referred to as **Nuclear Equations**. From the equation above, there are two products which have been formed on the right hand side and these products are called **Daughter nucleus** and **Alpha particle** respectively, and the original nucleus is called **Parent nucleus**.

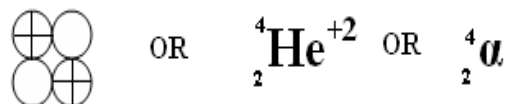
During the alpha decay:

- The top numbers balance on both sides of the equation i.e. ($226 = 222 + 4$), so that the nucleon number is unchanged.
- The bottom numbers also balance on both sides of the equation i.e. ($88 = 86 + 2$), so that the charge is conserved.
- A new element is formed with atomic number 2 less than before and the mass number 4 less than before.
- Checking on the periodic table the element with atomic number 86 is called Radon (Rn). So the Radium (Ra) nucleus has undergone alpha decay to form a Radon (Rn) nucleus and alpha particle i.e.



+ alpha particle (Helium nucleus)

The alpha particle is the **helium-4 nucleus** and can be shown as follows:



Work:

Plutonium (Pu) -240 has atomic number 94

and it undergoes alpha decay to form Uranium (U) -236.

- Write the nuclear equation for the decay.
- Identify the parent and daughter nuclei from the equation.

PROPERTIES OF ALPHA PARTICLES

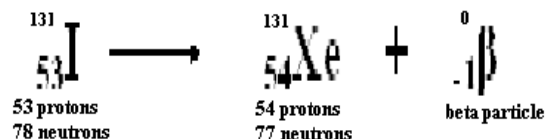
- Each alpha particle is 2 protons and 2 neutrons and it is identical to Helium-4 nucleus.
- Alpha particles have a relative charge of +2.
- Alpha particles have a relative bigger mass as compared to the other particles.
- Alpha particles have strong ionising effect, that is to say, alpha particles have a better ability to remove electrons from their shells in atoms. Remember alpha particles are positively charged and electrons are negatively charged so there is attraction.
- Alpha particles are not very penetrating; they are easily stopped by a thick sheet of paper, or by skin, or by a few centimetres of air.
- Alpha particles are deflected by magnetic and electric fields.
- Alpha particles have a speed of up to 0.1 x speed of light (one-tenth of the speed of light).

b) BETA PARTICLES (β)

During beta decay a neutron in the nucleus decays (breaks) into a proton and an electron. The electron leaves the nucleus at a high speed. The radioactive particle is the **electron**. The number of nucleons does not change, but we have got one proton more than before decay.

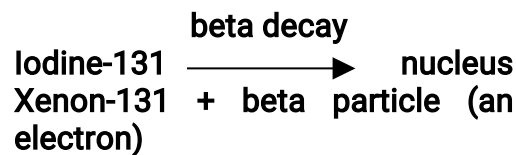
For example, Iodine – 131 (atomic number 53), decays by beta emission. There is addition of proton in the

nucleus making the atomic number to be 54 i.e.



This is a nuclear equation for beta minus decay. During this type of decay:

- The top numbers balance on both sides of the equation ($131 = 131 + 0$), so the nucleon number is conserved.
- The bottom numbers balance on both sides of the equation [$53 = 54 + (-1)$], so that the charge is conserved.
- A new element is formed, with an atomic number 1 more than before.
- Checking at the periodic table the element with an atomic number 54, it is found to be Xenon-131 nucleus as indicated in the equation before:



(these are decay products)

Work:

Radium (Ra) – 228 with atomic number 88 undergoes beta decay to form Actinium – 228 (Ac). Write a nuclear equation to show the decay process and identify the parent and daughter nuclei.

PROPERTIES OF BETA PARTICLES

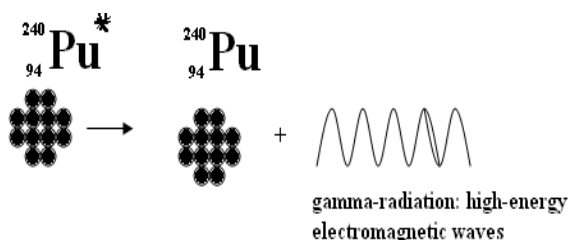
- The beta particle is an electron.
- Beta particles have a relative charge of -1.
- Beta particles have low mass.
- Beta particles have weak ionising effect.

- Beta particles are penetrating, but stopped by a few millimetres of aluminium or any other metal.
- Beta particles are deflected by magnetic and electric fields.
- Beta particles have a speed of up to $0.9 \times$ speed of light (nine-tenth of the speed of light).

c) GAMMA RADIATION (γ)

- When a gamma radiation occurs, high energy electromagnetic waves are emitted from the atomic nucleus. These waves are photons (small amount of energy released at a time), which have got higher frequency. A gamma radiation can happen after an alpha decay or beta decay.
- With some isotopes, the emissions of these alpha and beta decays leave the protons and neutrons in an excited arrangement. As the protons and neutrons rearrange to become more stable, they lose energy. This energy is emitted as a burst of gamma radiation/ray.
- Gamma emission by itself causes no change in mass number or atomic number of the nucleus.

The diagram below shows gamma decay process of Plutonium-240 forming Plutonium-240 with its both mass and atomic numbers unchanged:



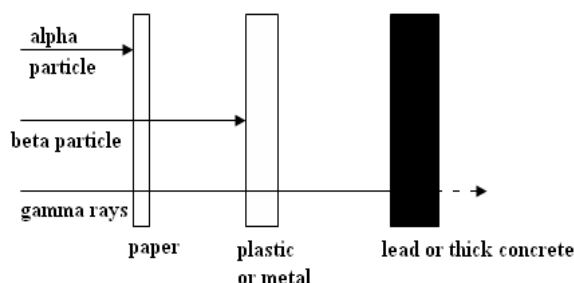
PROPERTIES OF GAMMA RAYS

- They are electromagnetic waves similar to X-rays.
- Gamma rays have a speed of light.
- Gamma rays have a very weak ionising effect, that is, gamma rays

have a very little ability to remove electrons from their paths in the atoms.

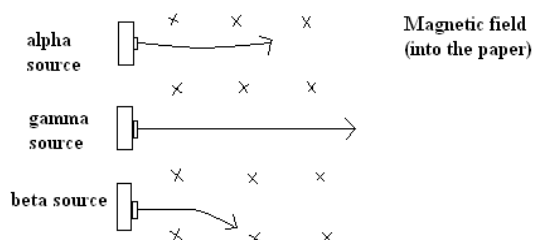
- Gamma rays are very penetrating; never completely blocked or stopped, though lead and thick concrete can reduce the intensity of penetration.
- Gamma rays are not deflected by magnetic or electric fields.
- Gamma rays have no mass and no charge.

DIAGRAM SHOWING PENETRATING DISTANCES OF THE THREE DECAYS:



THE EFFECT OF ELECTRIC AND MAGNETIC FIELDS

Like any stream of moving charged particles, alpha and beta rays are deflected by electric and magnetic fields. Beta particles show much greater deflections than alpha particles because of their lower mass. As stated already, gamma rays are not deflected by a magnetic field.

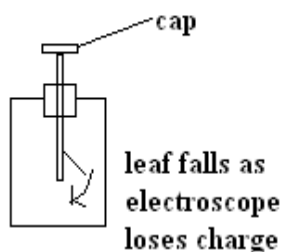


DETECTING NUCLEAR RADIATION

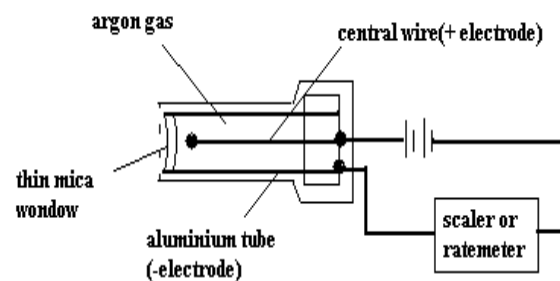
Most methods of detecting alpha, beta and gamma rays are based on the fact that these radiations have an ionising effect.

- **Leaf electroscope:** Alpha particles can be detected using a charged electroscope. The source is brought near the cap of a charged leaf electroscope, and the air around the

cap becomes ionised. Ions with a charge opposite to that on the electroscope are attracted to the cap where they neutralise some charge on it. The leaf falls as a result. In effect, the ionised air conducts charge away from the electroscope. The more intense the radiation, the faster the leaf falls. The method is suitable for detecting beta and gamma radiation as these cause insufficient ionisation of the air.



- **Geiger-Muller tube:** the tube contains argon gas at low pressure. A p.d. of about 400V is applied between the central wire electrode and the aluminium wall of the tube. The end of the tube is sealed by a mica 'window' thin enough to allow alpha particles to pass into the tube as well as beta and gamma radiation. When a charged particle or gamma radiation enters the tube, the argon gas becomes ionised. This triggers a whole avalanche of ions between the electrodes. For a brief moment, the gas conducts and a pulse of current flows in the circuits. The circuit includes either a **scaler** or a **ratemeter**. A scaler counts the pulses and shows the total on a display. A ratemeter indicates the number of pulses or counts per second. The complete apparatus is often called a **Geiger counter**.



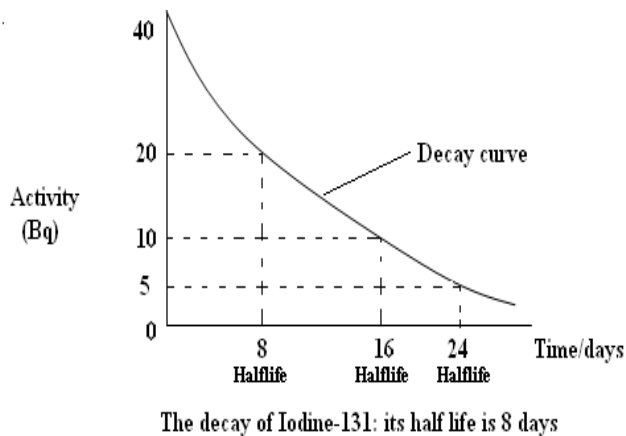
- **Solid-state detector:** The incoming radiation ionises atoms in a reverse-biased *p-n* junction diode. This makes the diode conduct and a pulse of current flows in the circuit as a result. As with GM tube, current pulses are detected using a scaler or a ratemeter.
- **Ionisation chambers/detectors**
- **Photographic plates**
- **Scintillation counter**
- **Spark counter**

HALF LIFE AND ACTIVITY

In radioactive sample, the average number of disintegrations per second is called the **Activity**. The SI unit of activity is the **Becquerel** (Bq). An activity of, say, 100Bq means that 100 nuclei are disintegrating per second.

Half life is the time required for the activity of a radioactive material to be reduced to half its original value. All radionuclide have a particular half life, some of which very long, while others are extremely short. For example, Uranium-238 has a long half life, 4.5×10^9 years. In contrast, Carbon-11 has a half life of only 20 minutes.

The graphs showing this decay are called **Decay curves**.



Example:

When a source is placed in front of a GM tube and scaler, the initial count rate after the background count has been deducted is 4000. After twenty minutes the count rate after deduction of the background count is 125. What is the half life of the source?

First method:

It is necessary to find out how many half lives there are in 20 minutes. Thus

1 st	half life	count rate	falls from	4000 to 2000
2 nd	"	"	"	2000 to 1000
3 rd	"	"	"	1000 to 500
4 th	"	"	"	500 to 250
5 th	"	"	"	250 to 125

Thus the activity has halved five times during the 20 minute interval and the half life is
 $20/5 = 4$ minutes.

Second method:

Original count rate $A_0 = 4000$

Number of half lives = n

Final count rate $A_n = 125$

$$A_0 = 2^n$$

$$A_n$$

$$4000 = 2^n$$

$$125$$

$$32 = 2^n$$

$$2^5 = 2^n$$

$$5 = n$$

Half life = $20/5 = 4$ minutes.

Work:

- The table below shows how the activity of a sample of Iodine-127 varies with time.

Time (min)	Activity (Bq)
0	75.8
10	57.4
20	43.5
30	33.0
40	25.0
50	19.0
60	14.4
70	10.9
80	8.3

- Plot a graph of activity against time.
 - From your graph find the time taken for the activity to drop from
 - 60 to 30 Bq
 - 40 to 20 Bq.
 - What is the half life of a sample of Iodine-128?
- When a source of radiation is placed in front of a GM counter, the initial count rate is 128. After 16 minutes, the count rate is 8. Calculate the half life of the source. (MANEB 2008)
 - Given that the half life of Iodine-128 is 25 minutes. If the activity of a sample its sample is 800Bq, what would you expect the activity to be after:
 - 25 minutes?
 - 50 minutes?
 - 100 minutes?

NUCLEAR ENERGY

The protons and neutrons in a nucleus possess nuclear potential energy – or **nuclear energy** for short. When alpha or beta particles are emitted by a radioactive isotope, they collide with surrounding atoms and make them move faster. In other words, the temperature rises as nuclear energy is transformed into thermal energy (heat). During radioactive decay, the energy released per atom is around a million times greater than that given off during a chemical change

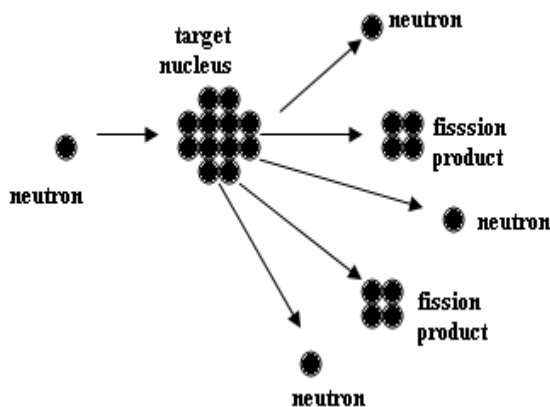
such as burning. However, rates of decay are usually very slow. Much faster decay can be produced in some cases by bombarding nuclei with other particles. Neutrons make the most effective missiles; being uncharged, they are not repelled by nuclei, can penetrate and disrupt them relatively easily.

Any process in which a particle penetrates a nucleus and changes it in some way is called a **nuclear reaction**.

Nuclear fission

When a neutron is made to strike and penetrates a nucleus, the nucleus becomes highly unstable and splits into two nuclei; shooting out two or three neutrons as it does so. This splitting process is what we call **nuclear fission**. Nuclear fission is an example of chain reaction and the fragments are thrown apart as **energy** is released.

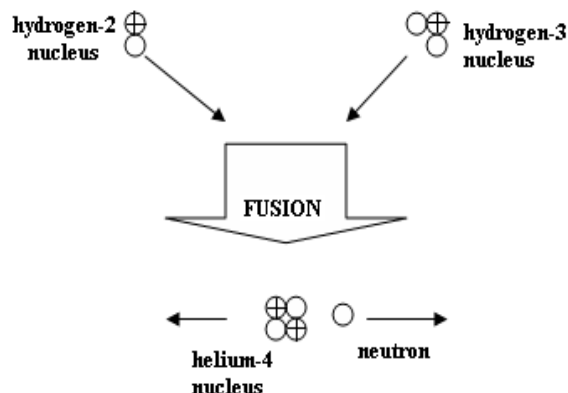
The number of nuclei undergoing fission multiplies rapidly, and an enormous quantity of energy is released in a short space of time.



Nuclear fusion

Nuclear fusion is the process by which multiple nuclei join together to form a heavier nucleus. Nuclei have to collide at very high speeds for fusion to occur, otherwise the repulsion between their charges prevents them from coming close enough to combine. It is accompanied by a release or absorption of energy. This reaction produces far more energy for a given mass of material than any fission reaction. The diagram below shows a

fusion reaction between two nuclei to form one heavier nucleus.



APPLICATIONS OF RADIOACTIVE SUBSTANCES

- Radioisotopes can be used to estimate the amount of wear in bearings. If the radioisotope is impregnated into bearings, the fine bearing filings are radioactive and are carried away by the oil. If a sample of the oil containing the filings is tested for radioactivity, the amount of wear can be estimated from the results.
- Leaks in pipes may be traced by introducing a small quantity of a radioisotope into the fluid in a pipe. A radiation detector can be used to determine where the radioisotope is escaping.
- Radiotherapy: Cobalt-60 emits high energy γ – radiation. This is used in cancer therapy instead of the more elaborate high energy X-radiation. These gamma rays can penetrate deep into the body and kill living cells. So a highly concentrated beam from Cobalt-60 source can be used to kill cancer cells. Treatment like this is called **radiotherapy**.
- Radioisotopes are widely used in industry, e.g. to check that the thickness of a material being produced is constant. As the material passes between the radioactive source and a counter, any variation in

thickness causes a change in the count rate (if the thickness decreases, the count rate increases) and irregularities can be pinpointed.

- Carbon-dating: Radioactive substances can be used to find the age of organic materials such as wood and cloth.
- Radioactive substances can be used in sterilisation as well as food preservation.
- Tracers: Radioactive substances are used as tracers i.e. checking the function of body organs. For example, to check thyroid function, a patient drinks a liquid containing Iodine-123, a gamma emitter.

Note: Naturally occurring radioactive substances have high nucleon numbers. It is

possible to make artificial radioactive substances by bombarding lighter nuclides with alpha-particles, protons or neutrons. The radioactive substances produced in this manner are known as **radioisotopes**.

DANGERS OF RADIOACTIVE SUBSTANCES

Some of the effects on humans of exposure to large doses, or prolonged small doses, of radiations are

- a. burns

- b. leukaemia (cancer of the blood)
- c. sterility (inability to produce children)
- d. some children born with serious abnormalities
- e. damage to the blood may lower resistance to normal diseases.

PRECAUTIONS WHEN HANDLING RADIOACTIVE MATERIALS

The growth and use of radioactive products has increased considerably since about 1930. Radioactive sources have become part of normal school equipment and although the sources are very weak, it is essential to take stringent safety precautions:

- a. The sources should only be handled by forceps provided and never touched by hand.
- b. They should never be pointed towards a person.
- c. Food should not be taken where the sources are being used, because it may become contaminated.
- d. Never smoke near a radioactive source.
- e. The user should wear rubber gloves, and hands should be washed after the sources have been put away safely.
- f. Do not stay much long within the area where there is radioactive substances because the longer the exposure time, the greater the risk.

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